
Imperfections and self testing in prepare-and-measure quantum key distribution

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Abstract

Quantum key distribution (QKD) protocols are intended to allow cryptographic keys to be generated and distributed in a way that is provably secure based on inherent limitations, such as the no-cloning principle, imposed by quantum mechanics. This unique advantage compared with classical cryptography comes with an added difficulty: key bits in QKD protocols are encoded in analogue quantum states and their preparation is consequently subject to the usual imprecisions inevitable in any real world experiment. The negative impact of such imprecisions is illustrated for the BB84 QKD protocol. Following this, the main part of this thesis is concerned with the incorporation of such imprecisions in security proofs of the BB84 and two semi-device-independent protocols against the class of collective attacks. On a technical level, by contrast with the vast majority of security proofs developed since the turn of the century, in which recasting the protocol into an equivalent entanglement-based form features heavily in the analysis, the main results obtained here are approached directly from the prepare-and-measure perspective and in particular the connection with the no-cloning theorem and an early security proof by Fuchs *et al.* against the class of individual attacks is emphasised.

This thesis also summarises, as an appendix, a separate project which introduces and defines a hierarchy of polytopes intermediate between the local and no-signalling polytopes from the field of Bell nonlocality.

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Foreword

This thesis reports research I was involved in during doctoral studies undertaken at the Laboratoire d'information quantique at the Université libre de Bruxelles during the period October 2010 – October 2014. I had the opportunity to work in a well-connected research group in quantum information theory with capable colleagues and where I was granted considerable autonomy to pursue research ideas that attracted my interest.

Doctoral research is not conducted in isolation. I would like to thank my current and former colleagues and office mates of the past four years – Jon Silman, Ross Duncan, Fred Ezerman, Manas Patra, Olmo Nieto Silleras, Cédric Bamps, and Damián Pitalúa-García – for various combinations of interesting and insightful discussions, keeping me motivated to get this thesis completed, and sometimes encouraging me to maintain some appearance of a social life. My co-supervisor, Stefano Pironio, followed my progress the most closely since I shared his office during the first year as a PhD student. As well as the sheer amount I learned through him simply by osmosis, I am also indebted to him for guiding me through some of the “tools of the trade” as a beginning researcher, notably navigating the publication process and refereeing. My thesis supervisor, Serge Massar, introduced me to the field of quantum information and encouraged me to start this PhD in the first place.

The final version of this thesis includes amendments recommended by the jury members: Nicolas Cerf, Pascal Kockaert, Nicolas Brunner, and Antonio Acín, as well as Serge Massar and Stefano Pironio.

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Chapter 1

Introduction

1.1 Quantum key distribution

1.1.1 General background

Quantum key distribution (QKD) [1, 2] is an approach to the problem of generating and distributing cryptographic keys for use in data encryption in a way that can be proved secure based on limitations inherent to quantum physics. Since its original proposal by Charles H. Bennett and Gilles Brassard in 1984 [3], QKD has emerged as one of the most promising practical applications exploiting features of quantum physics and among the most mature subfields of quantum information theory. QKD systems have been commercially available since around 2004 and are, at the time of writing, offered by at least four companies [4–7].

An overview of the motivation for QKD can be found in [8]. Briefly, in the setting considered, two spatially separated parties, traditionally called “Alice” and “Bob” in the literature, wish to be able to communicate privately. There are already various (classical) encryption schemes widely in use today, such as the RSA protocol commonly used to implement public key cryptography, however their security is subject to assumptions about an adversary’s technological and computational power and/or unproven conjectures concerning the difficulty of solving certain mathematical problems (e.g., in the case of RSA, prime factorisation). There are realistic circumstances in which this state of affairs could be considered unsatisfactory, particularly where long term secrecy is a requirement, as this requires encryption that can reasonably be expected to remain secure against even *future* technology

of a priori unknown capability. If, for instance, a message encrypted today is to remain secret for a period of, say, fifty years, the encryption scheme must remain practically unbreakable following any technological breakthroughs that may occur in the next half century, which may include the development of scalable quantum computers capable of, for instance, implementing Shor’s prime factorisation algorithm [9].

A resolution is to use an unconditionally secure encryption algorithm such as the one-time pad, in which successive bits of the message to be encrypted (if expressed in binary) are xored with the corresponding bits of a sufficiently long key. This, however, substitutes one problem for another: such unconditionally secure algorithms require an encryption key of the same length as the message to be encrypted, and a new encryption key must be generated and distributed – securely – every time a new message is to be transmitted.

The problem of regular and practical distribution of secret keys is where QKD is targeted. In its simplest description, the intent of a QKD protocol is that a cryptographic key randomly generated by one party (“Alice”) can be transmitted to a second distant party (“Bob”) in such a way that tampering by an adversary (usually called “Eve” in the literature) can be detected. This is achieved by Alice encoding her key bits on different nonorthogonal quantum states in such a way that any attempt to extract information by an eavesdropper will, with very high probability, disturb the transmission of information from Alice to Bob in a visible and detectable way. Since the distributed key should be random and in itself meaningless, failure of this test – indicating that an eavesdropper may have learned information about the key – is not fatal: Alice and Bob simply abort the protocol and may attempt to distribute a new key at some later time.

Exploiting fundamental physical limitations implied by quantum physics, such as the measurement-disturbance tradeoff or the no-cloning principle [10, 11], the use of quantum systems promises levels of security unachievable with any classical system when applied to the accomplishment of cryptographic tasks. With these benefits, however, comes an added difficulty. Unlike classical protocols intended for execution on a digital computing device and whose security is a property of a mathematically-defined algorithm, quantum protocols are specifications of physical systems that need to be implemented. The security of a real QKD implementation is thus dependent on the implementation closely matching the theoretical specification, and a gap between theory and practice can result in the security being compromised. This was most strikingly demonstrated by proof-of-principle hacks of commercial QKD systems [12, 13] at the beginning of the decade, which exploited specific properties of the components used that could allow an adversary to remotely tamper with or outright control their behaviour.

Implementation flaws permitting such hacks can in principle be fixed by improving the implementation to better correspond to the theoretical specification. Not all imperfections can be addressed in this way, however. Any QKD system, as with any experiment, will always be subject to finite precision of the implementation. In particular, the states prepared and the quantum measurements performed will never be exactly those required by the theoretical specification, the channel between Alice and Bob will never be perfectly noiseless or lossless, and Bob’s measurement devices will never have perfect detection efficiency. Such imperfections must thus be expected and accepted to some degree in any real QKD system and it is at the level of the theoretical security analysis that they must be accounted for.

1.1.2 Contribution and outline of this thesis

The main part of this thesis is concerned with the problem of state and measurement imprecisions in the case of the BB84 QKD protocol, the original protocol proposed by Bennett and Brassard in 1984 [3]. A second, more conceptual motivation was the development of techniques allowing the security of BB84-like protocols to be understood more directly from the prepare-and-measure perspective; this is by contrast with the majority of security analyses since around the year 2000 which recast the protocol under consideration into an entanglement-based form as a first step in the proof. In particular, the techniques that will be introduced in chapter 3 were originally inspired by an early security proof by Fuchs *et al.* [14] of the prepare-and-measure BB84 protocol against a restricted class of attacks called *individual attacks*, and it will be shown that security proofs against the larger class of *collective attacks* can be developed in a similar style.

The remainder of this chapter consists of an introduction to the aspects of QKD relevant to this thesis. This is not intended as a general introduction to QKD, which is already the subject of dedicated review articles [1, 2]. Experimental advances (covered in [15]) and protocols other than the BB84 protocol are not, for the most part, discussed here. The intent is rather to motivate and to place the main results of this thesis in context. Section 1.2 introduces and contrasts the prepare-and-measure and entanglement-based versions of the BB84 protocol and discusses both the correspondence between them and the limitations of this correspondence. Section 1.3 briefly discusses implementation imperfections. The emphasis is on the consideration of channel noise, which is standard in essentially all modern security proofs, and source state and measurement imprecisions, which comparatively few authors have studied. Section 1.4 is a brief introduction to the security of the BB84 protocol. The section begins with the no-cloning prin-

ciple as it was originally formulated by Wootters and Zurek [10] and Dieks [11] in 1982, which can be considered the intuition behind the security of the prepare-and-measure BB84 protocol. This is contrasted with the principle of monogamy of entanglement, which can be considered the basis for the security of the entanglement-based variant. Following this, a simplified derivation of the Fuchs *et al.* [14] security bound against individual attacks is given in the notation used later in this thesis and its connection with the no-cloning theorem is commented on. The problem of proving security against collective attacks is introduced and, finally, unconditional security (which will not be a goal in this thesis) is briefly commented on. Finally, this introductory chapter includes two appendices summarising a few useful definitions and relations relevant to this thesis. The material should be familiar to anyone with a previous background in quantum information theory and is, for the most part, covered in textbooks and lecture notes on the subject, such as [16–18].

The main results of this thesis are collected into three chapters:

Chapter 2 first demonstrates the necessity of accounting for source state and measurement alignment imprecisions in practical QKD security proofs. This is achieved by demonstrating, by means of a numerical optimisation, the existence of attacks that would allow an adversary to learn more about the key in the presence of alignment imprecisions than existing security proofs where these are not accounted for would imply. The chapter is based on a published article, Ref. [19].

Chapter 3 introduces security proof techniques for the prepare-and-measure BB84 protocol that can account for source state and measurement imprecisions against the class of collective attacks. The approach followed demonstrates that, at least for collective attacks, both known and new security results can be derived from the prepare-and-measure perspective in a relatively straightforward way. The main result is a key rate closely resembling one derived by Marøy *et al.* [20] for a BB84 implementation in which the source emits four arbitrary pure states (which may span a four-dimensional Hilbert space) and in which the detectors are left uncharacterised. The key rate can be further improved if a preprocessing procedure called *local randomisation*, proposed in [21], is applied, and the result is shown to be tight with the source characterisation used. As secondary results, a more specific key rate is derived for arbitrary qubit source states and a further improved key rate is derived if Bob’s measurements are additionally assumed two-dimensional. The chapter is drawn from Refs. [22, 23],

Chapter 4 explores two ways in which the BB84 protocol can be modified

to add a degree of self certification. The protocols differ in whether either Alice or Bob perform measurements intended to estimate a CHSH-type correlator, and their security against collective attacks is proved subject to the assumption of a two-dimensional source. The problem is inspired by device-independent QKD [24, 25] and a proof-of-principle prepare-and-measure semi-device-independent protocol [26]. An early version of part of this work is reported in a conference proceeding [27]; the remainder of this work is the subject of an article currently in preparation [28] at the time of writing.

In addition, a self-contained appendix summarises work that went beyond the theme – the security of prepare-and-measure BB84-like protocols – of the main part of this thesis:

Appendix A is more exploratory in nature and defines and investigates a class of polytopes in the space of joint probability distributions intermediate between the local and no-signalling polytopes from the field of Bell nonlocality. This work is the subject of an article in preparation [29] at the time of writing.

The publications in question are

- [19] **E. Woodhead** and S. Pironio, “*Effects of preparation and measurement misalignments on the security of the Bennett-Brassard 1984 quantum-key-distribution protocol*”, [Phys. Rev. A](#) **87**, 032315 (2013).
- [22] **E. Woodhead**, “*Quantum cloning bound and application to quantum key distribution*”, [Phys. Rev. A](#) **88**, 012331 (2013).
- [23] **E. Woodhead**, “*Tight asymptotic key rate for the Bennett-Brassard 1984 protocol with local randomization and device imprecisions*”, [Phys. Rev. A](#) **90**, 022306 (2014).
- [27] **E. Woodhead**, C. C. W. Lim, and S. Pironio, “*Semi-device-independent QKD Based on BB84 and a CHSH-Type Estimation*”, [Theory of Quantum Computation, Communication, and Cryptography](#), Lecture Notes in Computer Science, vol. 7582 (Springer, Berlin, Heidelberg, 2013), pp. 107–115.

The articles in preparation (titles provisional) are

- [28] **E. Woodhead** and S. Pironio, “*Secrecy in prepare-and-measure CHSH games with a qubit bound*”.
- [29] **E. Woodhead**, J. Silman, and S. Pironio, “*Partially deterministic polytopes*”.

1.2 The BB84 protocol

1.2.1 Prepare-and-measure version

In the BB84 protocol [3], illustrated in figure 1.1, Alice possesses a source capable of emitting one of the four qubit states $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$, where $|0\rangle$ and $|1\rangle$ are orthogonal and

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad (1.1)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \quad (1.2)$$

The pair $\{|0\rangle, |1\rangle\}$ is called the “z basis”, the states being eigenstates of the Pauli z operator

$$\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|. \quad (1.3)$$

Similarly, the states in the set $\{|+\rangle, |-\rangle\}$ are eigenstates of the Pauli x operator

$$\begin{aligned} \sigma_x &= |0\rangle\langle 1| + |1\rangle\langle 0| \\ &= |+\rangle\langle +| - |-\rangle\langle -| \end{aligned} \quad (1.4)$$

and are collectively called the “x basis”.

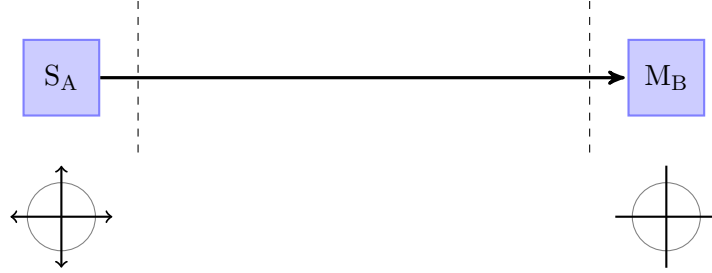


Figure 1.1: The BB84 protocol. Alice possesses a source (S_A) which can prepare any of the four BB84 states, $|0\rangle$, $|1\rangle$, $|+\rangle$, or $|-\rangle$, which are transmitted to Bob. Bob’s measurement device (M_A) can measure the received states either in the σ_z basis or in the σ_x basis.

The execution of the protocol consists of the following steps:

1. Alice transmits a random sequence of these qubits to Bob, recording which state and basis was used each time.

2. Upon reception of each qubit, Bob randomly measures in either the σ_z or σ_x basis, recording both the choice of basis and the result obtained each time.
3. Alice and Bob publicly reveal which bases they used and discard the cases where they used different bases.
4. Alice and Bob sacrifice and publicly reveal a randomly selected subset of their results. These are used to estimate the average error rates δ_z and δ_x in their z- and x-basis results.

At the end of this procedure, Alice and Bob each have two (random) bit strings $\mathbf{Z}_A$ and $\mathbf{X}_A$, and $\mathbf{Z}_B$ and $\mathbf{X}_B$, which are their versions of the z- and x-basis keys. From the publicly revealed information in step 4, Alice and Bob obtain an estimate of the joint prior probability distributions $p_{AB}^{(z)}(a, b)$ and $p_{AB}^{(x)}(a, b)$, $a, b \in \{0, 1\}$, where the x-basis results $+$ and $-$ can be taken to correspond to 0 and 1. The error rates are defined in terms of these by

$$\delta_z = p_{AB}^{(z)}(0, 1) + p_{AB}^{(z)}(1, 0), \quad (1.5)$$

$$\delta_x = p_{AB}^{(x)}(0, 1) + p_{AB}^{(x)}(1, 0). \quad (1.6)$$

In the cases where Bob used the same basis as Alice, their results should be perfectly correlated. In particular, if Alice chose between the two states in each basis equiprobably, one should have

$$p_{AB}^{(z)}(0, 0) = p_{AB}^{(z)}(1, 1) = p_{AB}^{(x)}(0, 0) = p_{AB}^{(x)}(1, 1) = 1/2 \quad (1.7)$$

and $\delta_z = \delta_x = 0$.

The key feature of the BB84 protocol from which security can be guaranteed is Alice's use of two conjugate (z and x) bases to encode the information transmitted to Bob. Because the four possible source states are nonorthogonal, no quantum measurement can perfectly distinguish between them, and an adversary attempting to gain information about the key this way will inevitably introduce errors that will reveal their presence. For example, if an adversary attempted to learn the z-basis key bits by measuring in the σ_z basis, the same operation would completely destroy any information about whether Alice transmitted $|+\rangle$ or $|-\rangle$ in the cases where the x basis was used and the adversary's tampering would be revealed in the form of errors between Alice's and Bob's x-basis key bits.

It is possible to prove that if the x-basis error rate δ_x is zero, then an adversary can have only negligible information about the z-basis key bits, and vice versa. As a first attempt at a security criterion, however, this is

insufficient: real world experimental implementations are never perfect and a nonzero error rate is a practical inevitability. By contrast, because the intent of QKD is provable security based only on the laws of physics (as opposed to security against an adversary limited by contemporary technology), for the purpose of security analysis a nonzero error rate must always be regarded as evidence of an adversary's presence.

Because the error rate observed in a BB84 implementation will always be nonzero, then, in practice one will never be able to rule out the presence of an adversary who may have obtained partial information about the key bits. Practically by definition, Alice and Bob will also very likely not share the same key. This was remedied with the proposal by Bennett, Brassard, and Robert of incorporating *privacy amplification* [30] as well as error correction into the definition of the protocol:

5. If the error rates are not too high, Alice and Bob extract a (generally shorter) secret key by error correction and privacy amplification. Otherwise, the protocol is aborted.

The purpose of the additional postprocessing is to allow Alice and Bob to extract a final, generally shorter, key in which the errors are corrected (Alice and Bob should share the same final key) and which is secret (an eavesdropper should have no information about the final key).

Whether and how such postprocessing can be done is a subject of research in itself. Fortunately, sufficient criteria have been derived which reduce the problem to evaluating or bounding measures of relative randomness or information shared between Alice, Bob, and Eve. This allows one to investigate the security of a QKD protocol without the need to concern oneself with the details of classical postprocessing. We will mainly use a criterion for the key rate credited to Devetak and Winter [31], which gives a simple expression for the key rate extractable by one-way¹ postprocessing in the asymptotic limit under the assumption of an individually and identically repeated attack by the eavesdropper (a similar result was obtained by Kraus, Gisin, and Renner [21, 32, 33]). In the earlier part of this thesis we will also use an older result by Csiszár and Körner [34] from classical information theory, which holds for a weaker security definition.

It should be noted that there is more than one variant of the BB84 protocol that follows the basic procedure outlined here. In the original proposal,

¹This is a type of postprocessing scheme in which only one party transmits a checksum to the other. Two-way postprocessing schemes, by contrast, may involve multiple rounds of public discussion both ways.

for instance, Alice and Bob both select equiprobably between the z and x bases, in which case the bases will be mismatched and the results discarded half the time. In an alternative version proposed in [35], Alice and Bob use one basis (e.g., the z basis) the vast majority of the time for the actual key generation, and only occasionally use the complementary (e.g., x) basis for the purpose of testing for the presence of an eavesdropper. In this version, the fraction of results used for key generation can be made arbitrarily close to 1. Other variants of the BB84 protocol add additional steps to those listed above. It is sometimes suggested that Alice and Bob should agree on and apply a random permutation to their key bits before the postprocessing is applied, which can simplify certain security proofs [33]. A more involved example concerns the common case of a practical QKD system in which the key bits are encoded on different photon (e.g., polarisation) states and an ideal single-photon source is approximated by weak laser pulses attenuated to the point that less than one photon is emitted on average in each pulse. In such an implementation there is always some probability that a given pulse will contain two or more photons in the same state, one of which an eavesdropper could intercept without introducing any visible disturbance (the photon-number-splitting attack [36, 37]). The decoy-state technique [38, 39], which was proposed to mitigate this vulnerability, requires that Alice select randomly between different pulse intensities during the course of the protocol, allowing additional tests of the quantum channel.

1.2.2 Entanglement-based version

The BB84 protocol also exists in an entanglement-based version, which was proposed by Bennett, Brassard and Mermin in 1992 [40] following a scheme based on the use of entangled states proposed by Ekert [41]. In this version of the protocol, Alice and Bob would ideally share a number of quantum systems each in the entangled state

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B) \quad (1.8)$$

which could, for instance, be distributed by a source located midway between them, and now both Alice and Bob choose randomly between performing σ_z and σ_x measurements. Note that the entangled state also be expressed in the x basis as

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|+\rangle_A|+\rangle_B + |-\rangle_A|-\rangle_B), \quad (1.9)$$

meaning that Alice and Bob should detect perfect correlations in both the cases where they both measure in the z basis and when they both measure in the x basis. For the purpose of security analysis, the source of entangled states is untrusted (the eavesdropper is assumed to be in control of it). In

this case, the eavesdropper may “attack” the protocol by preparing and distributing a tripartite state $|\psi\rangle_{ABE} \in \mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_E$ in which Eve’s part may be entangled with Alice’s and Bob’s. The estimation of the z- and x-basis error rates is intended to detect such an attack.

1.2.3 Correspondence

There is a well known equivalence between the entanglement-based and prepare-and-measure versions of the BB84 protocol, pointed out in [40], that is based on the following observations. First, in the prepare-and-measure version, one way that Alice could both randomly choose and prepare either of the z- or x-basis states is by preparing an entangled $|\Phi^+\rangle$ state in her lab and measuring one part of the state in either the σ_z or σ_x bases. This would project the second part of the state randomly onto one of the σ_z or σ_x eigenstates, respectively, which can then be transmitted to Bob. Second, in this implementation, Alice could just as well transmit the second part of the state to Bob *before* measuring σ_z or σ_x on her part. Third, finally, it can only be advantageous to Eve if Eve is granted control of the source of entangled states rather than Alice, which is recovers the entanglement-based version of the protocol. Specifically, this is because if Alice is in possession of the source of Φ^+ states, the best Eve could achieve with a unitary attack on the part transmitted to Bob is to transform the initial Φ^+ state to a tripartite state of the form

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|0\rangle_A|0\rangle_{BE} + |1\rangle_A|1\rangle_{BE}) \quad (1.10)$$

for some orthogonal states $|0\rangle_{BE}, |1\rangle_{BE} \in \mathcal{H}_B \otimes \mathcal{H}_E$, which is still a Φ^+ state, while if Eve is in possession of the source she could substitute any tripartite state $|\psi\rangle_{ABE}$ as her attack. It follows that a security proof of the entanglement-based BB84 protocol would also imply the security of the prepare-and-measure version of the protocol.

To some extent, the converse may also hold. The reason for this is that any tripartite state $|\psi\rangle_{ABE}$ in which $\mathcal{H}_A$ is two dimensional can be decomposed in the form

$$|\psi\rangle_{ABE} = \sqrt{p}|0\rangle_A|\alpha\rangle_{BE} + \sqrt{p'}|1\rangle_A|\alpha'\rangle_{BE}, \quad (1.11)$$

with $\sqrt{p}|\alpha\rangle_{BE} = (\langle 0|_A \otimes \mathbb{1}_{BE})|\psi_{ABE}\rangle$ and $\sqrt{p'}|\alpha'\rangle_{BE} = (\langle 1|_A \otimes \mathbb{1}_{BE})|\psi_{ABE}\rangle$, where $\mathbb{1}_{BE}$ is the identity operator acting on $\mathcal{H}_B \otimes \mathcal{H}_E$, such that $|\alpha\rangle_{BE}$ and $|\alpha'\rangle_{BE}$ are normalised and $p + p' = 1$. The same state can also be expressed in an analogous form in terms of the σ_x -basis states,

$$|\psi\rangle_{ABE} = \sqrt{q}|+\rangle_A|\beta\rangle_{BE} + \sqrt{q'}|-\rangle_A|\beta'\rangle_{BE}, \quad (1.12)$$

with

$$\sqrt{q}|\beta\rangle = \sqrt{\frac{p}{2}}|\alpha\rangle + \sqrt{\frac{p'}{2}}|\alpha'\rangle, \quad (1.13)$$

$$\sqrt{q'}|\beta'\rangle = \sqrt{\frac{p}{2}}|\alpha\rangle - \sqrt{\frac{p'}{2}}|\alpha'\rangle. \quad (1.14)$$

These relations imply constraints between the probability coefficients p , p' , q , and q' and the inner products $\langle\alpha|\alpha'\rangle$ and $\langle\beta|\beta'\rangle$,

$$q = \frac{1}{2} + \sqrt{pp'} \operatorname{Re}[\langle\alpha|\alpha'\rangle], \quad (1.15)$$

$$q' = \frac{1}{2} - \sqrt{pp'} \operatorname{Re}[\langle\alpha|\alpha'\rangle], \quad (1.16)$$

and

$$\sqrt{qq'}\langle\beta|\beta'\rangle = \frac{p-p'}{2} - i\sqrt{pp'} \operatorname{Im}[\langle\alpha|\alpha'\rangle]. \quad (1.17)$$

Note that the probability coefficients can be estimated by Alice in the entanglement-based version. In the typical case where $p = p' = q = q' = 1/2$ – which Alice could verify – the constraints above simplify to

$$|\beta\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle + |\alpha'\rangle), \quad (1.18)$$

$$|\beta'\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle - |\alpha'\rangle), \quad (1.19)$$

and

$$\operatorname{Re}[\langle\alpha|\alpha'\rangle] = \operatorname{Re}[\langle\beta|\beta'\rangle] = 0, \quad (1.20)$$

$$\operatorname{Im}[\langle\alpha|\alpha'\rangle] = \operatorname{Im}[\langle\beta|\beta'\rangle]. \quad (1.21)$$

The only difference with the situation considered in the prepare-and-measure BB84 version is that the inner products $\langle\alpha|\alpha'\rangle$ and $\langle\beta|\beta'\rangle$ may have a nonzero imaginary part. A security proof of the prepare-and-measure BB84 protocol may thus also imply the security of the entanglement-based version if the normally assumed orthogonality of the z- and x-basis source states is never used in the security proof.

While the correspondence described above holds for the prepare-and-measure version of the BB84 protocol as it was described in section 1.2.1, constraints of the type described above mean that the correspondence may no longer hold for generalised versions of the protocol. In general, if Alice performs the positive operator-valued measure (POVM) $\{\Pi_a\}_a$ on her part of an initial state ρ_{ABE} , the part shared by Bob and Eve is projected onto a state $\rho_{\text{BE}}^{(a)}$ with probability p_a given by

$$p_a \rho_{\text{BE}}^{(a)} = \operatorname{Tr}_A[\Pi_a \rho_{\text{ABE}}]. \quad (1.22)$$

Using the defining property $\sum_a \Pi_a = \mathbb{1}_A$ for any POVM, the average over projected states is

$$\begin{aligned} \sum_a p_a \rho_{\text{BE}}^{(a)} &= \sum_a \text{Tr}_A [\Pi_a \rho_{\text{ABE}}] \\ &= \text{Tr}_A [\rho_{\text{ABE}}] \\ &= \rho_{\text{BE}} , \end{aligned} \tag{1.23}$$

which is the same regardless of the measurement performed (as one should expect from the no-signalling principle). Applied to the entanglement-based version of the BB84 protocol and in terms of the notation introduced above, this implies that the relation

$$p|\alpha\rangle\langle\alpha| + p'|\alpha'\rangle\langle\alpha'| = q|\beta\rangle\langle\beta| + q'|\beta'\rangle\langle\beta'| , \tag{1.24}$$

called *basis independence* in [42], must necessarily hold between the projected z- and x-basis states, even if the states are prepared by more general measurements than σ_z and σ_x .

1.2.4 Alternatives to BB84

The BB84 protocol was the first QKD protocol to be proposed and remains one of the simplest and most studied in the literature and is the protocol that the majority of this thesis will be concerned with. Here, we summarise a few other notable protocols and approaches to QKD that have been proposed since 1984.

In addition to BB84, notable “traditional” schemes include B92 [43], the six-state protocol [44, 45], and the SARG04 protocol [46]. These are similarly based on the use of nonorthogonal source states and have various tradeoffs compared with BB84. The B92 protocol, proposed by Bennett in 1992, can be considered the minimal QKD protocol – only two nonorthogonal source states are used to encode Alice’s key bits – but has very low tolerance to noise. The six-state protocol is identical to the BB84 protocol with the difference that Alice and Bob both use the σ_y basis in addition to the σ_z and σ_x bases; the additional basis makes the implementation more complicated but permits a more thorough characterisation of the channel which slightly improves the six-state protocol’s tolerance to noise compared with BB84. SARG04 is intended to be more robust in implementations where the source imperfectly approximates a single-photon source. The protocol is based on identical hardware to the BB84 protocol – Alice’s source ideally prepares the same BB84 source states and Bob performs the same σ_x and σ_z measurements – but uses a different and more sophisticated sifting procedure than

BB84’s public reveal of basis choices. Other approaches, including protocols which use continuous degrees of freedom for encoding such as continuous-variable QKD (CV-QKD) and the coherent one-way (COW) protocol, can be found in [2].

In the last decade, alternative proposals have appeared which aim to minimise the assumptions needed to guarantee security, usually by introducing some degree of self-testing as part of a protocol. The most ambitious such proposal is so-called *device-independent* QKD, in which the detection of Bell-nonlocal correlations is used to certify the security and correct functioning of a protocol [25]; in this case, security is no longer dependent on any explicit characterisation of the devices. Intermediate approaches between traditional and fully device-independent QKD also exist. These include semi-device-independent QKD [26], in which the security of a prepare-and-measure scheme depends only on the assumption of a dimension bound on the devices, and measurement-device-independent QKD [47], a “reverse entanglement” scheme based on an (a priori untrusted) entangling measurement.

1.3 Implementation imperfections

Since its original proposal three decades ago, QKD implementations have advanced from lab demonstrations over less than a metre to long-distance experiments over ranges of a few hundred kilometres [15]. Theoretical analyses have progressed from the early consideration of simple intercept-resend attacks [48] to finite-key [49] unconditional security proofs based on universally composable [50, 51] security definitions. The gap between theory and practice, however, remains problematic (see [52] for a discussion published a few years ago). This section briefly discusses two types of imperfection – channel/detection noise and state imprecisions – that affect real QKD implementations. It should be stressed that, unlike the implementation flaws revealed by the hacking attacks [12, 13] cited earlier, these are inevitable to some degree in any real QKD system and, consequently, simply building a better implementation will not eliminate them entirely.

1.3.1 Channel/detection noise

Historically, the first imperfection to receive explicit attention was the inevitability of a nonzero noise rate, with the proposal of privacy amplification mentioned in the previous section. Essentially, the idea is that if the noise

rate is sufficiently small, the worst case is that an adversary may have obtained a limited amount of information about the key to be distributed. Provided this information is limited, it may nevertheless be possible to extract a shorter key in which the adversary’s information (if present) is effectively erased (“privacy amplification”) and relative errors between Alice’s and Bob’s versions of the key are removed (“error correction”, which will be necessary practically by definition if we are expecting any nonzero noise rate). Consideration of noise has long been standard and expected in theoretical work; the main purpose of a modern security proof of a given QKD protocol is to determine whether a secret key can be extracted given that a certain noise rate has been observed (usually assuming *all* of the observed noise is due to an adversary’s tampering), and, if so, give an explicit lower bound on the length of the key that can safely be extracted using known classical protocols for error correction and privacy amplification.

As early examples, two security results for the BB84 protocol that account for noise are cited here that will be relevant later in this thesis. The first is the key rate

$$r = h\left(\frac{1}{2} + \sqrt{\delta(1-\delta)}\right) - h(\delta) \quad (1.25)$$

derived by Fuchs, Gisin, Griffiths, Niu, and Peres in 1997 [14] (hereafter the “FGGNP rate”, for convenience). In (1.25), the quantity δ is the average error rate observed during the execution of the protocol, h is a function called the *binary entropy* and is defined by $h(x) = -x \log(x) - (1-x) \log(1-x)$, and here and throughout this thesis, we use $\log$ to denote the logarithm function in base 2 such that the final result is a quantity expressed in bits. The second is the Shor-Preskill key rate

$$r = 1 - 2h(\delta) \quad (1.26)$$

which was first derived by Shor and Preskill in the year 2000 [53] and is similarly expressed in terms of the average error rate δ and the binary entropy function h . In both cases, the *key rate* r (which is never greater than 1 here) is the average number of key bits that can securely be extracted per signal, and in the asymptotic limit, following one-way error correction and privacy amplification. In both cases, $r = 1$ if $\delta = 0$, indicating, as one might expect, that the key transmitted by Alice to Bob is already secure (and no further postprocessing is necessary) if no errors are observed which could be attributed to the presence of an eavesdropper. Both key rates become 0 at a certain threshold error rate (specifically $\delta \approx 14.64\%$ and $\delta \approx 11.00\%$, for the FGGNP and Shor-Preskill rates, respectively). This can roughly be thought of as the point where an eavesdropper may have learned as much information about Alice’s initial version of the key as Bob did, beyond which the extraction of a secret key by one-way postprocessing is no longer safe.

The reason we could quote *two* key rates for the BB84 protocol is that (1.25) and (1.26) were derived based on different underlying security definitions. Specifically, the authors of [14] considered a restricted class of attacks – called *individual attacks* in the literature – in which the eavesdropper is assumed to attack each quantum state in transit from Alice to Bob individually and identically and immediately measures each state to obtain a result. The eavesdropper’s end result is their best guess of the key. The final key, following the postprocessing, is “secure” in the sense that Alice and Bob share a uniformly random key that is completely uncorrelated from Eve’s final guess of the key. The Shor-Preskill rate, by contrast, was obtained as the result of a so-called *unconditional* security proof, meaning that Eve’s attack is no longer assumed to be individually and identically performed on each transmitted state. More importantly, Eve is also allowed to delay her measurement indefinitely, for instance until after the postprocessing is applied, and may even wait to find out what the key is used for before deciding which measurement to perform. In this case, the final key is secure in the stronger sense that it is uncorrelated with any *quantum* information Eve might possess.

It should be noted that the gap between the threshold error rates of 11.00% and 14.64% is not fully understood. In particular, it was found in [21, 32] that the lower bound of 11% could be increased to around 12.41% if Alice adds additional random noise to her version of the key (a preprocessing procedure called *local randomisation*). Smith *et al.* have shown that this can be further increased to 12.92% with more sophisticated preprocessing [54]. In the case of two-way postprocessing, the threshold error rate is known to be bounded between 20% and 25% [55, 56].

1.3.2 State imprecisions

A limitation of both the FGGNP and Shor-Preskill rates cited above is that they are derived assuming that the source states and/or measurement projection operators exactly satisfy the BB84 relations described in section 1.2, i.e., in some suitable basis they exactly coincide with the σ_z and σ_x eigenstates. As a result, the cited security results are not robust in the face of source and/or measurement imprecisions, and the question arises as to how they generalise in the case of source states and measurements that deviate from the ideal BB84 relations. This is a main theme of this thesis.

By contrast with channel noise, comparatively few theoretical security analyses have explicitly considered state and measurement imprecisions and it was only earlier this year that such imprecisions were given explicit consider-

ation in an experimental work [57]. Security analyses of the entanglement-based BB84 protocol (or the prepare-and measure protocol with a basis-independence assumption) can be found in [42, 49, 58–60] which partially or fully relax the assumptions made about Alice’s and/or Bob’s measurements. In particular, Ref. [59] gives a generalisation of the Shor-Preiskill key rate which holds for arbitrary imprecisions on both sides in the asymptotic limit, derived based on an entropic tradeoff relation dependent on a parameter characterising Alice’s measurement. This approach was adapted to the case of finite statistics in [49, 60]. An early security analysis of the prepare-and-measure BB84 protocol which relaxes the basis-independence assumption can be found in [61]. A later approach by Koashi [62], who considered a source emitting arbitrary states and a perfect detector on Bob’s side, was modified by Marøy, Lydersen, and Skaar [20] to obtain a security result in which Alice’s source emits arbitrary states and Bob’s device is left uncharacterised.

One of the main results of chapter 3 will be a comparatively simple derivation of a key rate closely resembling the Marøy *et al.* key rate, complemented with a demonstration of its optimality for the particular source characterisation used (and, technically, in the Devetak-Winter security framework that will be introduced in section 1.4.5).

1.4 Security of the BB84 protocol

1.4.1 The no-cloning theorem

The no-cloning theorem asserts that one cannot construct a cloning machine capable of making multiple perfect copies of arbitrary input quantum states, i.e., there is no physical system consistent with quantum physics capable of implementing the operation

$$|\psi\rangle \mapsto |\psi\rangle|\psi\rangle \quad (1.27)$$

that works for all input states without knowledge of the state in advance. The impossibility of perfect state cloning is obviously closely connected to the security of QKD, in that if an eavesdropper could make perfect copies of the quantum states in transit from Alice to Bob, they could learn the entire key by measuring their copy (if necessary, after the bases are revealed) without introducing any disturbance.

The original proof of the no-cloning theorem, by Wootters and Zurek [10] and Dieks [11], is by a counterexample based on essentially the same scenario

encountered in the BB84 protocol. Specifically, one considers a hypothetical cloning machine designed to output perfect copies of the z-basis states, according to

$$|0\rangle_A \mapsto |0\rangle_B |0\rangle_E, \quad (1.28)$$

$$|1\rangle_A \mapsto |1\rangle_B |1\rangle_E, \quad (1.29)$$

in which $|0\rangle_B, |1\rangle_B \in \mathcal{H}_B$ and $|0\rangle_E, |1\rangle_E \in \mathcal{H}_E$ are orthonormal. Such a cloner is in principle allowed in quantum physics as it satisfies unitarity, i.e., $\langle 0|0\rangle_A = \langle 0|0\rangle_B \langle 0|0\rangle_E = 1$, $\langle 1|1\rangle_A = \langle 1|1\rangle_B \langle 1|1\rangle_E = 1$, and $\langle 0|1\rangle_A = \langle 0|1\rangle_B \langle 0|1\rangle_E = 0$. Applying the relations

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad (1.30)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \quad (1.31)$$

linearity of quantum operations however implies that the same cloning machine necessarily transforms the x-basis states to

$$|+\rangle_A \mapsto \frac{1}{\sqrt{2}}(|0\rangle_B |0\rangle_E + |1\rangle_B |1\rangle_E), \quad (1.32)$$

$$|-\rangle_A \mapsto \frac{1}{\sqrt{2}}(|0\rangle_B |0\rangle_E - |1\rangle_B |1\rangle_E), \quad (1.33)$$

which differ from $|+\rangle_B |+\rangle_E$ and $|-\rangle_B |-\rangle_E$, respectively. Furthermore, both Bob and Eve receive the maximally mixed density operator $\frac{1}{2}\mathbb{1} = \frac{1}{2}|+\rangle\langle+| + \frac{1}{2}|-\rangle\langle-|$ regardless of whichever of $|+\rangle_A$ or $|-\rangle_A$ is used as the input. A cloner designed to output perfect duplicate copies of the two z-basis states would thus inevitably fail to make duplicate copies of the x-basis states, to the point that both Bob and Eve would be completely unable to distinguish between the two possible input x states.

1.4.2 Monogamy of entanglement

In the entanglement-based version of the BB84 protocol, one considers the worst-case situation in which Alice, Bob, and Eve share a tripartite state $|\psi\rangle_{ABE}$. Alice and Bob perform σ_z and σ_x measurements on their part $\rho_{AB} = \text{Tr}_E[|\psi\rangle\langle\psi|_{ABE}]$ of this state and estimate the z- and x-basis error rates, whose expectation values can be expressed as

$$\delta_z = \frac{1}{2} - \frac{1}{2}\langle\sigma_z \otimes \sigma_z\rangle_{\rho_{AB}}, \quad (1.34)$$

$$\delta_x = \frac{1}{2} - \frac{1}{2}\langle\sigma_x \otimes \sigma_x\rangle_{\rho_{AB}}, \quad (1.35)$$

where in general the expectation value of an operator is given by $\langle A \rangle_\rho = \text{Tr}[A\rho]$. In the ideal, noiseless situation, the security of the entanglement-based protocol can be understood in the following way. It is possible to

show that the only quantum state that reproduces $\delta_z = 0$ and $\delta_x = 0$ is the maximally entangled pure state $|\Phi^+\rangle_{AB} = (|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B)/\sqrt{2}$. Essentially the only possibility in case is that Alice, Bob, and Eve shared a state of the form $|\Psi\rangle_{ABE} = |\Phi^+\rangle_{AB} \otimes |\chi\rangle_E$, with Eve completely uncorrelated with Alice and Bob. This is an example of a general property of entangled states called the *monogamy of entanglement*: if Alice's and Bob's systems are maximally entangled with one another, which in the BB84 protocol is certified by verifying that $\delta_z = \delta_x = 0$, then neither can be entangled at all with a system in Eve's possession. This rules out that Eve can learn any information from her system about Alice's or Bob's key bits.

A simple way to see that Alice and Bob must share a Φ^+ state if $\delta_z = \delta_x = 0$ is to consider the sum

$$\delta_z + \delta_x = 1 - \frac{1}{2}\langle\sigma_z \otimes \sigma_z\rangle_{\rho_{AB}} - \frac{1}{2}\langle\sigma_x \otimes \sigma_x\rangle_{\rho_{AB}}, \quad (1.36)$$

which one can rearrange to

$$\frac{1}{2}\langle W \rangle_{\rho_{AB}} = 1 - \delta_z - \delta_x \quad (1.37)$$

for the expectation value of the *entanglement witness*

$$W = \sigma_z \otimes \sigma_z + \sigma_x \otimes \sigma_x. \quad (1.38)$$

The entanglement witness W has the two nondegenerate nonzero eigenvalues 2 and -2 , associated respectively with the entangled eigenstates

$$|\Phi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B), \quad (1.39)$$

$$|\Psi^-\rangle_{AB} = \frac{1}{\sqrt{2}}(|0\rangle_A|1\rangle_B - |1\rangle_A|0\rangle_B). \quad (1.40)$$

Put differently, W has the spectral decomposition $W = 2(|\Phi^+\rangle\langle\Phi^+|_{AB} - |\Psi^-\rangle\langle\Psi^-|_{AB})$. It follows that the ideal noiseless situation $\delta_z = \delta_x = 0$, which is equivalent to the maximal expectation value $\langle W \rangle_{\rho_{AB}} = 2$ for the entanglement witness W , can only be obtained with the corresponding eigenstate $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|_{AB}$.

1.4.3 Attack models

As mentioned in section 1.3.1, useful security proofs of the BB84 protocol account for channel noise which, from the point of view of security analysis, is treated as a side effect of tampering by an eavesdropper. Accounting for noise, the ultimate goal of security analysis is to produce a so-called

“unconditional” security proof, i.e., a proof that a cryptographic key can be extracted by error correction and privacy amplification and guaranteed secure against any attack allowed by quantum physics that an eavesdropper may have implemented that is compatible with some given observed error rate. Due to the difficulty of this problem, many security analyses consider intermediate, more restricted classes of attacks in which the eavesdropper is not granted unlimited power to tamper with the channel. There are two such classes of attacks that we will be concerned with that appear regularly in the literature: *individual* and *collective* attacks. In both cases, the security analysis is restricted to an i.i.d. (individually and identically distributed) problem, in that the adversary is assumed to intercept each state transmitted from Alice to Bob separately and in exactly the same way, as illustrated in figure 1.2.

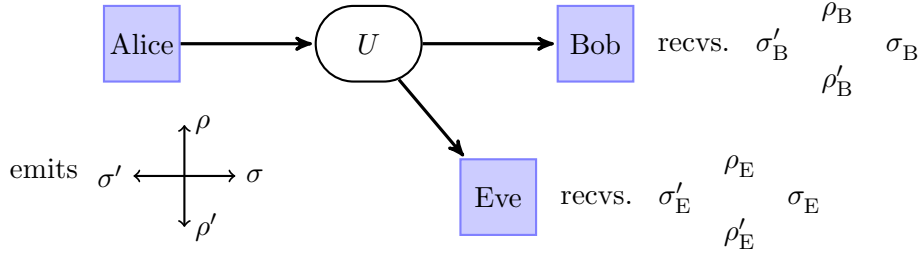


Figure 1.2: The i.i.d. unitary attack model common to both individual and collective attacks. Alice’s source may emit any among the four BB84 states $\rho = |0\rangle\langle 0|$, $\rho' = |1\rangle\langle 1|$, $\sigma = |+\rangle\langle +|$, or $\sigma' = |-\rangle\langle -|$. Eve applies some fixed unitary operation $U: \mathcal{H} \supseteq \mathcal{H}_A \rightarrow \mathcal{H}_B \otimes \mathcal{H}_E$ to each state individually emitted by Alice. Following the attack, Bob and Eve respectively receive the corresponding state $\rho_B, \rho'_B, \sigma_B$, or σ'_B ; and $\rho_E, \rho'_E, \sigma_E$, or σ'_E ; depending on which state Alice emitted.

The two attack classes, and the difference between them, can be summarised as follows:

- In an *individual* attack, Eve intercepts each state emitted by Alice separately and applies a unitary operation with the intent of partially cloning it. Eve then measures her intercepted part of each state, again individually and identically, and records a classical result that will serve as her best guess of Alice’s corresponding key bit.
- In a *collective* attack, Eve attacks each state unitarily and individually and identically, as in the individual attack scenario, but no assumption is made on her subsequent measurement. Unlike the individual attack scenario, Eve may delay her measurement until the very end of

the protocol (even after error correction and privacy amplification are applied) and may perform any joint measurement on the full collection of intercepted states.

In an unconditional security proof, also called a security proof against a *general* or *coherent* attack, all restrictions on Eve's allowed attack are removed. In the following subsections, we discuss individual and collective attacks, and the security of the BB84 protocol against these classes of attacks, in more detail.

1.4.4 Security against individual attacks

In this section, we give a simplified derivation of the FGGNP rate, already mentioned in section 1.3.1, which was first obtained by Fuchs *et al.* as a security bound for the BB84 protocol against individual attacks. This will serve to introduce some of the notations and techniques that will be used throughout the remainder of this thesis.

As mentioned earlier, in the individual attack scenario, an adversary is assumed to attack and measure her intercepted part of each quantum state transmitted from Alice to Bob individually and identically and before the postprocessing is applied. For simplicity, we will consider the case where only the z-basis results are used to generate the key. In this case, the correlation between Alice's, Bob's, and Eve's z-basis key bits before the postprocessing is described by the n -fold product of a joint probability distribution $p_{\text{ABE}}(a, b, e)$, $a, b \in \{0, 1\}$ which depends on the unitary attack and on Bob's and Eve's measurements.

The starting ingredient is a security criterion credited to Csiszár and Körner [34] stating that, in the asymptotic limit, a secret key can be extracted by one-way postprocessing from Alice to Bob at a rate which can be expressed as the difference between two conditional Shannon entropies associated with the probability distribution $p_{\text{ABE}}(a, b, e)$:

$$r = H(Z_A | Z_E) - H(Z_A | Z_B). \quad (1.41)$$

In general, the conditional Shannon entropy $H(X | Y)$ associated with two random variables X and Y is defined by

$$H(X | Y) = H(XY) - H(Y), \quad (1.42)$$

with the Shannon entropies $H(XY)$ and $H(Y)$ in turn defined in terms of the associated probability distributions by

$$H(XY) = - \sum_{xy} p_{XY}(x, y) \log(p_{XY}(x, y)), \quad (1.43)$$

$$H(Y) = - \sum_y p_Y(y) \log(p_Y(y)). \quad (1.44)$$

Intuitively, the conditional entropy $H(Z_A | Z_E)$ is a measure of how random Alice's record of z-basis bits is from Eve's perspective, measuring the average number of key bits that can be extracted by privacy amplification. $H(Z_A | Z_B)$ is similarly a measure of how random Alice's record is from Bob's perspective, and quantifies the key loss due to error correction. The final key $\mathbf{k}_A$ after the postprocessing, of length $n \approx rN$ where N is the initial number of z bits, is secure in the sense that the joint probability distribution is of the form

$$p_{ABE}(\mathbf{k}_A, \mathbf{k}_B, \mathbf{e}) \approx \left(\frac{1}{n} \delta_{\mathbf{k}_A, \mathbf{k}_B}\right) p_E(\mathbf{e}) \quad (1.45)$$

where

$$\delta_{\mathbf{k}_A, \mathbf{k}_B} = \begin{cases} 1 & : \mathbf{k}_A = \mathbf{k}_B \\ 0 & : \mathbf{k}_A \neq \mathbf{k}_B \end{cases} \quad (1.46)$$

is the Kronecker delta, the approximation approaching an equality in the limit $N \rightarrow \infty$.

The problem now is to obtain a lower bound on the Csiszár-Körner rate (1.41). The conditional entropy $H(Z_A | Z_B)$ presents no difficulty as it is a function of the joint probability distribution $p_{AB}(a, b)$ associated with Alice and Bob's results. This can simply be estimated directly, though for simplicity and anticipating that the relative errors would usually be symmetric we will replace it with $h(\delta_z)$, where we recall that the binary entropy function is $h(x) = -x \log(x) - (1-x) \log(1-x)$ and $\delta_z = p_{AB}(0, 1) + p_{AB}(1, 0)$ is the z-basis error rate. The less trivial problem is to derive a lower bound for $H(Z_A | Z_E)$, as this depends on the joint probability distribution $p_{AE}(a, e)$ which is a priori unknown. In the following we will show that the conditional entropy is lower bounded in terms of the x-basis error rate by $H(Z_A | Z_E) \geq h\left(\frac{1}{2} + \sqrt{\delta_x(1 - \delta_x)}\right)$. Combining these, we will have obtained the lower bound

$$r \geq h\left(\frac{1}{2} + \sqrt{\delta_x(1 - \delta_x)}\right) - h(\delta_z) \quad (1.47)$$

for the Csiszár-Körner rate. The expression (1.25) given in section 1.3.1 is the same rate in the special case where the error rates are the same, with $\delta = \delta_z = \delta_x$. In this case, the FGGNP rate becomes 0 for the threshold error rate $\delta = \frac{1}{2} - \frac{1}{2\sqrt{2}} \approx 14.64\%$.

For the purpose of evaluating the conditional entropy, it will be convenient note that it can alternatively be expressed as

$$\begin{aligned}
H(X | Y) &= - \sum_{x,y} p_{XY}(x, y) \log(p_{XY}(x, y)) + \sum_x p_Y(y) \log(p_Y(y)) \\
&= - \sum_{x,y} p_{XY}(x, y) [\log(p_{XY}(x, y)) - \log(p_Y(y))] \\
&= - \sum_{x,y} p_{XY}(x, y) \log(p_{X|Y}(x | y)) \\
&= \sum_y p_Y(y) \left[- \sum_x p_{X|Y}(x | y) \log(p_{X|Y}(x | y)) \right] \\
&= \sum_y p_Y(y) H(X | y), \tag{1.48}
\end{aligned}$$

where we used that $p_Y(y) = \sum_x p_{XY}(x, y)$ to obtain the second line and that $p_{XY}(x, y) = p_{X|Y}(x | y)p_Y(y)$ to obtain the third and fourth lines. This establishes that the conditional entropy $H(X | Y)$ is simply the average Shannon entropy of X conditioned on Y . We note that this allows for a simple derivation of an upper bound for $H(Z_A | Z_B)$ in terms of δ_z :

$$\begin{aligned}
H(Z_A | Z_B) &= \sum_b p_B(b) H(Z_A | b) \\
&= p_B(0) h(p_{A|B}(1 | 0)) + p_B(1) h(p_{A|B}(0 | 1)) \\
&\leq h(p_B(0) p_{A|B}(1 | 0) + p_B(1) p_{A|B}(0 | 1)) \\
&= h(\delta_z), \tag{1.49}
\end{aligned}$$

with the inequality on the third line following from the well known (and easily verified) property of concavity of the binary entropy function. The upper bound $H(Z_A | Z_B) \leq h(\delta_z)$ confirms that $h(\delta_z)$ can safely be substituted in place of $H(Z_A | Z_B)$ in the expression above for the Csiszár-Körner rate.

We now turn to the main problem of obtaining a lower bound on the conditional entropy $H(Z_A | Z_E)$ between Alice and Eve. We begin by applying (1.48) to reexpress the entropy as

$$\begin{aligned}
H(Z_A | Z_E) &= \sum_e p_E(e) H(Z_A | e) \\
&= \sum_e p_E(e) h(p_{A|E}(a | e)) \tag{1.50}
\end{aligned}$$

(for either value of a , since $h(p_{A|E}(0 | e)) = h(p_{A|E}(1 | e))$). For the purpose of obtaining a lower bound, it will be convenient to introduce a new variable $D_{z|e}$ such that

$$H(Z_A | Z_E) = \sum_e p_E(e) h\left(\frac{1}{2} + \frac{1}{2} D_{z|e}\right). \tag{1.51}$$

The quantity $D_{z|e}$, called the “information gain” in [14], is defined by

$$D_{z|e} = |p_{A|E}(0 | e) - p_{A|E}(1 | e)|, \quad (1.52)$$

which, using that $p_{A|E}(0 | e) + p_{A|E}(1 | e) = 1$, rearranges to

$$\max(p_{A|E}(0 | e), p_{A|E}(1 | e)) = \frac{1}{2} + \frac{1}{2}D_{z|e}. \quad (1.53)$$

The goal now is to determine the tradeoff between $D_{z|e}$ and the x-basis error rate δ_x for any unitary attack. Since a unitary operation preserves the relative relations (inner products) between states, it is possible and will be convenient to simply treat the source Hilbert space $\mathcal{H}_A$ as if it were a subspace of the joint subspace shared by Bob and Eve, i.e., $\mathcal{H}_A \subset \mathcal{H}_B \otimes \mathcal{H}_E$. Calling the density operators corresponding to the z-basis states $\rho = |0\rangle\langle 0|$ and $\rho' = |1\rangle\langle 1|$, the states received by Eve are the partial traces $\rho_E = \text{Tr}_B[\rho]$ and $\rho'_E = \text{Tr}_B[\rho']$. The (conditional) probability of Eve obtaining the result e , of corresponding POVM element M_e , is then given by

$$p_{E|A}(e | 0) = \text{Tr}[M_e \rho_E] \quad \text{or} \quad p_{E|A}(e | 1) = \text{Tr}[M_e \rho'_E], \quad (1.54)$$

depending on which of the z-basis states Alice sent. Assuming that Alice selects equiprobably between them, such that $p_A(0) = p_A(1) = 1/2$, $D_{z|e}$ can be developed as

$$\begin{aligned} p_E(e)D_{z|e} &= |p_{AE}(0, e) - p_{AE}(1, e)| \\ &= \left| \frac{1}{2} \text{Tr}[M_e \rho_E] - \frac{1}{2} \text{Tr}[M_e \rho'_E] \right| \\ &= \left| \frac{1}{2} \text{Tr}[(\mathbb{1}_B \otimes M_e)Z] \right| \\ &= \left| \frac{1}{2} \langle + | \mathbb{1}_B \otimes M_e | - \rangle + \frac{1}{2} \langle - | \mathbb{1}_B \otimes M_e | + \rangle \right| \\ &= |\text{Re}[\langle + | \mathbb{1}_B \otimes M_e | - \rangle]| \end{aligned} \quad (1.55)$$

where $Z = \rho - \rho' = |0\rangle\langle 0| - |1\rangle\langle 1| = |+\rangle\langle -| + |-\rangle\langle +|$. In this way, we have explicitly introduced the x-basis source states into the expression for $D_{z|e}$. Representing Bob’s x-basis measurement by the POVM $\{F, F'\}$, the x-basis measurement can also be explicitly introduced using that $\mathbb{1}_B = F + F'$, with the result

$$p_E(e)D_{z|e} = |\text{Re}[\langle + | F \otimes M_e | - \rangle] + \text{Re}[\langle + | F' \otimes M_e | - \rangle]|. \quad (1.56)$$

This result can be upper bounded by

$$\begin{aligned} p_E(e)D_{z|e} &\leq |\langle + | F \otimes M_e | - \rangle| + |\langle + | F' \otimes M_e | - \rangle| \\ &\leq \sqrt{\langle + | F \otimes M_e | + \rangle} \sqrt{\langle - | F \otimes M_e | - \rangle} \\ &\quad + \sqrt{\langle + | F' \otimes M_e | + \rangle} \sqrt{\langle - | F' \otimes M_e | - \rangle}, \end{aligned} \quad (1.57)$$

where the second line follows from applying the Cauchy-Schwarz inequality to, e.g., the inner product of $\sqrt{F} \otimes \sqrt{M_e}|+\rangle$ and $\sqrt{F} \otimes \sqrt{M_e}|-\rangle$. (Because the operators F , F' , and M_e are Hermitian and positive semidefinite as POVM elements, their square roots are well defined.) Note that the result has the form $\sqrt{a}\sqrt{b} + \sqrt{c}\sqrt{d}$; any expression of this type can be viewed as a scalar product and, again by the Cauchy-Schwarz inequality, can be upper bounded by either $\sqrt{a+c}\sqrt{b+d}$ or $\sqrt{a+d}\sqrt{b+c}$. Applying this,

$$\begin{aligned} p_E(e)D_{z|e} &\leq \sqrt{\langle +|F \otimes M_e|+ \rangle + \langle -|F' \otimes M_e|- \rangle} \\ &\quad \times \sqrt{\langle -|F \otimes M_e|- \rangle + \langle +|F' \otimes M_e|+ \rangle} \\ &= \sqrt{\text{Tr}[(F \otimes M_e)\sigma] + \text{Tr}[(F' \otimes M_e)\sigma']} \\ &\quad \times \sqrt{\text{Tr}[(F \otimes M_e)\sigma'] + \text{Tr}[(F' \otimes M_e)\sigma]}, \end{aligned} \quad (1.58)$$

where we have introduced the density operators $\sigma = |+\rangle\langle +|$ and $\sigma' = |-\rangle\langle -|$ for the x-basis states. We note that, for the four terms appearing under the square roots, the sum is proportional to the probability of Eve obtaining the result e :

$$\begin{aligned} p_E(e) &= \frac{1}{2} \text{Tr}[(F \otimes M_e)\sigma] + \frac{1}{2} \text{Tr}[(F' \otimes M_e)\sigma] \\ &\quad + \frac{1}{2} \text{Tr}[(F \otimes M_e)\sigma'] + \frac{1}{2} \text{Tr}[(F' \otimes M_e)\sigma']. \end{aligned} \quad (1.59)$$

We now introduce a variable $\delta_{x|e}$ defined such that

$$p_E(e)\delta_{x|e} = \frac{1}{2} \text{Tr}[(F' \otimes M_e)\sigma] + \frac{1}{2} \text{Tr}[(F \otimes M_e)\sigma'], \quad (1.60)$$

$$p_E(e)(1 - \delta_{x|e}) = \frac{1}{2} \text{Tr}[(F \otimes M_e)\sigma] + \frac{1}{2} \text{Tr}[(F' \otimes M_e)\sigma']. \quad (1.61)$$

The quantity $\delta_{x|e}$ can be interpreted as the rate at which Alice and Bob would detect errors in the x basis conditioned on Eve obtaining the result e , if Eve had measured the POVM $\{M_e\}$. A property that will be important is that they average to the x-basis error rate:

$$\sum_e p_E(e)\delta_{x|e} = \frac{1}{2} \text{Tr}[F'\sigma_B] + \frac{1}{2} \text{Tr}[F\sigma'_B] = \delta_x. \quad (1.62)$$

Applying (1.60) and (1.61) to (1.58), we find that

$$p_E(e)D_{z|e} \leq \sqrt{2p_E(e)\delta_{x|e}} \sqrt{2p_E(e)(1 - \delta_{x|e})}, \quad (1.63)$$

which simplifies to an upper bound on $D_{z|e}$ that depends only on $\delta_{x|e}$:

$$D_{z|e} \leq 2\sqrt{\delta_{x|e}(1 - \delta_{x|e})}. \quad (1.64)$$

We now return to the conditional entropy. Explicitly inserting the upper bound (1.64) for $D_{z|e}$ into the lower bound (1.50) for the conditional Shannon entropy,

$$H(Z_A | Z_E) \geq \sum_e p_E(e)h\left(\frac{1}{2} + \sqrt{\delta_{x|e}(1 - \delta_{x|e})}\right). \quad (1.65)$$

Finally, using that the function $x \mapsto h(\frac{1}{2} + \sqrt{x(1-x)})$ is convex, we obtain the desired lower bound

$$H(Z_A | Z_E) \geq h(\frac{1}{2} + \sqrt{\delta_x(1-\delta_x)}), \quad (1.66)$$

for the conditional entropy, from which the FGGNP rate bound (1.47) above follows.

At this point, we have proved the security of the BB84 protocol against the class of individual attacks under the assumption that the source states satisfy the ideal BB84 relations, which was used in (1.55). A worthwhile remark is that the end result holds independently of Bob's measurements: the conditional entropy $H(Z_A | Z_B)$ is simply a function of the joint probability associated with Alice's and Bob's z-basis bits independently of the measurement performed, while in the derivation of the lower bound on $H(Z_A | Z_E)$ we used only that Bob's x-basis measurement is an unspecified binary-outcome POVM $\{F, F'\}$. The authors of [14] also explicitly derived a family of optimal unitary attacks and measurement for which the Csiszár-Körner rate coincides with the FGGNP bound, demonstrating that the bound is in fact tight.

Note that while we used the BB84 relations $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ in order to obtain the fourth line of (1.55), we never actually used the orthogonality relation $\langle 0|1\rangle = 0$, and in particular the derivation of the FGGNP rate still holds if $\langle 0|1\rangle$ is allowed a nonzero imaginary part. Following the discussion in section 1.2.3, the same rate still holds for the entanglement-based version of the BB84 protocol.

To end this section, we remark on a connection between the derivation given here and the no-cloning theorem as it was outlined in section 1.4.1. First, a corollary of (1.64) that was pointed out in [14] is that

$$\sum_e p_E(e) D_{z|e} \leq 2\sqrt{\delta_x(1-\delta_x)}, \quad (1.67)$$

which follows because the function $x \mapsto 2\sqrt{x(1-x)}$ is concave. The left-hand side can be expressed as

$$\begin{aligned} \sum_e p_E(e) D_{z|e} &= \sum_e |p_{AE}(0, e) - p_{AE}(1, e)| \\ &= \sum_e \frac{1}{2} |p_{E|A}(e | 0) - p_{E|A}(e | 1)|. \end{aligned} \quad (1.68)$$

The second line coincides with the definition of the total variation distance (or statistical distance) between two probability distributions, which we denote by $D(p_{E|0}, p_{E|1})$ for the probability distributions $p_{E|0}$ and $p_{E|1}$ of elements $p_{E|A}(e | 0)$ and $p_{E|A}(e | 1)$ respectively. In terms of Eve's marginals

of the z-basis states and the POVM $\{M_e\}$, its quantum value is

$$D(p_{E|0}, p_{E|1}) = \sum_e \frac{1}{2} |\text{Tr}[M_e(\rho_E - \rho'_E)]|. \quad (1.69)$$

The maximum of (1.69) over all POVMs $\{M_e\}$ is a distance between the density operators ρ_E and ρ'_E , which we denote by $D(\rho_E, \rho'_E)$, called the *trace distance*. The lowest possible x-basis error rate δ_x , with the minimisation taken over all POVMs $\{F, F'\}$ Bob could perform, can likewise be expressed in terms of the trace distance between Bob's marginals σ_B and σ'_B of the x-basis states by $\delta_x = \frac{1}{2} - \frac{1}{2}D(\sigma_B, \sigma'_B)$. Since the tradeoff relation (1.67) holds regardless of the measurements performed by Bob and Eve, it holds for the optimal measurements for which $D(p_{E|0}, p_{E|1}) = D(\rho_E, \rho'_E)$ and $\delta_x = \frac{1}{2} - \frac{1}{2}D(\sigma_B, \sigma'_B)$. Substituting these into (1.67) and rearranging, we obtain the alternative expression

$$D(\rho_E, \rho'_E)^2 + D(\sigma_B, \sigma'_B)^2 \leq 1 \quad (1.70)$$

for the tradeoff relation with the explicit appearance of the measurement operators removed. If we define the operators $Z = \rho - \rho'$ and $X = \sigma - \sigma'$, the result can also be expressed as

$$\frac{1}{4}\|Z_E\|_1^2 + \frac{1}{4}\|X_B\|_1^2 \leq 1 \quad (1.71)$$

in terms of an operator norm $\|\cdot\|_1$ called the *trace norm*, which the trace distance is typically defined in terms of. The counterexample used as a proof of the no-cloning theorem in section 1.4.1 is captured by the fact that if $\frac{1}{2}\|X_B\|_1 = 1$, then (1.71) implies $\frac{1}{2}\|Z_E\|_1 = 0$, i.e., if Bob can perfectly distinguish between the two x-basis states emitted by Alice, then Eve has no ability to distinguish between the two z-basis states. Conversely, if $\frac{1}{2}\|Z_E\|_1 = 1$, i.e., if Eve attacks in such a way as to be able to perfectly distinguish the z-basis states, then $\frac{1}{2}\|X_B\|_1 = 0$, i.e., Bob will be unable to distinguish between the x states and the error rate δ_x will be 1/2.

1.4.5 Security against collective attacks

As mentioned in section 1.4.3, the class of collective attacks (first defined by Biham *et al.* [63]) is defined similarly to the class of individual attacks in that an adversary is still assumed to attack the states in transit from Alice to Bob individually and identically. The important difference is that, in a collective attack, Eve is allowed to store her part of the quantum state (“quantum side information”) indefinitely in a quantum memory. In particular, Eve may perform any collective measurement on all the quantum states acquired via the unitary attacks and may delay her measurement until

after the postprocessing is applied. For this class of attacks, a lower bound on the asymptotic secret key rate extractable by one-way postprocessing is given by the Devetak-Winter rate [31]

$$r = H(Z_A | E) - H(Z_A | Z_B), \quad (1.72)$$

The Devetak-Winter rate can be considered the analogue of the Csiszár-Körner rate which applies to the class of individual attacks. The difference is that the conditional Shannon entropy $H(Z_A | Z_E)$ which appeared in the Csiszár-Körner rate is now replaced by the conditional von Neumann entropy $H(Z_A | E)$. This is defined by

$$H(Z_A | E) = S(\tau_{ZE}) - S(\tau_E), \quad (1.73)$$

where the von Neumann entropy is generally defined by $S(\rho) = -\text{Tr}[\rho \log(\rho)]$ and (1.73) is evaluated on the classical-quantum state

$$\tau_{ZE} = p_A(0)|0\rangle\langle 0|_Z \otimes \rho_E + p_A(1)|1\rangle\langle 1|_Z \otimes \rho'_E, \quad (1.74)$$

where the orthogonal states $|0\rangle_Z$ and $|1\rangle_Z$ denote the state of a classical register in Alice's possession and ρ_E and ρ'_E are Eve's partial traces of the z-basis states emitted by Alice with probabilities $p_A(0)$ and $p_A(1)$ respectively, as in the previous subsection. The state (1.74) describes the correlation between Alice's record of which z-basis state was transmitted and the corresponding quantum state that Eve has managed to acquire, and replaces the joint probability distribution $p_{AE}(a, e)$ of the previous subsection (which is no longer necessarily assumed to exist at all). The final key is secure in the stronger sense that the classical-quantum state $\tau_{K_A K_B E}$ describing the correlation between Alice's and Bob's final keys and Eve's quantum side information has the approximate form

$$\tau_{K_A K_B E} \approx \left(\sum_{\mathbf{k} \in \{0,1\}^n} \frac{1}{n} |\mathbf{k}\rangle\langle \mathbf{k}|_{K_A} \otimes |\mathbf{k}\rangle\langle \mathbf{k}|_{K_B} \right) \otimes \sigma_E, \quad (1.75)$$

with the approximation again becoming an equality in the asymptotic limit.

If the Devetak-Winter rate (1.72) is minimised for the prepare-and-measure BB84 protocol over all possible unitary attacks (or all pre-measurement tripartite states $|\psi\rangle_{ABE}$ for the entanglement-based version) for fixed error rates δ_z and δ_x , the result is the Shor-Preskill rate

$$r \geq 1 - h(\delta_x) - h(\delta_z). \quad (1.76)$$

(The version $r = 1 - 2h(\delta)$ quoted in section 1.3.1 is a special case with $\delta_z = \delta_x = \delta$.) The minimising unitary attack is, in fact, the same as the unitary part of the optimal attack that was derived by Fuchs *et al.* for the Csiszár-Körner rate.

There are several derivations of the Shor-Preskill rate as a security bound for the BB84 protocol. The original derivation, by Shor and Preskill, was derived based on results from the theory of entanglement purification and quantum error correction codes [53]. The first derivation as a lower bound on the Devetak-Winter or a similar rate was by Renner, Gisin, and Kraus [32]. Here, we highlight a particularly simple derivation based on a tradeoff relation,

$$H(X_A | B) + H(Z_A | E) \geq 1, \quad (1.77)$$

conjectured by Renes and Boileau [64] and later proved by Berta *et al.* [59] which holds following σ_x and σ_z measurements on the $\mathcal{H}_A$ part of any tripartite density operator ρ_{ABE} acting on $\mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_E$. This allows the conditional entropy $H(Z_A | E)$ appearing in the Devetak-Winter rate to be bounded in terms of quantities estimable by Alice and Bob working in cooperation:

$$\begin{aligned} r &= H(Z_A | E) - H(Z_A | Z_B) \\ &\geq 1 - H(X_A | B) - H(Z_A | Z_B) \\ &\geq 1 - H(X_A | X_B) - H(Z_A | Z_B) \\ &\geq 1 - h(\delta_x) - h(\delta_z). \end{aligned} \quad (1.78)$$

Note that, in each case, the Shor-Preskill rate was derived from the entanglement-based perspective, i.e., the key rate was derived for the entanglement-based version of the BB84 protocol and uses the equivalence explained in section 1.2.2 in order to claim the result as a security bound for the prepare-and-measure version. In chapter 3, we will investigate how key rates can be derived directly from the prepare-and-measure perspective, in a style similar to the derivation of the FGGNP rate given in section 1.4.4, which will include the Shor-Preskill key rate as a special case.

1.4.6 Unconditional security

In an unconditional security proof, no assumptions are made about an adversary's attack. In the asymptotic limit, the security of the BB84 protocol against general attacks is known to be the same as for collective attacks, i.e., the Shor-Preskill rate is still known to apply as a security bound. Its original derivation by Shor and Preskill was, in fact, in the context of an unconditional security proof. (Mayers, who is usually credited with the first proof of unconditional security of the BB84 protocol, derived a bound on the tolerable channel noise that was lower than the Shor-Preskill bound of about 11% [65]).

For entanglement-based QKD protocols and prepare-and-measure protocols that satisfy the basis-independence condition, security against collective attacks is known to imply unconditional security with the same key rate in the asymptotic limit. Security proofs based on the Devetak-Winter bound or a similar result typically establish this via the exponential quantum de Finetti theorem [66] or the related postselection technique [67] for a version of the BB84 protocol in which Alice and Bob apply a random permutation to their raw key bits. Note that such a step is necessary for security proofs based on the Devetak-Winter rate, which was itself derived assuming an identical and independent distribution of the underlying shared state.

The reduction to collective attacks for prepare-and-measure protocols has not to date received such explicit consideration. As such, the results derived in chapters 3 and 4, which are based on the Devetak-Winter bound, are given as security bounds applicable to collective attacks and the question of whether or how they translate to unconditional security proofs will not be explicitly addressed here.

1.A Comparing quantum states

The similarity or distinguishability of two pure quantum states $|\psi\rangle$ and $|\phi\rangle$ is naturally characterised by their inner product $\langle\phi|\psi\rangle$. There is more than one possible way to generalise this concept to density operators, each with different uses. In this section we define and describe two ways of comparing quantum states, the *trace distance* and the *fidelity*, which are widely used in quantum information theory and which will frequently be used in this thesis. Both can be defined in terms of an operator norm called the *trace norm*.

1.A.1 The trace norm

Definition

The trace norm of a linear operator $A: \mathcal{H} \rightarrow \mathcal{H}'$, noted $\|A\|_1$, is defined by

$$\|A\|_1 = \text{Tr}[|A|], \quad (1.79)$$

with $|A|$ in turn defined by $|A| = \sqrt{A^\dagger A}$. Note that $A^\dagger A$ is positive semidefinite, i.e.,

$$\langle\psi|A^\dagger A|\psi\rangle \geq 0, \quad \forall |\psi\rangle \in \mathcal{H}; \quad (1.80)$$

consequently its square root is well defined as the unique positive semidefinite operator such that $\sqrt{A^\dagger A} \sqrt{A^\dagger A} = A^\dagger A$.

Alternative definitions

The trace norm admits a couple of useful equivalent alternative expressions. By the singular value decomposition theorem, there is an orthonormal basis $\{|k\rangle\}$ of $\mathcal{H}$ and an orthonormal basis $\{|k'\rangle\}$ of $\mathcal{H}'$ in which the operator A takes the expression $A = \sum_k s_k |k'\rangle\langle k|$, where the s_k are real and non-negative. In terms of this factorisation, $|A| = \sum_k s_k |k\rangle\langle k|$, from which we find

$$\|A\|_1 = \sum_k s_k, \quad (1.81)$$

i.e., the trace norm of an operator is simply the sum of its singular values.

The operator A can also be expressed in terms of $|A|$ by $A = U|A|$ (its polar decomposition), where the change of basis is achieved with the unitary $U = \sum_k |k'\rangle\langle k|$. Applied to (1.79), this means that there always exists a unitary U such that

$$\|A\|_1 = \text{Tr}[UA]. \quad (1.82)$$

It is possible to prove that the unitary operation above maximises the right-hand side of (1.82), from which we obtain a second expression for the trace norm:

$$\|A\|_1 = \max_U |\text{Tr}[UA]|. \quad (1.83)$$

The upper bound $|\text{Tr}[UA]| \leq \|A\|_1$ is a special case of a more general inequality. Specifically, if A and B are two linear operators (with the same domain and codomain), then

$$|\text{Tr}[A^\dagger B]| \leq \sum_k s_k t_k, \quad (1.84)$$

where $\{s_k\}$ and $\{t_k\}$ are respectively the singular values of A and B , ordered such that $s_k \geq s_{k+1}$ and $t_k \geq t_{k+1}$.

Basic properties

It is easy to see from the various expressions given here that the trace norm is indeed a matrix norm. For instance the condition $\|A\|_1 = 0 \Rightarrow A = 0$ is

evident from (1.82), while the property $\|A + B\|_1 \leq \|A\|_1 + \|B\|_1$ follows easily from (1.83):

$$\begin{aligned}
\|A + B\|_1 &= \max_U |\text{Tr}[U(A + B)]| \\
&\leq \max_U (|\text{Tr}[UA]| + |\text{Tr}[UB]|) \\
&\leq \max_U |\text{Tr}[UA]| + \max_U |\text{Tr}[UB]| \\
&= \|A\|_1 + \|B\|_1.
\end{aligned} \tag{1.85}$$

From (1.81), it is also clear that $\|A\|_1 = \|A^\dagger\|_1$.

Hermitian operators

In the case where A is Hermitian, i.e., $\mathcal{H}' = \mathcal{H}$ and $A^\dagger = A$, its trace norm is simply the sum of the absolute values of its eigenvalues. If $A = \sum_k a_k |k\rangle\langle k|$ is a diagonalised expression for A , with $\{|k\rangle\}$ forming an orthonormal basis, then $|A| = \sum_k |a_k| |k\rangle\langle k|$ and

$$\|A\|_1 = \sum_k |a_k|. \tag{1.86}$$

The operator U appearing in (1.82) is also Hermitian in this case, and can be obtained by $U = P - Q$ where, for instance P and Q can be defined by

$$P = \sum_{k, a_k \geq 0} |k\rangle\langle k|, \tag{1.87}$$

$$Q = \sum_{k, a_k < 0} |k\rangle\langle k|. \tag{1.88}$$

Defined this way, $A = U|A|$. Note that $P^2 = P$, $Q^2 = Q$, $PQ = QP = 0$, and $P + Q = \mathbb{1}$, i.e., U is simply the difference between two orthogonal projectors corresponding to the positive and negative eigenvalue subspaces of A . For A Hermitian, then, its trace norm can be identified by

$$\|A\|_1 = \max_U \text{Tr}[UA], \tag{1.89}$$

where, the maximisation this time is taken over the set of Hermitian unitaries $U = U^\dagger$. Note that it is no longer necessary to take the absolute value, as $\text{Tr}[UA]$ is always real in this case and if U is unitary then its negation $-U$ is also a unitary operator. Inserting $U = 2P - \mathbb{1}$, where P is a projector, the trace norm can equivalently be obtained by

$$\frac{1}{2}\|A\|_1 = \max_P \text{Tr}[PA] - \frac{1}{2} \text{Tr}[A], \tag{1.90}$$

with the maximisation over all projectors acting on $\mathcal{H}$. Finally, we note that the result is unaffected if the maximisation is extended over the set of all POVM elements:

$$\frac{1}{2}\|A\|_1 = \max_M \text{Tr}[MA] - \frac{1}{2}\text{Tr}[A], \quad (1.91)$$

with $M^\dagger = M$ and $0 \leq M \leq 1$. To see this, it is sufficient to verify that for any POVM element M , one can construct a projection operator P such that $\text{Tr}[MA] \leq \text{Tr}[PA]$. Expressing M in its spectral decomposition $M = \sum_k m_k |k\rangle\langle k|$, with $0 \leq m_k \leq 1$, the trace becomes

$$\text{Tr}[MA] = \sum_k m_k \langle k|A|k\rangle. \quad (1.92)$$

For each value of k , $\langle k|A|k\rangle$ is either positive, in which case $m_k \langle k|A|k\rangle \leq \langle k|A|k\rangle$, or $\langle k|A|k\rangle$ is negative or zero, in which case $\langle k|A|k\rangle \leq 0$ by definition. It follows that the sum in (1.92) is upper bounded by $\sum_k p_k \langle k|A|k\rangle$, with

$$p_k = \begin{cases} 1 & : \langle k|A|k\rangle > 0 \\ 0 & : \langle k|A|k\rangle \leq 0 \end{cases}, \quad (1.93)$$

and $\text{Tr}[MA] \leq \text{Tr}[PA]$ with $P = \sum_k p_k |k\rangle\langle k|$.

1.A.2 The trace distance

Definition

The trace distance between two density operators ρ and σ is defined by

$$D(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1. \quad (1.94)$$

For pure states $\psi = |\psi\rangle\langle\psi|$ and $\phi = |\phi\rangle\langle\phi|$, the trace distance reduces to a function

$$D(\psi, \phi) = \sqrt{1 - |\langle\psi|\phi\rangle|^2} \quad (1.95)$$

of the inner product.

Basic properties of the trace distance

The trace distance satisfies the general properties of a distance measure, most of which follow from properties of the trace norm discussed above. For instance,

$$D(\rho, \sigma) = 0 \Rightarrow \|\rho - \sigma\|_1 = 0 \Rightarrow \rho - \sigma = 0 \Rightarrow \rho = \sigma. \quad (1.96)$$

From the property $\|A + B\|_1 \leq \|A\|_1 + \|B\|_1$, we obtain

$$D(\rho, \tau) = \frac{1}{2} \|(\rho - \sigma) + (\sigma - \tau)\|_1 \leq D(\rho, \sigma) + D(\sigma, \tau) \quad (1.97)$$

and

$$D(\rho, \sigma) \leq \frac{1}{2} \|\rho\|_1 + \frac{1}{2} \|\sigma\|_1 = \frac{1}{2} \text{Tr}[\rho] + \frac{1}{2} \text{Tr}[\sigma]. \quad (1.98)$$

The latter bound is attained if and only if ρ and σ are orthogonal. According to the expression (1.94) for the trace norm of a Hermitian operator, there exists a projection operator P for which

$$D(\rho, \sigma) = \text{Tr}[P(\rho - \sigma)] - \frac{1}{2} \text{Tr}[\rho - \sigma]. \quad (1.99)$$

Requiring $D(\rho, \sigma) = \frac{1}{2} \text{Tr}[\rho + \sigma]$, we extract

$$\text{Tr}[P\rho] - \text{Tr}[P\sigma] = \text{Tr}[\rho]. \quad (1.100)$$

Because $\text{Tr}[P\rho] \leq \|\rho\|_1 = \text{Tr}[\rho]$ and $\text{Tr}[P\sigma] = \text{Tr}[P\sigma P] \geq 0$ (because $P\sigma P$ is positive semidefinite), (1.100) implies $\text{Tr}[P\rho P] = \text{Tr}[\rho]$ and $\text{Tr}[P\sigma P] = 0$. Similarly, for the projector $Q = \mathbb{1} - P$, $\text{Tr}[Q\rho Q] = 0$ and $\text{Tr}[Q\sigma Q]$. Working with ρ and using that $Q\rho Q \geq 0$, $\text{Tr}[Q\rho Q] = 0$ implies $Q\rho Q = 0$, which in turn implies $Q\sqrt{\rho} = \sqrt{\rho}Q = 0$. Multiplying again by $\sqrt{\rho}$ and reinserting $Q = \mathbb{1} - P$, we obtain

$$P\rho P = \rho, \quad (1.101)$$

$$Q\rho Q = 0. \quad (1.102)$$

Similarly for σ ,

$$P\sigma P = 0, \quad (1.103)$$

$$Q\sigma Q = \sigma. \quad (1.104)$$

ρ and σ are thus orthogonal in the sense that their support is on orthogonal subspaces.

If ρ and σ are properly normalised, i.e., if $\text{Tr}[\rho] = \text{Tr}[\sigma] = 1$, then the factor of $1/2$ in (1.94) guarantees $D(\rho, \sigma) \leq 1$, again with equality if and only if ρ and σ are orthogonal.

Finally, a useful property of the trace distance is that it can only decrease following partial tracing, i.e., if $\rho_A = \text{Tr}_B[\rho_{AB}]$ and $\sigma_A = \text{Tr}_B[\sigma_{AB}]$, then

$$D(\rho_A, \sigma_A) \leq D(\rho_{AB}, \sigma_{AB}). \quad (1.105)$$

This is easily seen by expressing the trace distance as a maximisation over projection operators:

$$\begin{aligned}
D(\rho_A, \sigma_A) &= \max_{P_A} \text{Tr}[P_A(\rho_A - \sigma_A)] \\
&= \max_{P_A} \text{Tr}[(P_A \otimes \mathbb{1}_B)(\rho_{AB} - \sigma_{AB})] \\
&\leq \max_{P_{AB}} \text{Tr}[P_{AB}(\rho_{AB} - \sigma_{AB})] \\
&= D(\rho_{AB}, \sigma_{AB}).
\end{aligned} \tag{1.106}$$

Relevance to state discrimination

The trace distance has a well known operational significance in the context of state discrimination. Specifically, suppose one wishes to distinguish between two density operators ρ and σ , drawn with equal probability, i.e., $p(\rho) = p(\sigma) = 1/2$, with a binary outcome measurement described by a POVM $\{M, N\}$. If M is intended to detect the state ρ and N the state σ , the average probability of correctly guessing the state (called the “guessing probability”) is given by

$$\begin{aligned}
P_{\text{guess}} &= p(\rho, M) + p(\sigma, N) \\
&= \frac{1}{2}p(M | \rho) + \frac{1}{2}p(N | \sigma) \\
&= \frac{1}{2} \text{Tr}[M\rho] + \frac{1}{2} \text{Tr}[N\sigma] \\
&= \frac{1}{2} + \frac{1}{2} \text{Tr}[M(\rho - \sigma)],
\end{aligned} \tag{1.107}$$

where we inserted $N = \mathbb{1} - M$ to obtain the final line. From the expression (1.91) for the trace norm, and noting that $\text{Tr}[\rho - \sigma] = 0$, we obtain a version of the Helstrom bound [68]:

$$P_{\text{guess}} \leq \frac{1}{2} + \frac{1}{2}D(\rho, \sigma). \tag{1.108}$$

The lower bound $P_{\text{guess}} \geq \frac{1}{2} - \frac{1}{2}D(\rho, \sigma)$ can be obtained in a similar way.

In QKD, it is more conventional to express security bounds in terms of the error rate δ , i.e., the average probability of *incorrectly* identifying the state. In anticipation of this, we note that the Helstrom bound can equivalently be given for the error rate:

$$\frac{1}{2} - \frac{1}{2}D(\rho, \sigma) \leq \delta \leq \frac{1}{2} + \frac{1}{2}D(\rho, \sigma), \tag{1.109}$$

which can be obtained by essentially the same derivation as the Helstrom bound for P_{guess} given above, or simply by using that $\delta = 1 - P_{\text{guess}}$. Note that the lower and upper bounds in (1.109) can be rearranged to give

$$D(\rho, \sigma) \geq |1 - 2\delta|. \tag{1.110}$$

Expressed this way, the average error rate provides a lower bound on the trace distance between ρ and σ .

If the states ρ and σ are drawn with prior probabilities $p(\rho) = p$ and $p(\sigma) = q = 1 - p$, the Helstrom bound generalises to

$$\frac{1}{2} - \frac{1}{2}\|p\rho - q\sigma\|_1 \leq P_{\text{guess}} \leq \frac{1}{2} + \frac{1}{2}\|p\rho - q\sigma\|_1 \quad (1.111)$$

and

$$\|p\rho - q\sigma\|_1 \geq |1 - 2P_{\text{guess}}|. \quad (1.112)$$

1.A.3 The fidelity

Definition

The fidelity is a comparison measure between density operators that was introduced by Josza [69]. Unlike the trace distance, it does not have an obvious operational significance, however it will prove useful as an intermediate quantity in certain derivations. In this thesis, for two density operators ρ and σ we will take the fidelity to be defined by

$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1. \quad (1.113)$$

It should be noted that it is sometimes the square of (1.113) that is called the “fidelity” in the literature. By contrast with the trace distance, the fidelity is a measure of “closeness” of two states in the sense that $F(\rho, \sigma) = 1$ if $\rho = \sigma$ and $F(\rho, \sigma) = 0$ if ρ and σ are orthogonal. If $\psi = |\psi\rangle\langle\psi|$ and $\phi = |\phi\rangle\langle\phi|$ are pure states, the fidelity reduces to the inner product $F(\psi, \phi) = |\langle\psi|\phi\rangle|$.

Uhlmann’s theorem

An important alternative definition of the fidelity is given by Uhlmann’s theorem, which states that

$$F(\rho, \sigma) = \max_{\psi, \phi} |\langle\psi|\phi\rangle|, \quad (1.114)$$

with the maximisation taken over all purifications $|\psi\rangle$ and $|\phi\rangle$ of the density operators ρ and σ , i.e., subject to the constraints $\rho_A = \text{Tr}_B[|\psi\rangle\langle\psi|_{AB}]$ and $\sigma_A = \text{Tr}_B[|\phi\rangle\langle\phi|_{AB}]$ if ρ and σ are taken to act on the Hilbert space $\mathcal{H}_A$ and $|\psi\rangle$ and $|\phi\rangle$ are vectors in an extended Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$.

Uhlmann's theorem can be proved in the following way. We express ρ and σ in terms of their spectral decompositions $\rho = \sum_k p_k |\alpha_k\rangle\langle\alpha_k|$ and $\sigma = \sum_j q_j |\beta_j\rangle\langle\beta_j|$, which we use to define the (unnormalised) state $|\Phi^+\rangle = \sum_k |\alpha_k\rangle|k\rangle$ for any arbitrary basis $\{|k\rangle\}$ of an auxiliary Hilbert space and the unitary $V = \sum_k |\beta_k\rangle\langle\alpha_k|$. In terms of these, note that

$$|\psi\rangle = \sqrt{\rho} \otimes \mathbb{1} |\Phi^+\rangle, \quad (1.115)$$

$$|\phi\rangle = (\sqrt{\sigma}V) \otimes \mathbb{1} |\Phi^+\rangle, \quad (1.116)$$

are purifications of ρ and σ , respectively, which expand to their Schmidt decompositions. In terms of these, for the fidelity we obtain

$$F(\rho, \sigma) = \max_U |\text{Tr}[U \sqrt{\rho} \sqrt{\sigma}]| \quad (1.117)$$

$$= \max_U |\text{Tr}[U \sqrt{\rho} \sqrt{\sigma} V]| \quad (1.118)$$

$$= \max_U |\langle\Phi^+|(\sqrt{\rho} \sqrt{\sigma} V) \otimes U^\dagger |\Phi^+\rangle| \quad (1.119)$$

$$= \max_U |\langle\psi| \mathbb{1} \otimes U |\phi\rangle|. \quad (1.120)$$

Since $\mathbb{1} \otimes U |\phi\rangle$ is still a purification of σ for any unitary U , and all purifications of a density operator are related by unitaries on the auxiliary (purifying) Hilbert space, the final line is simply a maximisation over all purifications of ρ and σ .

Relation to the trace distance

Uhlmann's theorem can be used to derive an upper bound [70],

$$D(\rho, \sigma) \leq \sqrt{1 - F(\rho, \sigma)^2}, \quad (1.121)$$

on the trace distance in terms of the fidelity. To see this, we take $|\psi\rangle$ and $|\phi\rangle$ to be purifications of ρ and σ such that $F(\rho, \sigma) = |\langle\psi|\phi\rangle|$. Using that the trace distance can only decrease as a result of partial tracing,

$$D(\rho, \sigma) \leq D(\psi, \phi) = \sqrt{1 - |\langle\psi|\phi\rangle|^2} = \sqrt{1 - F(\rho, \sigma)^2}. \quad (1.122)$$

1.B Miscellaneous tools

1.B.1 Swap trick

Given a maximally entangled bipartite state

$$|\Phi^+\rangle = \frac{1}{\sqrt{N}} \sum_{k=1}^N |k\rangle_A |k\rangle_B \in \mathcal{H}_A \otimes \mathcal{H}_B \quad (1.123)$$

and any operator A acting on $\mathcal{H}_A$, a useful identity is

$$A \otimes \mathbb{1} |\Phi^+\rangle = \mathbb{1} \otimes A^\top |\Phi^+\rangle. \quad (1.124)$$

The transpose in (1.124) is taken relative to the basis in which the $|\Phi^+\rangle$ state is defined. Specifically, if the operator A has matrix elements $\langle j|A|k\rangle$, then $A^\top$ is taken to be the operator such that $\langle j|A^\top|k\rangle = \langle k|A|j\rangle$. The proof proceeds by inserting the identity $\mathbb{1}_A = \sum_j |j\rangle\langle j|_A$ and rearranging terms:

$$\begin{aligned} \sqrt{N} A \otimes \mathbb{1} |\Phi^+\rangle &= \left(\sum_j |j\rangle\langle j|_A \right) A \otimes \mathbb{1} \left(\sum_k |k\rangle_A |k\rangle_B \right) \\ &= \sum_{jk} |j\rangle_A \langle j|A|k\rangle_A |k\rangle_B \\ &= \sum_{jk} |j\rangle_A |k\rangle_B \langle k|A^\top|j\rangle_B \\ &= \sum_j \mathbb{1} \otimes A^\top |j\rangle_A |j\rangle_B. \end{aligned} \quad (1.125)$$

A related identity, for the trace of the product of two operators, is

$$\text{Tr}[AB] = N \langle \Phi^+ | A \otimes B^\top | \Phi^+ \rangle, \quad (1.126)$$

which can be obtained by a similar derivation as (1.124):

$$\begin{aligned} \text{Tr}[AB] &= \sum_k \langle k|AB|k\rangle \\ &= \sum_{jk} \langle k|A|j\rangle \langle j|B|k\rangle \\ &= \sum_{jk} \langle k|A|j\rangle_A \langle k|B^\top|j\rangle_B \\ &= \left(\sum_k \langle k|_A \langle k|_B \right) A \otimes B^\top \left(\sum_j |j\rangle_A |j\rangle_B \right) \\ &= N \langle \Phi^+ | A \otimes B^\top | \Phi^+ \rangle. \end{aligned} \quad (1.127)$$

The relations above can be useful in problems involving a maximisation over a given set of operators, especially where the set as a whole is invariant under transposition. For instance, the transpose of a Hermitian operator is still Hermitian and the transpose of a unitary operator is still unitary.

1.B.2 Schmidt decomposition

It is possible to show that any bipartite pure state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ admits an expression of the form

$$|\psi\rangle = \sum_k \lambda_k |k\rangle_A |k\rangle_B, \quad (1.128)$$

in which $\{|k\rangle_A\}$ and $\{|k\rangle_B\}$ are, respectively, orthonormal bases of $\mathcal{H}_A$ and $\mathcal{H}_B$ and all the coefficients λ_k are real and nonnegative.

The Schmidt decomposition can be arrived at in the following way: first, we find the marginal density operator $\rho = \text{Tr}_B[|\psi\rangle\langle\psi|]$, which we express in terms of its spectral decomposition $\rho = \sum_k p_k |k\rangle\langle k|_A$. We then expand $|\psi\rangle$ in the basis $\{|k\rangle_A\}$ composed of its eigenstates, obtaining

$$|\psi\rangle = \left(\sum_k |k\rangle\langle k|_A \otimes \mathbb{1}_B \right) |\psi\rangle = \sum_k |k\rangle_A |\tilde{k}\rangle_B, \quad (1.129)$$

where $|\tilde{k}\rangle_B = (\langle k|_A \otimes \mathbb{1}_B) |\psi\rangle$. Finally evaluating ρ again,

$$\rho = \sum_{jk} |j\rangle\langle k|_A \langle \tilde{k} | \tilde{j} \rangle_B, \quad (1.130)$$

which only recovers the spectral decomposition of ρ if $\langle \tilde{j} | \tilde{k} \rangle_B = p_k \delta_{jk}$. The Schmidt decomposition is obtained by substituting $|\tilde{k}\rangle_B = \sqrt{p_k} |k\rangle_B$ and $\sqrt{p_k} = \lambda_k$.

A corollary of the derivation of the Schmidt decomposition given here is that all purifications of a density operator ρ , i.e., pure states $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ such that $\rho = \text{Tr}_B[|\psi\rangle\langle\psi|]$, differ only by unitary operations acting on the auxiliary Hilbert space. If $|\psi\rangle$ and $|\psi'\rangle$ are two purifications of ρ , according to the derivation given above they can be expressed in terms of their Schmidt decompositions $|\psi\rangle = \sum_k \sqrt{p_k} |k\rangle_A |k\rangle_B$ and $|\psi'\rangle = \sum_k \sqrt{p_k} |k\rangle_A |k'\rangle_B$ for the same set $\{|k\rangle_A\}$ of eigenstates of ρ . $|\psi\rangle$ and $|\psi'\rangle$ are then related by $|\psi'\rangle = \mathbb{1} \otimes U |\psi\rangle$ for the unitary $U = \sum_k |k'\rangle\langle k|_B$. Conversely, if $|\psi\rangle$ is a purification of ρ and U a unitary operation, then $|\psi'\rangle = \mathbb{1} \otimes U |\psi\rangle$ is still a purification: $\text{Tr}_B[|\psi'\rangle\langle\psi'|] = \text{Tr}_B[U |\psi\rangle\langle\psi| U^\dagger] = \text{Tr}_B[|\psi\rangle\langle\psi|] = \rho$.

1.B.3 von Neumann trace inequality

For A, B Hermitian, the trace of their product is upper bounded by

$$\text{Tr}[AB] \leq \sum_k a_k b_k, \quad (1.131)$$

in which a_k and b_k are the eigenvalues of A and B respectively, listed in decreasing order such that $a_k \geq a_{k+1}$ and $b_k \geq b_{k+1}$. This can be obtained by substituting spectral decompositions $A = \sum_j a_j |\alpha_j\rangle\langle\alpha_j|$ and $B = \sum_k b_k |\beta_k\rangle\langle\beta_k|$ for A and B :

$$\begin{aligned}\text{Tr}[AB] &= \sum_{jk} a_j b_k |\langle\alpha_j|\beta_k\rangle|^2 \\ &= \sum_{jk} a_j S_{jk} b_k \\ &= \mathbf{a}^\top \mathbf{S} \mathbf{b},\end{aligned}\tag{1.132}$$

where we have introduced the matrix $\mathbf{S}$ of elements $S_{ij} = |\langle\alpha_j|\beta_k\rangle|^2$, and $\mathbf{a}$ and $\mathbf{b}$ are the column matrices of elements a_j and b_k , respectively. Note that $\sum_j S_{jk} = \sum_k S_{jk} = 1$ and $S_{jk} \geq 0$, meaning that $\mathbf{S}$ is a doubly stochastic matrix and thus (by the Birkhoff-von Neumann theorem) can be expressed as a convex sum of permutation matrices

$$\mathbf{S} = \sum_{\sigma} p_{\sigma} \mathbf{\Pi}_{\sigma},\tag{1.133}$$

with $p_{\sigma} \geq 0$, $\sum_{\sigma} p_{\sigma} = 1$, and the permutation matrix $\mathbf{\Pi}_{\sigma}$ is of elements $\Pi_{jk}^{(\sigma)} = \delta_{j,\sigma(k)}$. Consequently,

$$\begin{aligned}\text{Tr}[AB] &= \sum_{\sigma} p_{\sigma} \mathbf{a}^\top \mathbf{\Pi}_{\sigma} \mathbf{b} \\ &\leq \max_{\sigma} \mathbf{a}^\top \mathbf{\Pi}_{\sigma} \mathbf{b} \\ &= \max_{\sigma} \sum_{jk} \delta_{j,\sigma(k)} a_j b_k \\ &= \max_{\sigma} \sum_k a_{\sigma(k)} b_k \\ &= \sum_k a_k b_k.\end{aligned}\tag{1.134}$$

Because the spectrum of a Hermitian operator is a unitary invariant (in the sense that A and $UAU^\dagger$ have the same eigenvalues), an alternative expression is

$$\max_U \text{Tr}[AUBU^\dagger] = \sum_k a_k b_k.\tag{1.135}$$

The equality $\text{Tr}[AUBU^\dagger] = \sum_k a_k b_k$ is clearly attainable with $U = \sum_k |\alpha_k\rangle\langle\beta_k|$, for which $UBU^\dagger$ is diagonal in the same basis as A .

More generally for, operators $A, B: \mathcal{H} \rightarrow \mathcal{H}'$, a similar inequality credited to John von Neumann [71] holds in terms of their singular values $\{s_j\}$ and

$\{t_k\}$:

$$|\mathrm{Tr}[A^\dagger B]| \leq \sum_k s_k t_k, \quad (1.136)$$

with the singular values similarly listed in decreasing order. The analogue of (1.135) is

$$\max_{U,V} |\mathrm{Tr}[A^\dagger U B V^\dagger]| = \sum_k s_k t_k, \quad (1.137)$$

with the maximisation taken over unitaries $U: \mathcal{H} \rightarrow \mathcal{H}$ and $V: \mathcal{H}' \rightarrow \mathcal{H}'$.

A simple corollary of the von Neumann trace inequality is that

$$|\mathrm{Tr}[AB]| \leq \|A\|_1 \|B\|_\infty, \quad (1.138)$$

with the ∞ -norm defined by $\|B\|_\infty = \lim_{p \rightarrow \infty} \mathrm{Tr}[|B|^p]^{1/p}$, i.e., the maximum singular value of B .

Chapter 2

Impact of device imprecisions on security

The main purpose of this chapter is to demonstrate the negative impact of source and measurement imprecisions on the security of the BB84 protocol. This is achieved by numerically minimising the key rate against an adversary restricted to individual attacks, for which necessary and sufficient conditions are known, which is shown to decrease in cases where Alice's and Bob's bases are allowed to differ from σ_z and σ_x . With the exception of section 2.A, the work presented here was earlier reported in [19], on which this chapter is based. The project itself originated from a loose collaboration with the authors of [72, 73], who similarly explored the effect of imperfections in the alignment of measurement bases on the characterisation of quantum resources through quantum state tomography and entanglement witnesses. In the context of this thesis, this chapter motivates and establishes the necessity of the work detailed in chapters 3 and 4.

While working on the project, it was initially attempted to derive the the key rate analytically. An early partial result in this direction is included as appendix 2.A at the end of this chapter, as it may be of interest in itself.

2.1 Introduction

In the BB84 protocol, Alice emits one of two states $|\alpha\rangle$ or $|\alpha'\rangle$, which we collectively call the “z basis”, or one of the two states $|\beta\rangle$ or $|\beta'\rangle$, which we

collectively call the “x basis”. The other party (“Bob”) then randomly measures each qubit he receives in one of two bases $\{|A\rangle, |A'\rangle\}$ or $\{|B\rangle, |B'\rangle\}$. As discussed in section 1.2.1, in its ideal formulation, the states $\{|\alpha\rangle, |\alpha'\rangle\}$ and $\{|\beta\rangle, |\beta'\rangle\}$ prepared by Alice are supposed to constitute orthogonal bases, i.e., the states should satisfy the orthogonality relations

$$\langle\alpha|\alpha'\rangle = \langle\beta|\beta'\rangle = 0. \quad (2.1)$$

Furthermore, the two bases on Alice’s and on Bob’s sides are supposed to be mutually unbiased, i.e., they should differ exactly by an angle of $\pi/2$, or satisfy the relations¹

$$|\beta\rangle = \frac{1}{\sqrt{2}}[|\alpha\rangle + |\alpha'\rangle], \quad (2.2a)$$

$$|\beta'\rangle = \frac{1}{\sqrt{2}}[|\alpha\rangle - |\alpha'\rangle], \quad (2.2b)$$

and

$$|B\rangle = \frac{1}{\sqrt{2}}[|A\rangle + |A'\rangle], \quad (2.3a)$$

$$|B'\rangle = \frac{1}{\sqrt{2}}[|A\rangle - |A'\rangle]. \quad (2.3b)$$

While existing security proofs for BB84 routinely account for channel noise, most security analyses take for granted that the source states and measurement precisely satisfy the conditions (2.1), (2.2), and (2.3). As was stressed in the introduction, imprecisions are inevitable in a realistic implementation of the protocol. For instance, the measurement of a polarisation qubit may not be more precise than, say, 2° or 4° (on the Bloch sphere) due to the intrinsic uncertainty of the polarisation rotator used. Such imperfections may allow an eavesdropper to gain more information about the shared key than existing security proofs would imply. While some theoretical security analyses have relaxed these assumptions, such as those cited in section 1.3.2, they were generally not applied to experimental QKD demonstrations at the time the work outlined here was originally undertaken [74] and it is only very recently that such imprecisions were given explicit consideration in an experimental project [57].

The main purpose of this chapter is to illustrate the negative impact of state and measurement alignment imprecisions on the performance of a QKD protocol, specifically in the case of the BB84 protocol with one-way post-processing. Instead of deriving a new security proof, the aim is rather to

¹In addition, in the ideal formulation, the bases on Alice’s and Bob’s sides are usually taken to be perfectly aligned, i.e., $|\alpha\rangle = |A\rangle$, $|\alpha'\rangle = |A'\rangle$, $|\beta\rangle = |B\rangle$, and $|\beta'\rangle = |B'\rangle$. This is a non issue from the point of view of security analysis, however, as any relative misalignment between Alice’s and Bob’s bases can be absorbed into the adversary’s unitary attack.

demonstrate an explicit advantage gained by an eavesdropper. For this purpose, the analysis is restricted to individual attacks for which necessary and sufficient conditions for security are known, for which we optimise over all possible unitary individual attacks in the presence of imperfections.

Note that this approach contrasts with existing security proofs which account for device imprecisions, such as those mentioned in section 1.3.2. Such security proofs derive lower bounds on the secret key rate, i.e., they bound the security “from below” (ruling out possible successful attacks by an eavesdropper below a certain threshold) and is not a priori clear whether and to what extent any reduction in the key rate is genuine and not an artefact of a suboptimal security proof or overly demanding security definition. Since the key rates given in this chapter were obtained by minimising over an adversary’s possible individual attacks, i.e., security was bounded “from above”, they represent an *upper* bound on security in that nonideal BB84 implementations of the type considered here are proved *insecure* beyond a certain threshold. Any reduction in the key rate below known security bounds proved for an ideal BB84 implementation thus indicates a genuine advantage that could be exploited by an eavesdropper.

For simplicity, the analysis is restricted to the case where the states emitted by Alice still form two orthonormal bases as in (2.1). (Any deviation from (2.1) can only reinforce the effects of imperfections that will be illustrate here.) We suppose, however, that Alice’s preparation and Bob’s measurement bases are not exactly mutually unbiased, but that they differ by angles φ_A and φ_B , respectively, different from $\pi/2$. That is, we suppose instead of (2.2) and (2.3) that

$$|\beta\rangle = \cos\left(\frac{\varphi_A}{2}\right)|\alpha\rangle + \sin\left(\frac{\varphi_A}{2}\right)|\alpha'\rangle, \quad (2.4a)$$

$$|\beta'\rangle = \cos\left(\frac{\varphi_A}{2}\right)|\alpha'\rangle - \sin\left(\frac{\varphi_A}{2}\right)|\alpha\rangle, \quad (2.4b)$$

and

$$|B\rangle = \cos\left(\frac{\varphi_B}{2}\right)|\alpha\rangle + \sin\left(\frac{\varphi_B}{2}\right)|\alpha'\rangle, \quad (2.5a)$$

$$|B'\rangle = \cos\left(\frac{\varphi_B}{2}\right)|\alpha'\rangle - \sin\left(\frac{\varphi_B}{2}\right)|\alpha\rangle. \quad (2.5b)$$

It is clear that such errors will in general reduce the security of BB84. For example, in the extreme case where the two bases accidentally coincide ($\varphi_A, \varphi_B = 0$), an eavesdropper could perfectly clone the states sent by Alice without revealing her presence. In this chapter, we will demonstrate more generally a reduction in the extractable secret key rate of the BB84 protocol against individual attacks, for a given error rate, when $\varphi_A, \varphi_B \neq \pi/2$.

Though the reduction in the key rate that we observe is small for deviations from the ideal situation expected in realistic implementations, the results,

given in section 2.2, nevertheless show that Alice and Bob can erroneously conclude that they have established a secure key if the inevitable experimental errors in the alignment of the bases are not taken into account. Technical details concerning the derivation of the results presented in section 2.2 are deferred to section 2.3.

2.2 Results

2.2.1 Problem definition

As explained in section 1.4.4, in the case of one-way communication from Alice to Bob, the asymptotic key rate secure against individual attacks is given by the Csiszár-Körner bound, which we express as

$$r = H(K_A | K_E) - H(K_A | K_B) \quad (2.6)$$

in terms of the conditional Shannon entropies $H(K_A | K_E)$ and $H(K_A | K_B)$ between Alice and Eve and between Alice and Bob, respectively. We will mainly consider the original version of the BB84 protocol in which Alice and Bob both use the z and x bases equiprobably and the sifted results from both bases are used to generate the final key. In order to accommodate this in (2.6) we formally treat the choice of basis (which is public knowledge) as part of the key, i.e. $K_A = (A, U)$ and $K_B = (B, V)$. In this case, for instance for the conditional entropy between Alice and Bob,

$$\begin{aligned} H(K_A | K_B) &= H(AU | BV) \\ &= H(ABUV) - H(BV) \\ &= H(A | B; UV) + H(BUV) - H(BV). \end{aligned} \quad (2.7)$$

Since we are considering only the results where Alice and Bob chose the same basis, for our purposes $V = U$, in which case (2.7) simplifies to

$$\begin{aligned} H(K_A | K_B) &= H(A | B; U) \\ &= \sum_u p_U(u) H(A | B; U = u) \\ &= p_z H(Z_A | Z_B) + p_x H(X_A | X_B), \end{aligned} \quad (2.8)$$

where p_z and p_x ($p_z + p_x = 1$) are the probabilities that Alice and Bob use the z and x bases, respectively. A similar result holds for $H(K_A | K_E)$. The result, as one might expect, is that the key rate is simply the average of the Csiszár-Körner rates in each basis:

$$r = (r_z + r_x)/2, \quad (2.9)$$

where

$$r_z = H(Z_A | Z_E) - H(Z_A | Z_B), \quad (2.10)$$

$$r_x = H(X_A | X_E) - H(X_A | X_B), \quad (2.11)$$

and we inserted explicitly $p_z = p_x = 1/2$.

Recall that, in the case of individual attacks, Eve performs the same unitary attack and measures her intercepted part of each state individually. Following Fuchs *et al.* [14], we also allow Eve to possess a quantum memory and to delay her measurements on the states in her possession until after the bases are revealed (but not until after the postprocessing). In this case, as was mentioned in 1.4.4, the asymptotic key rate under conditions (2.1), (2.2), (2.3) is given in terms of δ by

$$r = h\left(\frac{1}{2} + \sqrt{\delta(1-\delta)}\right) - h(\delta), \quad (2.12)$$

where $h = -x \log(x) - (1-x) \log(1-x)$ is the binary entropy function expressed in bits.

Our task is to minimise the expression (2.6) for a given average error rate δ using the preparation and measurement bases defined by (2.4) and (2.5) rather than the ideal ones. To simplify the analysis we will assume that the errors observed between Alice and Bob are symmetric, i.e.,

$$p_{AB}^{(z)}(0,1) = p_{AB}^{(z)}(1,0) = p_{AB}^{(x)}(0,1) = p_{AB}^{(x)}(1,0), \quad (2.13)$$

in which case the error rates in the z and x bases, δ_z and δ_x , are the same. Given our assumptions about the symmetries in the errors observed by Alice and Bob, $H(K_A | K_B)$ is a simple function of δ :

$$H(K_A | K_B) = H(Z_A | Z_B) = H(X_A | X_B) = h(\delta). \quad (2.14)$$

In general there need not be such symmetries in the joint probabilities $p_{AE}^{(u)}(a, e)$ shared between Alice and Eve, and $H(K_A | K_E)$ is accordingly more complicated. In each basis it will be convenient to parameterise these quantities in terms of an error Δ_u dependent on the basis $u \in \{z, x\}$ analogous to the error rate, and an offset v_u :

$$p_{AE}^{(u)}(0,0) = \frac{1}{2}(1 - \Delta_u - v_u), \quad (2.15a)$$

$$p_{AE}^{(u)}(0,1) = \frac{1}{2}(\Delta_u + v_u), \quad (2.15b)$$

$$p_{AE}^{(u)}(1,0) = \frac{1}{2}(\Delta_u - v_u), \quad (2.15c)$$

$$p_{AE}^{(u)}(1,1) = \frac{1}{2}(1 - \Delta_u + v_u). \quad (2.15d)$$

The inverse relations are $\Delta_u = p_{\text{AE}}^{(u)}(0, 1) + p_{\text{AE}}^{(u)}(1, 0)$ and $v_u = p_{\text{AE}}^{(u)}(0, 1) - p_{\text{AE}}^{(u)}(1, 0)$. The conditional entropy between Alice and Eve is given by

$$H(K_A | K_E) = \frac{1}{2} (H(Z_A | Z_E) + H(X_A | X_E)), \quad (2.16)$$

with $H(Z_A | Z_E)$ given by

$$\begin{aligned} H(Z_A | Z_E) = & h(\Delta_z) - h\left(\frac{1}{2} + v_z\right) \\ & + (1 - \Delta_z)h\left[\frac{1}{2}\left(1 + \frac{v_z}{1 - \Delta_z}\right)\right] + \Delta_z h\left[\frac{1}{2}\left(1 + \frac{v_z}{\Delta_z}\right)\right] \end{aligned} \quad (2.17)$$

and $H(X_A | X_E)$ given by a similar expression in terms of Δ_x and v_x .

Results for the numerical optimisation of this problem are presented in the next subsection. Details of the parameterisation and techniques employed are deferred to section 2.3.

2.2.2 Optimisation results

In numerically evaluating the key rate, it generally seemed to be the case, as one might expect, that the minimal key rate is found for a unitary interaction that gives Eve symmetric information about the bits in Alice's possession. In terms of the parameterisation introduced at the end of the previous section, this is the case where $v_z = v_x = 0$ and $\Delta_z = \Delta_x = \Delta$. The key rate is then a simple function of δ and Δ :

$$r = h(\Delta) - h(\delta). \quad (2.18)$$

Supported by a few test cases, this simplification was applied in the results that are now presented. (Note that even if Eve's optimal attack does not generally satisfy this symmetry, our results still represent an upper bound on the secure key rate, which conclusively shows that Eve can gain information by exploiting preparation and measurement imperfections with respect to the ideal case.)

Figure 2.1 is a plot of the optimised key rate as a function of δ for a few fixed values of $\varphi_A = \varphi_B = \varphi$. The values of φ used are 90° (the ideal case), 80° , and 70° . The latter two are the worst-case scenarios if there are absolute experimental errors of respectively 5° and 10° on the orientations of the bases both used by Alice and measured by Bob. That is, if Alice and Bob know, say, that their devices are accurate to within five degrees, i.e., $80^\circ \leq \varphi_A, \varphi_B \leq 90^\circ$, then the worst key rate that we have found corresponds to the situation $\varphi_A = \varphi_B = \varphi = 80^\circ$. The worst-case scenario is thus that the largest possible error on the orientation of the devices is systematic.

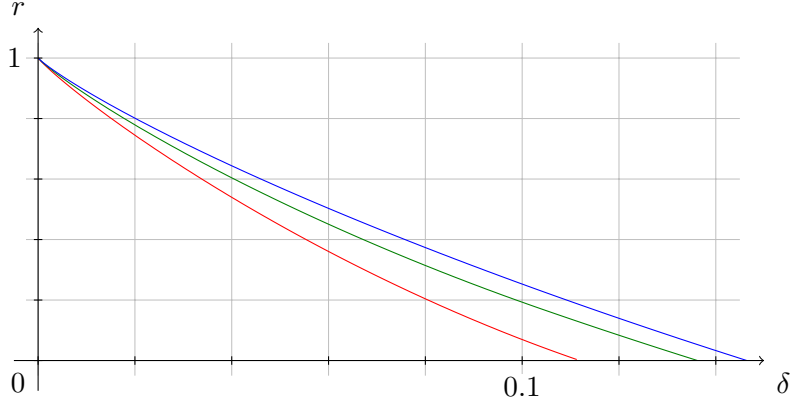


Figure 2.1: Variation of key rate (r) as a function of error rate (δ) for source and measurement-basis angles $\varphi = 90^\circ$, 80° , and 70° , corresponding to the worst-case scenarios for errors of 0° , 5° , and 10° respectively.

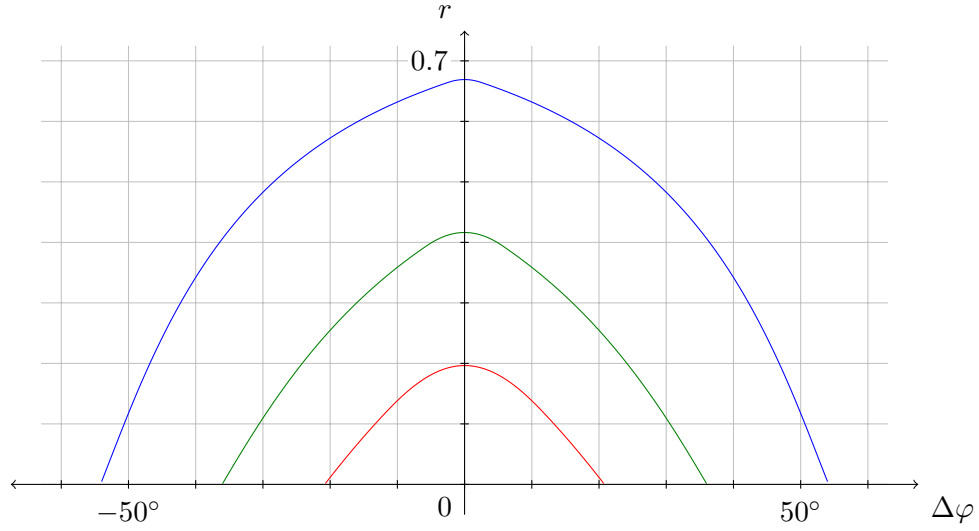


Figure 2.2: Variation of key rate (r) as a function of the deviation $\Delta\varphi = 90^\circ - \varphi$ from the ideal BB84 relations, for error rates $\delta = \frac{1}{4}\delta_0$, $\frac{1}{2}\delta_0$, and $\frac{3}{4}\delta_0$, where $\delta_0 \approx 0.1464$ is the upper secure bound on the error rate.

Figure 2.2 is a plot of the minimised key rate as a function of the deviation $\Delta\varphi = \pi/2 - \varphi$ from the ideal case, for error rates of $\frac{1}{4}\delta_0$, $\frac{1}{2}\delta_0$, and $\frac{3}{4}\delta_0$, where $\delta_0 = \frac{1}{2} - \frac{1}{4}\sqrt{2} \approx 0.1464$ is the threshold error rate for an ideal implementation.

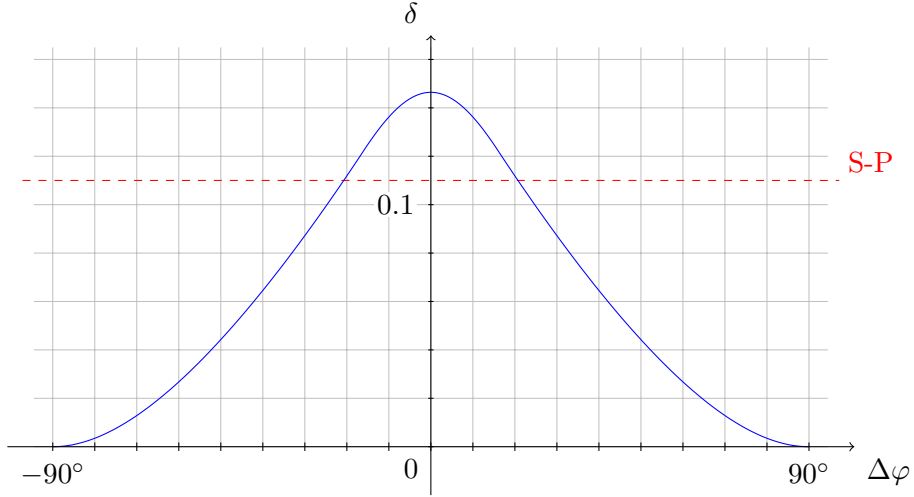


Figure 2.3: Maximum secure error rate – i.e., the error rate δ for which the key rate r becomes zero – as a function of the deviation $\Delta\varphi = 90^\circ - \varphi$ from the ideal BB84 relations. The horizontal dashed line (S-P) corresponds to the Shor-Preskill bound of about 0.11.

Finally, figure 2.3 is a plot of the upper secure bound on the error rate as a function of the deviation $\Delta\varphi = \pi/2 - \varphi$. The Shor-Preskill bound of around 0.11 [53], representing the best known threshold error rate below which an ideal BB84 implementation is known to be secure against arbitrary attacks, is added for comparison.

2.2.3 Discussion

Assuming that Alice and Bob observe errors that are symmetric, according to (2.13), using a combination of analytical and numerical techniques we have determined upper bounds on the key rate for preparation and measurement devices characterised by the misalignment angles φ_A and φ_B defined in (2.4) and (2.5). As soon as $\varphi_A, \varphi_B \neq \pi/2$, we find that these upper bounds are lower than the optimal key rate (2.12) for a given error rate δ , therefore showing that imperfections in the preparation and measurement devices can be exploited by an eavesdropper if they are not taken into account in the security proof. We also draw attention to the fact that the threshold error rate illustrated in figure 2.3 drops below the Shor-Preskill bound of about

0.11 for deviation angles larger than about 20.7° , demonstrating that the Shor-Preskill key rate is certainly insecure in this case.

The upper bounds that we have obtained correspond to the best individual attack that is symmetric, i.e., that satisfies $v_z = v_x = 0$ and $\Delta_z = \Delta_x = \Delta$. We have numerically verified in a few test cases that the best overall individual attack satisfies this symmetry condition. We thus expect our upper bounds on the key rate to actually correspond to the *optimal* key rates in the presence of imperfections of the type we consider.

If Alice and Bob know that their devices are accurate to within a given precision $\Delta\varphi$, they should assume, for the purpose of proving security, that their devices are characterised by the angles φ_A and φ_B compatible with this precision that yield the worst-case key rate. We verified in a few test cases that this happens for the smallest angles φ_A and φ_B consistent with the set error, at least in the case where the set error is the same on Alice's and Bob's devices. It is for this reason that the above figures are plotted for values of the angles satisfying $\varphi_A = \varphi_B = \varphi = \pi/2 - \Delta\varphi$.

All the results that we have presented here were obtained for the case where both bases are used to establish the secret key. One may also consider the variant of BB84 in which only one basis is used to generate the key [35]. In the ideal case, this results in a key rate that is asymptotically twice as high, as the sifting step, where half of the results are discarded, is no longer necessary. We have also adapted our analysis to this situation and have found that for high error rates the two-basis protocol results in a higher key rate than the single-basis one, suggesting that the former is more robust against alignment errors.

Finally, we remind the reader that the results are derived with the assumption that the states prepared by Alice form an orthonormal basis, i.e., satisfy (2.1). Relaxing this condition could only strengthen the effects of imperfections observed here.

2.3 Technical details

2.3.1 Eve's interaction

The model applied here is a straightforward adaptation of the one considered in [14] and summarised in section 1.4.4. In the worst-case scenario the eavesdropper (Eve) has replaced the quantum channel between Alice and

Bob with a lossless channel, before appending an ancilla to the state sent by Alice and applying a unitary operation with the intent of cloning the communication. We express the interaction as

$$|\alpha\rangle|\text{anc}\rangle \mapsto |\alpha\rangle_{\text{BE}}, \quad (2.19\text{a})$$

$$|\alpha'\rangle|\text{anc}\rangle \mapsto |\alpha'\rangle_{\text{BE}}, \quad (2.19\text{b})$$

in the z basis, and similarly

$$|\beta\rangle|\text{anc}\rangle \mapsto |\beta\rangle_{\text{BE}}, \quad (2.20\text{a})$$

$$|\beta'\rangle|\text{anc}\rangle \mapsto |\beta'\rangle_{\text{BE}}, \quad (2.20\text{b})$$

in the x basis. Unitarity and linearity of the interaction implies that the states appearing on the right-hand sides, which exist in the Hilbert space $\mathcal{H}_B \otimes \mathcal{H}_E$ of states accessible to Bob and Eve, satisfy the same relations as (2.4). Because of this, in the following we will simply identify the states shared by Bob and Eve with the corresponding source states.

In order to parameterise the interaction, we set

$$|\alpha\rangle = |A\rangle(|a\rangle + |b\rangle) + |A'\rangle(|c\rangle + |d\rangle), \quad (2.21\text{a})$$

$$|\alpha'\rangle = |A'\rangle(|a\rangle - |b\rangle) + |A\rangle(|c\rangle - |d\rangle), \quad (2.21\text{b})$$

and

$$|\beta\rangle = |B\rangle(|a'\rangle + |b'\rangle) + |B'\rangle(|c'\rangle + |d'\rangle), \quad (2.22\text{a})$$

$$|\beta'\rangle = |B'\rangle(|a'\rangle - |b'\rangle) + |B\rangle(|c'\rangle - |d'\rangle), \quad (2.22\text{b})$$

where $|a\rangle, |b\rangle, |c\rangle, |d\rangle \in \mathcal{H}_E$ are (not necessarily normalised) states accessible to Eve whose “metric” $\gamma_{ij} = \langle i|j\rangle$, $i, j \in \{a, b, c, d\}$ completely defines Eve’s interaction. Combining (2.21) and (2.22) with (2.4) and (2.5), we extract the relations

$$|a'\rangle = \cos(\bar{\varphi})|a\rangle + \sin(\bar{\varphi})|d\rangle, \quad (2.23\text{a})$$

$$|d'\rangle = \cos(\bar{\varphi})|d\rangle - \sin(\bar{\varphi})|a\rangle, \quad (2.23\text{b})$$

and

$$|b'\rangle = \cos(\varphi)|b\rangle + \sin(\varphi)|c\rangle, \quad (2.24\text{a})$$

$$|c'\rangle = \cos(\varphi)|c\rangle - \sin(\varphi)|b\rangle, \quad (2.24\text{b})$$

where we have set

$$\bar{\varphi} = \frac{\varphi_B - \varphi_A}{2}, \quad (2.25\text{a})$$

$$\varphi = \frac{\varphi_B + \varphi_A}{2}. \quad (2.25\text{b})$$

The problem now is to identify the metric γ_{ij} which will maximise the information Eve is able to gain about Alice's raw key. Note that this information also depends on the measurements Eve performs on her part of the states she shares with Bob. In general these will be positive operator-valued measures (POVMs) which are allowed to depend on the basis (since we allow Eve to possess a quantum memory). We call the POVMs $\{F_u, F'_u\}, u \in \{z, x\}$, with $F_u + F'_u = \mathbb{1}_E$. As will be explained in the next subsection, we will be able to eliminate the explicit appearance of the POVM elements in our optimisation problem.

2.3.2 Eve's error rate

As stated in the introduction to this section, we wish to minimise the extractable secret key rate, which involves maximising the mutual information $H(K_A | K_E)$. As a stepping stone to optimising this quantity we will consider the average error rate in Eve's inference of Alice's bits, Δ , first introduced in section 2.1, in (2.15). Working in a single basis u for now, this quantity is given by

$$\Delta_u = p_{AE}^{(u)}(0, 1) + p_{AE}^{(u)}(1, 0). \quad (2.26)$$

In general the conditional entropy depends on both this error Δ_u and the asymmetry v_u also introduced in (2.15), and is an increasing function as Δ_u approaches 1/2 for fixed v_u . Rather than attempting to *directly* optimise the mutual information in terms of Δ_u and v_u , we instead turn our attention to the combination

$$\Delta_u(\varepsilon_u) = (1 + \varepsilon_u)p_{AE}^{(u)}(0, 1) + (1 - \varepsilon_u)p_{AE}^{(u)}(1, 0). \quad (2.27)$$

In terms of Δ_u and v_u this is

$$\Delta_u(\varepsilon) = \Delta_u + \varepsilon_u v_u. \quad (2.28)$$

Optimising this quantity yields a v_u , dependent on the weighting parameter ε_u , and an optimal Δ_u given v_u . By varying ε_u one may hope to sweep the range of values of v_u and obtain a profile of minimised Δ_u as a function of ε_u . The motivation for this approach becomes apparent when we express $\Delta_u(\varepsilon_u)$ in terms of Eve's unitary attack and POVM elements.

In terms of Eve's interaction and measurement,

$$p_{AE}^{(z)}(0, 1) = \frac{1}{2} \text{Tr}[\rho_E F'_z], \quad p_{AE}^{(z)}(1, 0) = \frac{1}{2} \text{Tr}[\rho'_E F_z], \quad (2.29a)$$

$$p_{AE}^{(x)}(0, 1) = \frac{1}{2} \text{Tr}[\sigma_E F'_x], \quad p_{AE}^{(x)}(1, 0) = \frac{1}{2} \text{Tr}[\sigma'_E F_x], \quad (2.29b)$$

where we introduce the density operators $\rho = |\alpha\rangle\langle\alpha|$, $\rho' = |\alpha'\rangle\langle\alpha'|$, $\sigma = |\beta\rangle\langle\beta|$, and $\sigma' = |\beta'\rangle\langle\beta'|$, and e.g. $\rho_E = \text{Tr}_B[\rho]$ where Tr_B is the partial trace over $\mathcal{H}_B$, and $F_u, F'_u, u \in \{z, x\}$ are Eve's POVM elements as discussed at the end of the previous subsection. Substituting into (2.27) and using that $F'_u = \mathbb{1} - F_u$, we obtain, e.g. for $u = z$,

$$\Delta_z(\varepsilon_z) = \frac{1}{2}(1 + \varepsilon_z) - \frac{1}{2} \text{Tr}[(\rho_E - \rho'_E) + \varepsilon_z(\rho_E + \rho'_E)]F_z. \quad (2.30)$$

This expression is minimised by taking for F_z a projector which selects the positive eigenvalue part of the operator in the trace (the Helström bound), as discussed in section 1.A.1. The result of optimising over Eve's measurement is

$$\Delta_z(\varepsilon_z) = \frac{1}{2} - \frac{1}{4} \|(\rho_E - \rho'_E) + \varepsilon_z(\rho_E + \rho'_E)\|_1, \quad (2.31)$$

where for an arbitrary operator A we recall that the trace norm is defined by $\|A\|_1 = \text{Tr}[\sqrt{A^\dagger A}]$. This replaces the explicit appearance of Eve's POVM with an eigenvalue problem, leaving only an optimisation over Eve's interaction. Note that this would not be possible if we instead attempted to optimise Δ_u for fixed v_u , since in that case the POVM element F_z would appear explicitly in the constraint as well as in the expression to optimise.

Continuing with the z basis as an example, we now describe how we approach the problem of maximising $\Delta_z(\varepsilon_z)$ and how we extract the corresponding values of Δ_z and v_z . In terms of the four states $|a\rangle, |b\rangle, |c\rangle$ and $|d\rangle$ introduced earlier in order to parameterise the unitary attack,

$$\frac{1}{2}(\rho_E - \rho'_E) = |a\rangle\langle b| + |b\rangle\langle a| + |c\rangle\langle d| + |d\rangle\langle c|, \quad (2.32a)$$

$$\frac{1}{2}(\rho_E + \rho'_E) = |a\rangle\langle a| + |b\rangle\langle b| + |c\rangle\langle c| + |d\rangle\langle d|. \quad (2.32b)$$

In general our problem is to extract the eigenvalues of an operator A given its decomposition

$$A = A^{ij}|i\rangle\langle j| \quad (2.33)$$

in terms of the states $|i\rangle, i \in \{a, b, c, d\}$ (where we adopt the convention of summing over repeated indices). Explicitly decomposing a vector $|u\rangle$ on the same basis as $|u\rangle = u^i|i\rangle$, the action of A on $|u\rangle$ is

$$\begin{aligned} A|u\rangle &= A^{ij}|i\rangle\langle j|u^k|k\rangle \\ &= A^{ij}\gamma_{jk}u^k|i\rangle. \end{aligned} \quad (2.34)$$

It is not difficult to see that determining the eigenvalues and eigenstates of A is equivalent to determining the eigenvalues and eigenvectors of the matrix $\mathbf{A}\mathbf{\Gamma}$, where $\mathbf{A} = (A_{ij})$ and $\mathbf{\Gamma} = (\gamma_{ij})$. (This remains true even in the case where the vectors $\{|i\rangle\}$ are not linearly independent.) The matrix whose

eigenvalues we wish to determine may be expressed as $\mathbf{D} + \varepsilon \mathbf{\Gamma}$, where

$$\mathbf{D} = \begin{bmatrix} \gamma_{ba} & b^2 & \gamma_{bc} & \gamma_{dc} \\ a^2 & \gamma_{ab} & \gamma_{ac} & \gamma_{ad} \\ \gamma_{da} & \gamma_{db} & \gamma_{dc} & d^2 \\ \gamma_{ca} & \gamma_{cb} & c^2 & \gamma_{cd} \end{bmatrix}, \quad (2.35)$$

$$\mathbf{\Gamma} = \begin{bmatrix} a^2 & \gamma_{ab} & \gamma_{ac} & \gamma_{ad} \\ \gamma_{ba} & b^2 & \gamma_{bc} & \gamma_{dc} \\ \gamma_{ca} & \gamma_{cb} & c^2 & \gamma_{cd} \\ \gamma_{da} & \gamma_{db} & \gamma_{dc} & d^2 \end{bmatrix}, \quad (2.36)$$

and $a^2 = \gamma_{aa}$, and so on. Let the eigenvalues of this matrix be $\{\lambda_p\}$ and the corresponding (not necessarily normalised) eigenvectors be $\{\mathbf{v}_p\}$, such that

$$(\mathbf{D} + \varepsilon \mathbf{\Gamma}) \mathbf{v}_p = \lambda_p \mathbf{v}_p. \quad (2.37)$$

In terms of the set of corresponding eigenstates, the operator F_z has the expression

$$F_z = \sum_{\lambda_p > 0} \frac{|v_p\rangle \langle v_p|}{\langle v_p | v_p \rangle}, \quad (2.38)$$

where $|v_p\rangle = v_p^i |i\rangle$, $i \in \{a, b, c, d\}$ and the sum is over the indices p for which $\lambda_p > 0$. Using this and that the $|v_p\rangle$ are orthogonal, we obtain a matrix expression for the trace of an arbitrary operator A multiplied by F_z :

$$\begin{aligned} \text{Tr}[AF_z] &= \sum_{\lambda_p > 0} \frac{\langle v_p | A | v_p \rangle}{\langle v_p | v_p \rangle} \\ &= \sum_{\lambda_p > 0} \frac{\mathbf{v}_p^\dagger \mathbf{\Gamma} \mathbf{A} \mathbf{\Gamma} \mathbf{v}_p}{\mathbf{v}_p^\dagger \mathbf{\Gamma} \mathbf{v}_p}, \end{aligned} \quad (2.39)$$

The explicit expressions for Δ_z and v_z are

$$\Delta_z = \frac{1}{2} - \sum_{\lambda_p > 0} \frac{\mathbf{v}_p^\dagger \mathbf{\Gamma} \mathbf{D} \mathbf{v}_p}{\mathbf{v}_p^\dagger \mathbf{\Gamma} \mathbf{v}_p}, \quad (2.40)$$

$$v_z = \frac{1}{2} - \sum_{\lambda_p > 0} \frac{\mathbf{v}_p^\dagger \mathbf{\Gamma}^2 \mathbf{v}_p}{\mathbf{v}_p^\dagger \mathbf{\Gamma} \mathbf{v}_p}. \quad (2.41)$$

With Δ_z and v_z determined, we have an optimised value of $H(Z_A | Z_E)$ for fixed v_z , and all that remains is to optimise $H(Z_A | Z_E)$ over ε_z .

Finally, the generalisation when we consider two bases is straightforward: we will approach the optimisation of $H(K_A | K_E)$ by introducing three weighting parameters ε_z , ε_x , and ε , instead of one, optimising the quantity

$$\Delta(\varepsilon_z, \varepsilon_x, \varepsilon) = \frac{1}{2}(1 + \varepsilon)\Delta_z(\varepsilon_z) + \frac{1}{2}(1 - \varepsilon)\Delta_x(\varepsilon_x), \quad (2.42)$$

and then optimising $H(K_A, K_E)$ over $(\varepsilon_0, \varepsilon_1, \varepsilon)$.

2.3.3 Inherent error rate

All that remains now, before being able to optimise (2.42) over all of Eve's possible unitary interactions, is to determine the full set of constraints on the metric γ_{ij} , since not all metrics will represent a unitary interaction, and to determine the relationship between the metrics γ_{ij} and γ'_{ij} in the two bases (which depends only on the angles φ and $\bar{\varphi}$). This is done in the next subsection. Before this, we demonstrate that there is a minimum nonzero error rate if $\varphi_A \neq \varphi_B$ (in which case Alice and Bob's bases cannot be perfectly aligned). This is easily verified by expressing the error rate δ in terms of an orthogonal basis $\{|\gamma\rangle, |\gamma'\rangle\}$ intermediate between $\{|\alpha\rangle, |\alpha'\rangle\}$ and $\{|\beta\rangle, |\beta'\rangle\}$, and a basis $\{|C\rangle, |C'\rangle\}$ midway between $\{|A\rangle, |A'\rangle\}$ and $\{|B\rangle, |B'\rangle\}$. Specifically,

$$|\gamma\rangle = \cos\left(\frac{\varphi_A}{4}\right)|\alpha\rangle + \sin\left(\frac{\varphi_A}{4}\right)|\alpha'\rangle, \quad (2.43a)$$

$$|\gamma'\rangle = \cos\left(\frac{\varphi_A}{4}\right)|\alpha'\rangle - \sin\left(\frac{\varphi_A}{4}\right)|\alpha\rangle, \quad (2.43b)$$

and

$$|C\rangle = \cos\left(\frac{\varphi_B}{4}\right)|A\rangle + \sin\left(\frac{\varphi_B}{4}\right)|A'\rangle, \quad (2.44a)$$

$$|C'\rangle = \cos\left(\frac{\varphi_B}{4}\right)|A'\rangle - \sin\left(\frac{\varphi_B}{4}\right)|A\rangle. \quad (2.44b)$$

Setting

$$\Sigma_z = |\gamma\rangle\langle\gamma| - |\gamma'\rangle\langle\gamma'|, \quad (2.45a)$$

$$\Sigma_x = |\gamma\rangle\langle\gamma'| + |\gamma'\rangle\langle\gamma|, \quad (2.45b)$$

and

$$\sigma_z = |C\rangle\langle C| - |C'\rangle\langle C'|, \quad (2.46a)$$

$$\sigma_x = |C\rangle\langle C'| + |C'\rangle\langle C|, \quad (2.46b)$$

then with this choice of basis the expression we find for the error rate is

$$\begin{aligned} \delta &= \frac{1}{2} - \frac{1}{4} \cos\left(\frac{\varphi_A}{2}\right) \cos\left(\frac{\varphi_B}{2}\right) \text{Tr}[\Sigma_z(\sigma_z \otimes \mathbb{1}_E)] \\ &\quad - \frac{1}{4} \sin\left(\frac{\varphi_A}{2}\right) \sin\left(\frac{\varphi_B}{2}\right) \text{Tr}[\Sigma_x(\sigma_x \otimes \mathbb{1}_E)]. \end{aligned} \quad (2.47)$$

Clearly, $-2 \leq \text{Tr}[\Sigma_z(\sigma_z \otimes \mathbb{1}_E)] \leq 2$ and $-2 \leq \text{Tr}[\Sigma_x(\sigma_x \otimes \mathbb{1}_E)] \leq 2$, and we find the bound

$$\delta \geq \frac{1}{2} - \frac{1}{2} \max\{|\cos(\bar{\varphi})|, |\cos(\varphi)|\}, \quad (2.48)$$

with $\bar{\varphi}$ and φ defined as in (2.25) (this bound is also saturated, e.g. if Eve does not interfere with the channel, in which case $\Sigma_{z,x} = \sigma_{z,x}$). The corresponding upper bound is

$$\delta \leq \frac{1}{2} + \frac{1}{2} \max\{|\cos(\bar{\varphi})|, |\cos(\varphi)|\}. \quad (2.49)$$

2.3.4 Transformation and constraints

We now determine the full set of constraints on the metric elements γ_{ij} . First, we impose that the error rate is fixed at δ . This, combined with $\langle \alpha | \alpha \rangle = \langle \alpha' | \alpha' \rangle = 1$, imposes

$$a^2 + b^2 = 1 - \delta, \quad (2.50a)$$

$$c^2 + d^2 = \delta, \quad (2.50b)$$

and $\text{Re}[\gamma_{ab}] = \text{Re}[\gamma_{cd}] = 0$ for the z basis, with analogous constraints for the x basis. The components γ_{ab} , γ_{ac} , γ_{bd} , and γ_{cd} transform between the two bases according to

$$\begin{aligned} \gamma'_{ab} = & \cos(\bar{\varphi}) \cos(\varphi) \gamma_{ab} + \cos(\bar{\varphi}) \sin(\varphi) \gamma_{ac} \\ & + \sin(\bar{\varphi}) \cos(\varphi) \gamma_{db} + \sin(\bar{\varphi}) \cos(\varphi) \gamma_{dc}, \end{aligned} \quad (2.51a)$$

$$\begin{aligned} \gamma'_{ac} = & \cos(\bar{\varphi}) \cos(\varphi) \gamma_{ac} - \cos(\bar{\varphi}) \sin(\varphi) \gamma_{ab} \\ & + \sin(\bar{\varphi}) \cos(\varphi) \gamma_{dc} - \sin(\bar{\varphi}) \sin(\varphi) \gamma_{db}, \end{aligned} \quad (2.51b)$$

$$\begin{aligned} \gamma'_{db} = & \cos(\bar{\varphi}) \cos(\varphi) \gamma_{db} + \cos(\bar{\varphi}) \sin(\varphi) \gamma_{dc} \\ & - \sin(\bar{\varphi}) \cos(\varphi) \gamma_{ab} - \sin(\bar{\varphi}) \sin(\varphi) \gamma_{ac}, \end{aligned} \quad (2.51c)$$

$$\begin{aligned} \gamma'_{dc} = & \cos(\bar{\varphi}) \cos(\varphi) \gamma_{dc} - \cos(\bar{\varphi}) \sin(\varphi) \gamma_{db} \\ & - \sin(\bar{\varphi}) \cos(\varphi) \gamma_{ac} + \sin(\bar{\varphi}) \sin(\varphi) \gamma_{ab}. \end{aligned} \quad (2.51d)$$

The transformation from $[\gamma_{ab}, \gamma_{ac}, \gamma_{db}, \gamma_{dc}]^T$ to $[\gamma'_{ab}, \gamma'_{ac}, \gamma'_{db}, \gamma'_{dc}]^T$ can be expressed as

$$\begin{bmatrix} \cos(\bar{\varphi}) & \sin(\bar{\varphi}) \\ -\sin(\bar{\varphi}) & \cos(\bar{\varphi}) \end{bmatrix} \otimes \begin{bmatrix} \cos(\varphi) & \sin(\varphi) \\ -\sin(\varphi) & \cos(\varphi) \end{bmatrix}. \quad (2.52)$$

(2.51a) and (2.51d) together with the constraint $\text{Re}[\gamma_{ab}] = \text{Re}[\gamma_{cd}] = 0$ imply $\text{Re}[\gamma_{ac}] = \text{Re}[\gamma_{bd}] = 0$.

For a' and d' , we find

$$a'^2 = \cos(\bar{\varphi})^2 a^2 + \sin(\bar{\varphi})^2 d^2 + \sin(2\bar{\varphi}) \text{Re}[\gamma_{ad}], \quad (2.53a)$$

$$d'^2 = \cos(\bar{\varphi})^2 d^2 + \sin(\bar{\varphi})^2 a^2 - \sin(2\bar{\varphi}) \text{Re}[\gamma_{ad}], \quad (2.53b)$$

from which we immediately see that $a'^2 + d'^2 = a^2 + d^2$. From (2.53), and taking the real and imaginary parts of

$$\gamma'_{ad} = -\frac{1}{2} \sin(2\bar{\varphi})(a^2 - d^2) + \cos(\bar{\varphi})^2 \gamma_{ad} - \sin(\bar{\varphi})^2 \gamma_{da}, \quad (2.54)$$

we find

$$\delta'_{ad} = \cos(2\bar{\varphi}) \delta_{ad} + \sin(2\bar{\varphi}) \text{Re}[\gamma_{ad}], \quad (2.55a)$$

$$\text{Re}[\gamma'_{ad}] = \cos(2\bar{\varphi}) \text{Re}[\gamma_{ad}] - \sin(2\bar{\varphi}) \delta_{ad}, \quad (2.55b)$$

$$\text{Im}[\gamma'_{ad}] = \text{Im}[\gamma_{ad}], \quad (2.55c)$$

where $\delta_{ad} = \frac{a^2 - d^2}{2}$. Similarly, $b'^2 + c'^2 = b^2 + c^2$ and

$$\delta'_{bc} = \cos(2\varphi)\delta_{bc} + \sin(2\varphi) \operatorname{Re}[\gamma_{bc}], \quad (2.56a)$$

$$\operatorname{Re}[\gamma'_{bc}] = \cos(2\varphi) \operatorname{Re}[\gamma_{bc}] - \sin(2\varphi)\delta_{bc}, \quad (2.56b)$$

$$\operatorname{Im}[\gamma'_{bc}] = \operatorname{Im}[\gamma_{bc}], \quad (2.56c)$$

with $\delta_{bc} = \frac{b^2 - c^2}{2}$. Orthogonality of $|\alpha\rangle$ and $|\alpha'\rangle$ implies $\operatorname{Im}[\gamma_{bc}] = \operatorname{Im}[\gamma_{ad}]$.

We still require $a'^2 \leq 1 - \delta$ and $d'^2 \leq \delta$ individually, which impose

$$\cos(\bar{\varphi})^2 a^2 + \sin(\bar{\varphi})^2 d^2 + \sin(2\bar{\varphi}) \operatorname{Re}[\gamma_{ad}] \leq 1 - \delta, \quad (2.57a)$$

$$\cos(\bar{\varphi})^2 d^2 + \sin(\bar{\varphi})^2 a^2 - \sin(2\bar{\varphi}) \operatorname{Re}[\gamma_{ad}] \leq \delta. \quad (2.57b)$$

Equation (2.57a) is automatically satisfied, in the sense that there are no new restrictions on a^2 , d^2 , or $\operatorname{Re}[\gamma_{ad}]$, if $\delta \geq \frac{1}{2} - \frac{1}{2}|\cos(\bar{\varphi})|$. Equation (2.57b) is automatically satisfied if $\delta \leq \frac{1}{2} + \frac{1}{2}|\cos(\bar{\varphi})|$. Similarly, we automatically have $b'^2 \leq 1 - \delta$ and $c'^2 \leq \delta$ as long as $\frac{1}{2} - \frac{1}{2}|\cos(\varphi)| \leq \delta \leq \frac{1}{2} + \frac{1}{2}|\cos(\varphi)|$.

Finally, using $a'^2 + b'^2 = a^2 + b^2$ and $c'^2 + d'^2 = c^2 + d^2$, we obtain the constraint

$$\begin{aligned} & \sin(2\bar{\varphi}) \operatorname{Re}[\gamma_{ad}] + \sin(2\varphi) \operatorname{Re}[\gamma_{bc}] \\ &= \sin(\bar{\varphi})^2(a^2 - d^2) + \sin(\varphi)^2(b^2 - c^2). \end{aligned} \quad (2.58)$$

2.3.5 Optimisation

The plots given in figures 2.1 and 2.2 were generated by numerically maximising $\Delta = \Delta(\varepsilon_z = \varepsilon_x = \varepsilon = 0)$, defined by equation (2.42), using MATLAB's `fmincon` routine, over all metrics γ_{ij} respecting the constraints derived in the preceding subsection for the reported angles φ and values of δ and with $\bar{\varphi} = 0$, and calculating the corresponding value of $H(K_A | K_E)$. For simplicity, no systematic optimisation was performed over $(\varepsilon_z, \varepsilon_x, \varepsilon)$. Optimising over $(\varepsilon_z, \varepsilon_x, \varepsilon)$ in a few test cases generally supported the expectation that the minimal key rate would be obtained for the maximal value of Δ with a symmetric attack ($v_z = v_x = 0$ and $\Delta_z = \Delta_x$). Similarly, investigating test cases generally found that the minimal key rate, given a common error bound on the deviation of φ_A and φ_B from 90° , was obtained by setting both to the worst case such that $\varphi_A = \varphi_B = \varphi$ and $\bar{\varphi} = 0$. As a result, the key rates given in section 2.2.2 are an upper bound on the secure key rate (which is sufficient to demonstrate a degradation in performance) which we believe are very likely the optimal key rates.

The maximum tolerable error rates reported in figure 2.3 are those for which $\Delta = \delta$ for the angles φ considered, again with $\bar{\varphi} = 0$.

In addition to the key rates reported in section 2.2.2, we also similarly investigated the case in which only one basis is used to generate the key, by maximising only Δ_z . In this case, the resulting key rates (not accounting for sifting) are lower than those obtained for the case in which both bases are used, for the same parameters. This suggests that implementations of BB84 in which both bases are used to generate the key are likely to be more robust against implementation errors, as we alluded to in section 2.2.3.

2.A Partial analytic solution

It is possible to go some way toward deriving an analytical result for the optimal key rate in the case where $\varphi_A = \varphi_B = \varphi$. The steps described here were not employed in the numerical optimisation whose results are presented in section 2.2. The derivation is incomplete and only given here in case it later become useful.

Numerical optimisation suggests that $H(K_A | K_E)$ is minimised for the global minimum of Δ , which is found by setting $\varepsilon = 0$, and corresponds to $v = 0$. In this case the optimum satisfies $\gamma_{ab} = \gamma_{ac} = \gamma_{db} = \gamma_{dc} = 0$. This simplifies the problem sufficiently that an analytical expression for the eigenvalues of the matrix D may be obtained. With these simplifications,

$$D = \begin{bmatrix} 0 & b^2 & \gamma_{bc} & 0 \\ a^2 & 0 & 0 & \gamma_{ad} \\ \gamma_{da} & 0 & 0 & d^2 \\ 0 & \gamma_{cb} & c^2 & 0 \end{bmatrix}. \quad (2.59)$$

The characteristic polynomial associated with this matrix is

$$P(\lambda) = \lambda^4 - (a^2b^2 + c^2d^2 + 2\operatorname{Re}[\gamma_{ad}\gamma_{bc}^*])\lambda^2 + (a^2d^2 - |\gamma_{ad}|^2)(b^2c^2 - |\gamma_{bc}|^2). \quad (2.60)$$

The positive eigenvalues are

$$\lambda_{\pm}^2 = \frac{x \pm \sqrt{x^2 - 4y}}{2} \quad (2.61)$$

where

$$x = a^2b^2 + c^2d^2 + 2\operatorname{Re}[\gamma_{ad}\gamma_{bc}^*] \quad (2.62)$$

$$y = (a^2d^2 - |\gamma_{ad}|^2)(b^2c^2 - |\gamma_{bc}|^2). \quad (2.63)$$

The quantity we wish to optimise is the sum of the positive eigenvalues, which simplifies to

$$\lambda_+ + \lambda_- = [x + 2y^{1/2}]^{1/2}, \quad (2.64)$$

or, reintroducing the metric components,

$$\begin{aligned} (\lambda_+ + \lambda_-)^2 &= a^2b^2 + c^2d^2 + 2\operatorname{Re}[\gamma_{ad}\gamma_{bc}^*] \\ &\quad + 2\sqrt{(a^2d^2 - |\gamma_{ad}|^2)(b^2c^2 - |\gamma_{bc}|^2)}. \end{aligned} \quad (2.65)$$

When we switch basis, the only quantity to change is $\gamma_{bc} \mapsto -\gamma_{bc}^*$. Setting

$$X = a^2b^2 + c^2d^2 + 2\operatorname{Im}[\gamma_{ad}]\operatorname{Im}[\gamma_{bc}] + 2\sqrt{y}, \quad (2.66)$$

$$Y = 2\operatorname{Re}[\gamma_{ad}]\operatorname{Re}[\gamma_{bc}], \quad (2.67)$$

the average of the sum of the positive eigenvalues squared is

$$\frac{1}{4}(\sqrt{X+Y} + \sqrt{X-Y})^2 = \frac{X}{2} + \frac{\sqrt{X^2 - Y^2}}{2}, \quad (2.68)$$

which is clearly maximised for $\operatorname{Re}[\gamma_{ad}] = 0$ such that $Y = 0$, simplifying to just

$$\begin{aligned} X &= a^2b^2 + c^2d^2 + 2\operatorname{Im}[\gamma_{ad}]^2 \\ &\quad + 2\sqrt{(a^2d^2 - \operatorname{Im}[\gamma_{ad}]^2)(b^2c^2 - \operatorname{Re}[\gamma_{bc}]^2 - \operatorname{Im}[\gamma_{ad}]^2)}. \end{aligned} \quad (2.69)$$

since $\operatorname{Im}[\gamma_{ad}] = \operatorname{Im}[\gamma_{bc}]$. The derivative of this quantity is zero either where $\operatorname{Im}[\gamma_{ad}] = 0$ or $a^2d^2 = b^2c^2 - \operatorname{Re}[\gamma_{bc}]^2$ (in which case X is independent of $\operatorname{Im}[\gamma_{ad}]$). Setting $A^2 = a^2b^2$ and $B^2 = b^2c^2 - \operatorname{Re}[\gamma_{bc}]^2$, $\operatorname{Im}[\gamma_{ad}]$ is the global maximum wherever

$$\frac{A}{B} + \frac{B}{A} \geq 2, \quad (2.70)$$

which is always the case. Equation (2.69) thus simplifies to

$$X = a^2b^2 + c^2d^2 + 2ad\sqrt{b^2c^2 - \operatorname{Re}[\gamma_{bc}]^2}. \quad (2.71)$$

In terms of this parameter, Eve's error is $\Delta = \frac{1}{2} - \sqrt{X}$, and since the optimum always seems to be symmetric, the key rate is simply

$$r = h(\Delta) - h(\delta). \quad (2.72)$$

In the case we are considering, where $\bar{\varphi} = 0$, (2.58) reduces to an equality involving only $\operatorname{Re}[\gamma_{bc}]$, b , and c ,

$$\operatorname{Re}[\gamma_{bc}] = \frac{1}{2}\tan(\varphi)(b^2 - c^2) = \frac{1}{2}\tan(\varphi)\left(\frac{b}{c} - \frac{c}{b}\right)bc, \quad (2.73)$$

which can be substituted into (2.71). Combining this with the constraint $-bc \leq \text{Re}[\gamma_{bc}] \leq bc$, we obtain a constraint between b and c which rearranges to

$$|\csc(\varphi)| - |\cot(\varphi)| \leq \frac{b}{c} \leq |\csc(\varphi)| + |\cot(\varphi)|. \quad (2.74)$$

For at least some angles φ and error rates δ , an optimal or near-optimal attack seems to be obtained by saturating the constraint above, i.e. taking $\frac{b}{c} = |\csc(\varphi)| + |\cot(\varphi)|$, and taking the maximal value $c^2 = \delta$ for c . In this case, $d^2 = 0$, $b^2 = \lambda c^2$, $a^2 = 1 - \delta - \lambda_\varphi c^2$, and X simplifies to

$$X = a^2 b^2 = (1 - (1 + \lambda_\varphi)\delta) \lambda_\varphi \delta, \quad (2.75)$$

where we set

$$\lambda_\varphi = (|\csc(\varphi)| + |\cot(\varphi)|)^2 = \frac{1 + |\cos(\varphi)|}{1 - |\cos(\varphi)|}. \quad (2.76)$$

Eve's error rate is then

$$\Delta = \frac{1}{2} - \sqrt{\lambda_\varphi \delta (1 - (1 + \lambda_\varphi)\delta)}. \quad (2.77)$$

The corresponding key rate is

$$r = h\left(\frac{1}{2} + \sqrt{\lambda_\varphi \delta (1 - (1 + \lambda_\varphi)\delta)}\right) - h(\delta), \quad (2.78)$$

which seems to closely recover the numerically obtained rates well for angles φ less than about 80° . Solving for $\Delta = \delta$, the threshold error for this family of attacks is found by

$$\delta_0 = \frac{1 - |\cos(\varphi)|}{4 + 2|\sin(\varphi)|}. \quad (2.79)$$

Figures 2.4, 2.5, and 2.6 give an overview of how the analytic rate (2.78) compares to the rates that can be obtained numerically following the approach in this chapter. Figure 2.4 gives a side-by-side comparison between the analytic and numerically-derived key rate for a source and measurement-basis angle $\varphi = \alpha = \beta$ of 85° . The two rates match well for error rates below about $\delta \approx 0.07$. Figure 2.5 is a plot of the difference between the analytic and numerically-derived key rates for angles $\varphi = 90^\circ$, 85° , and 80° , illustrating how the two key rates begin to coincide for smaller angles. Finally, figure 2.6 gives a side-by-side comparison between analytic threshold error rate (2.79) and the numerically-derived threshold error rate already given in figure 2.3 for deviations $\Delta\varphi = 90^\circ - \varphi$ from the ideal angle of 90° . The curves essentially coincide for deviations $\Delta\varphi$ greater than about 18° , corresponding to $\varphi \lesssim 72^\circ$.

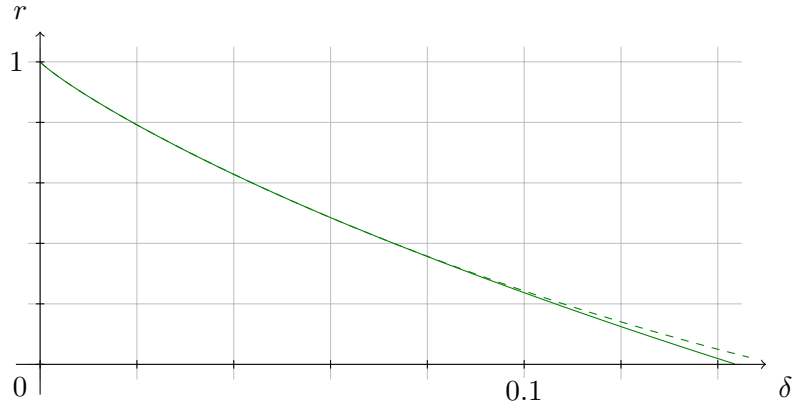


Figure 2.4: Numerically-derived (solid curve) and analytic (dashed curve) key rate (r) for a source and basis angle of $\varphi = 85^\circ$ as a function of the error rate δ .

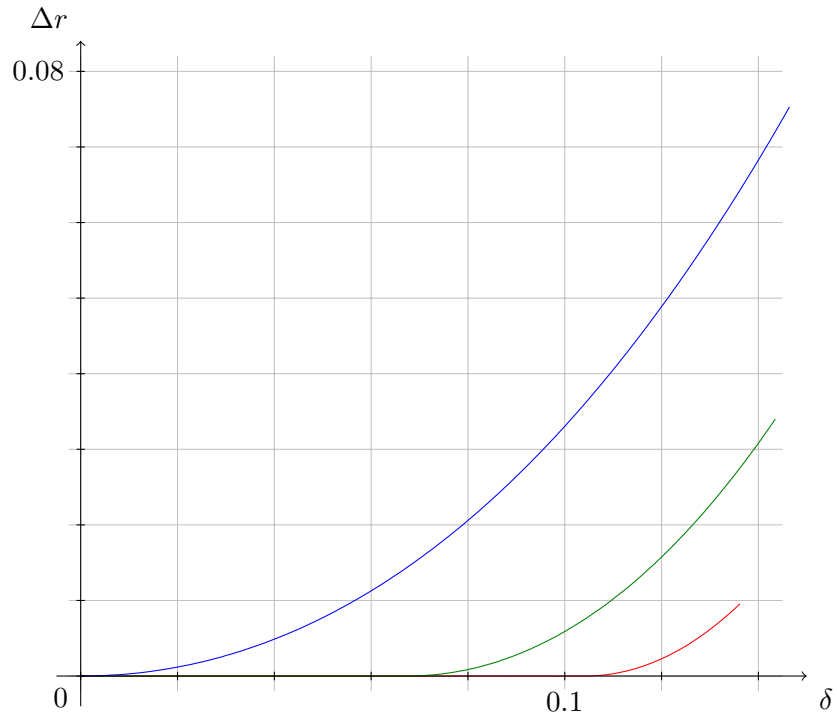


Figure 2.5: Difference (Δr) between analytic and numerically-derived key rates for source and measurement-basis angles of $\varphi = 90^\circ$, $\varphi = 85^\circ$, and $\varphi = 80^\circ$ as a function of error rate (δ).

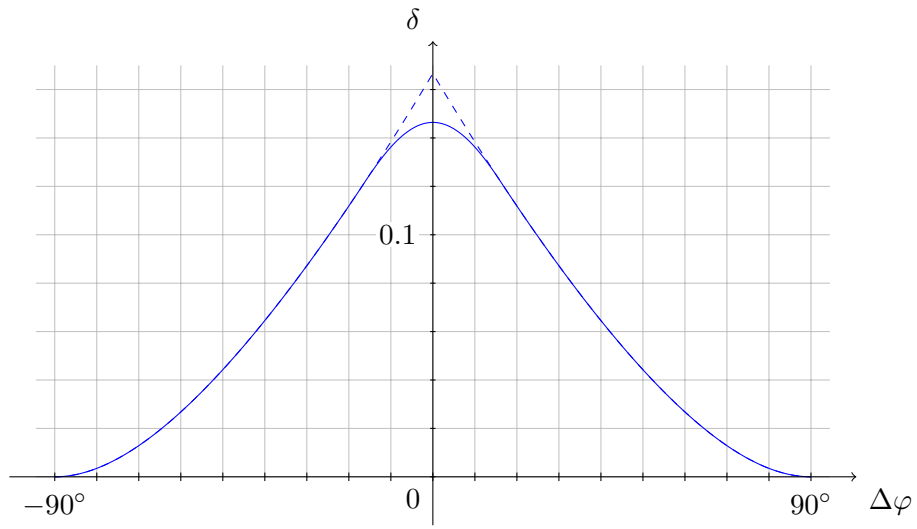


Figure 2.6: Threshold error rate (δ) obtained numerically (solid curve) and for the analytic key rate for deviations $\Delta\varphi = 90^\circ - \varphi$ from the ideal case.

Chapter 3

Security from cloning bounds

This chapter introduces techniques which can be applied to relatively straightforward security proofs of the BB84 protocol against collective attacks from the prepare-and-measure perspective. The approach can account for imprecisions in both the source states and Bob's measurements and can incorporate the local randomisation preprocessing procedure proposed in [21]. The main result is a key rate for a BB84 implementation in which the source emits four arbitrary pure states (which need not necessarily be qubits) and Bob's detector is left uncharacterised. The key rate is shown to be a tight bound on the Devetak-Winter rate with the source characterisation used. With the local randomisation omitted, the key rate closely resembles the Marøy *et al.* key rate [20].

In addition, more specific secondary results are derived for arbitrary qubit source states using an alternative source characterisation that will be more convenient for the semi-device-independent protocols studied in the next chapter. Specifically, a key rate is derived for a BB84 implementation emitting arbitrary qubit states with the detector left uncharacterised on Bob's side, and an improved key rate is derived in the case where both the source and detector are assumed two dimensional.

The exposition and results in this chapter are based on Refs. [22, 23].

3.1 Introduction

3.1.1 Outline

The previous chapter established the necessity of accounting for device imprecisions in QKD security proofs. This chapter introduces an approach that can account for such imprecisions in asymptotic security proofs of the prepare-and-measure BB84 protocol against collective attacks. In particular, the source states do not have to satisfy the basis-independence condition explained in section 1.2.3.

Most security proofs of the BB84 protocol, and especially those addressing collective or general attacks, have been developed from the entanglement-based perspective. In some cases where device imprecisions were considered, such as in [49], the result was claimed to account for source imprecisions in the prepare-and-measure scenario in full generality without noting the basis-independence limitation. Even the security proofs of the prepare-and-measure BB84 protocol [20, 61, 62] cited in section 1.3.2 that explicitly relax the basis-independence condition first recast the BB84 protocol into an entanglement-based form as a first step in the security analysis. The authors of [20, 61], for instance, considered a “virtual” entanglement-based protocol in which Alice’s bit and basis choices are both determined by the outcome of a single four-outcome measurement on an entangled “coin” state of the form

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|z\rangle|\Psi_z\rangle + |x\rangle|\Psi_x\rangle), \quad (3.1)$$

defined in terms of entangled states $|\Psi_z\rangle$ and $|\Psi_x\rangle$ of the form

$$|\Psi_z\rangle = \sqrt{p}|0\rangle_A|\alpha\rangle_{BE} + \sqrt{p'}|1\rangle_A|\alpha'\rangle_{BE}, \quad (3.2)$$

$$|\Psi_x\rangle = \sqrt{q'}|+\rangle_A|\beta\rangle_{BE} + \sqrt{q}|-\rangle_A|\beta'\rangle_{BE}. \quad (3.3)$$

In this virtual protocol, Alice’s measurement in the basis of states $|z\rangle|0\rangle$, $|z\rangle|1\rangle$, $|x\rangle|+\rangle$, and $|x\rangle|-\rangle$ determines which of the four states $|\alpha\rangle$, $|\alpha'\rangle$, $|\beta\rangle$, and $|\beta'\rangle$ is distributed to Bob (and Eve), and the security proof assumes a lower bound on the inner product $\langle\Psi_z|\Psi_x\rangle$. In this chapter, we will avoid this detour.

This chapter’s main result is a secret key rate for a BB84 implementation in which the source emits four given but arbitrary pure states and which is largely device-independent on Bob’s side, incorporating the local randomisation preprocessing procedure introduced in [21]. The key rate, both with and without local randomisation, is shown to be a tight lower bound on the Devetak-Winter rate with the particular source characterisation used. For

an ideal BB84 source, the previously known threshold error rates of 12.41% and 11%, with and without the preprocessing respectively, are recovered as special cases. The key rate without the preprocessing is similar to the key rate derived by Marøy, Lydersen, and Skaar [20], one of the few previous results that had accounted for arbitrary imprecisions in both the source and detector.

Additionally, partially in anticipation of the semi-device-independent protocols studied in the next chapter, a second key rate is derived in the more restricted case where the source is assumed to emit only arbitrary qubit states. If Bob’s detector measurements are also assumed two dimensional, the key-rate bound can be further improved to the point that, if no errors are observed, the key rate is 1.

3.1.2 Scenario and sketch of approach

Here we outline the type of scenario that we will consider and the general approach that will be followed in this chapter. We consider a BB84 implementation in which Alice possesses a source which, depending on a choice of bit and “basis”, emits one of four different quantum states which is sent to Bob. Similar to the notation in the previous chapter, we will consistently call the two “z-basis” states $|\alpha\rangle$ and $|\alpha'\rangle$, and the two “x-basis” states $|\beta\rangle$ and $|\beta'\rangle$ (note that this nomenclature is not intended to imply that these states are necessarily really the σ_z and σ_x eigenstates). We denote the corresponding density operators by $\rho = |\alpha\rangle\langle\alpha|$, $\rho = |\alpha'\rangle\langle\alpha'|$, $\sigma = |\beta\rangle\langle\beta|$, and $\sigma' = |\beta'\rangle\langle\beta'|$. Following a unitary attack by a potential eavesdropper (“Eve”), these states are contained in some Hilbert space $\mathcal{H}_A \subset \mathcal{H}_B \otimes \mathcal{H}_E$, where $\mathcal{H}_B$ and $\mathcal{H}_E$ respectively denote the Hilbert spaces accessible to Bob and Eve. Upon reception of the states emitted by Alice, Bob performs one of two binary-outcome measurements (a “z-basis” or “x-basis” measurement, though we do not assume Bob is necessarily measuring σ_z or σ_x) in order to compute his version of the raw key.

By contrast with the previous chapter, we consider the BB84 variant where only the z-basis results are used to generate the key. In this case, as explained in section 1.4.5, assuming an adversary restricted to collective attacks a lower bound on the key rate that can securely be extracted by one-way postprocessing is given by the Devetak-Winter bound. In the general approach followed in this chapter, this problem is broken into two parts.

Section 3.2 first describes how the conditional entropy can be lower bounded in various situations in terms of the fidelity $F(\rho_E, \rho'_E)$ between the parts of

the z-basis states accessible to the adversary. The simplest resulting bound on the key rate, for an unbiased source and without local randomisation, is

$$r \geq 1 - h\left(\frac{1}{2} + \frac{1}{2}F(\rho_E, \rho'_E)\right) - h(\delta_z), \quad (3.4)$$

in which the fidelity is defined by $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$, $h(x) = -x \log(x) - (1-x) \log(1-x)$ is the binary entropy, all logarithms are in base 2, and subscripts indicate partial tracing in the usual way (e.g., $\rho_E = \text{Tr}_B[\rho]$). The section also gives a generalisation of (3.4) if Alice randomly flips a fraction η of her key bits (the local randomisation preprocessing procedure) and derives a relation for the conditional min-entropy in terms of the trace distance which will be useful in the next chapter.

In this way, the problem of lower bounding the key rate is generally reduced to lower bounding the fidelity $F(\rho_E, \rho'_E)$, which is addressed in subsequent sections of this chapter. We use the fact that Alice and Bob working in cooperation can lower bound the trace distances $D(\rho_B, \rho'_B)$ and $D(\sigma_B, \sigma'_B)$ in terms of the observed bit error rates via the Helstrom bound (see section 1.A.2):

$$D(\rho_B, \rho'_B) \geq |1 - 2\delta_z|, \quad (3.5)$$

$$D(\sigma_B, \sigma'_B) \geq |1 - 2\delta_x|. \quad (3.6)$$

The lower bounds for the fidelity will be derived in terms of the trace distance $D(\sigma_B, \sigma'_B)$ (and, in section 3.5.2, $D(\rho_B, \rho'_B)$). Such bounds can be viewed as a characterisation of the degree to which quantum states can be cloned.

The bounds that will be derived in sections 3.3, 3.4, and 3.5 depend on relations between the source states $|\alpha\rangle$, $|\alpha'\rangle$, $|\beta\rangle$, and $|\beta'\rangle$ emitted by the source. In section 3.4 for instance, similar to the authors of [20, 61] we will consider the case where the four source states (which may span a four-dimensional Hilbert space in this case) are characterised by a “basis overlap” angle θ , which we define such that

$$\sqrt{1 + |\sin(\theta)|} = \frac{1}{2}|\langle\alpha|\beta\rangle + \langle\alpha'|\beta\rangle + \langle\alpha|\beta'\rangle - \langle\alpha'|\beta'\rangle| \quad (3.7)$$

wherever the right-hand side is greater than 1. The cloning bound we will obtain in this case is

$$F(\rho_E, \rho'_E) \geq f_\theta(D(\sigma_B, \sigma'_B)), \quad (3.8)$$

with the function f_θ defined by

$$f_\theta(x) = \begin{cases} |\sin(\theta)|x - |\cos(\theta)|\sqrt{1-x^2} & : x \geq |\cos(\theta)| \\ 0 & : x \leq |\cos(\theta)| \end{cases}, \quad (3.9)$$

which provides a lower bound on $F(\rho_E, \rho'_E)$ wherever the right-hand side of (3.7) is greater than 1. In this case, substituting (3.8) and $D(\sigma_B, \sigma'_B) \geq |1 - 2\delta_x|$ into (3.4), the resulting key rate is

$$r \geq 1 - h\left(\frac{1}{2} + \frac{1}{2}f_\theta(|1 - 2\delta_x|)\right) - h(\delta_z), \quad (3.10)$$

which coincides with the key rate given in [20] for this particular scenario. A generalisation of this key rate with local randomisation incorporated will also be shown to be tight. Section 3.5 will give fidelity bounds in the more restricted case of a source emitting arbitrary qubit states using a different source characterisation than (3.7).

3.2 Conditional entropy bounds

3.2.1 Asymptotic key-rate bound without preprocessing

In this subsection, we derive (3.4). The starting point is the Devetak-Winter bound

$$r \geq H(Z_A | E) - H(Z_A | Z_B). \quad (3.11)$$

As shown in section 1.4.4, the conditional Shannon entropy $H(Z_A | Z_B)$, which quantifies the key loss due to error correction, is upper bounded by $h(\delta_z)$ (with equality in the typical case of symmetric errors). The conditional von Neumann entropy $H(Z_A | E)$ quantifies the amount of key that can be extracted securely by privacy amplification, which we evaluate on the classical-quantum state

$$\tau_{ZE} = \frac{1}{2}(|0\rangle\langle 0|_Z \otimes \rho_E + |1\rangle\langle 1|_Z \otimes \rho'_E), \quad (3.12)$$

where ρ and ρ' are the z-basis states emitted by Alice, and $|0\rangle_Z$ and $|1\rangle_Z$ denote the state of a classical register in her possession.

In order to arrive at (3.4), we use that $H(Z_A | E) \geq H(Z_A | EE')$ for any extension of the state (3.12) to a larger Hilbert space $\mathcal{H}_Z \otimes \mathcal{H}_E \otimes \mathcal{H}_{E'}$ (formally this follows from the property of strong subadditivity of the von Neumann entropy [75]). In particular, we use this to replace ρ_E and ρ'_E in (3.12) with purifications $|\psi\rangle$ and $|\psi'\rangle \in \mathcal{H}_E \otimes \mathcal{H}_{E'}$. Applying the definition of the conditional von Neumann entropy, we find

$$\begin{aligned} H(Z_A | E) &\geq H(Z_A | EE') \\ &= 1 - S\left[\frac{1}{2}(|\psi\rangle\langle\psi| + |\psi'\rangle\langle\psi'|)\right] \\ &= 1 - h\left[\frac{1}{2}(1 + |\langle\psi|\psi'\rangle|)\right], \end{aligned} \quad (3.13)$$

where $S(\rho) = -\text{Tr}[\rho \log(\rho)]$. Since by Uhlmann's theorem we can choose $|\psi\rangle$ and $|\psi'\rangle$ such that $F(\rho_E, \rho'_E) = |\langle\psi|\psi'\rangle|$, we find

$$H(Z_A | E) \geq 1 - h\left(\frac{1}{2} + \frac{1}{2}F(\rho_E, \rho'_E)\right), \quad (3.14)$$

which completes the derivation of (3.4). Note that (3.14) is equivalent to an analogous bound on the Holevo quantity derived in [76].

The proof given here holds with the assumption that Alice emits the z states $|\alpha\rangle$ and $|\alpha'\rangle$ with equal probability. If we wish to account for the possibility of a biased source, as was considered in [20] for instance, then it is straightforward to adapt the derivation of (3.4). In particular, if the two z -basis states are emitted with probabilities $p = (1 + \varepsilon)/2$ and $p' = (1 - \varepsilon)/2$ respectively, (3.14) generalises to

$$H(Z_A | E) \geq h(p) - h\left[\frac{1}{2} + \frac{1}{2}\sqrt{\varepsilon^2 + (1 - \varepsilon^2)F(\rho_E, \rho'_E)^2}\right]. \quad (3.15)$$

The key rates obtained later will hold essentially unmodified if there is a bias in the x basis, provided the observed coincidence statistics are used to correctly bound the trace distance $D(\sigma_B, \sigma'_B)$. Note that this is a departure from the approach used by the authors of [20, 61, 62], where any bias in the source was included in their analogue of the source characterisation (3.7). This way we obtain a generally better result. For instance, for an otherwise ideal BB84 implementation and in the absence of errors, we obtain the best possible key rate of $r = h(p)$.

3.2.2 Incorporating local randomisation

If Alice applies local randomisation, i.e., flips a fraction η of her z -basis key bits, this has the effect of transforming the classical-quantum state (3.12) to

$$\begin{aligned} \tau_{ZE} = & \frac{1}{2}((1 - \eta)|0\rangle\langle 0|_Z + \eta|1\rangle\langle 1|_Z) \otimes \rho_E \\ & + \frac{1}{2}(\eta|0\rangle\langle 0|_Z + (1 - \eta)|1\rangle\langle 1|_Z) \otimes \rho'_E, \end{aligned} \quad (3.16)$$

which expresses that the z -basis states in Eve's possession are now correlated with a probabilistic mixture of Alice's (new) z -basis bits. In order to derive a lower bound on $H(Z_A | E)$, we first reexpress the classical-quantum state (3.16) as

$$\tau_{ZE} = \frac{1}{2}|0\rangle\langle 0|_Z \otimes \tilde{\rho}_E + \frac{1}{2}|1\rangle\langle 1|_Z \otimes \tilde{\rho}'_E, \quad (3.17)$$

where $\tilde{\rho}_E = \text{Tr}_B[\tilde{\rho}]$ and $\tilde{\rho}'_E = \text{Tr}_B[\tilde{\rho}']$, and we have set

$$\tilde{\rho} = (1 - \eta)\rho + \eta\rho', \quad (3.18)$$

$$\tilde{\rho}' = \eta\rho + (1 - \eta)\rho', \quad (3.19)$$

The conditional von Neumann entropy, evaluated directly on the classical-quantum state (3.17), simplifies to

$$H(Z_A | E) = 1 + \frac{1}{2}(S(\tilde{\rho}_E) + S(\tilde{\rho}'_E)) - S(\frac{1}{2}(\rho_E + \rho'_E)), \quad (3.20)$$

with $S(\rho) = -\text{Tr}[\rho \log(\rho)]$. Following the approach in the preceding subsection, we use that $H(Z_A | E) \geq H(Z_A | EE')$ for any extension $\tau_{ZEE'}$ of the state (3.17) to a larger Hilbert space in order to replace ρ_E and ρ'_E with purifications $|\psi\rangle$ and $|\psi'\rangle$ chosen such that $F(\rho_E, \rho'_E) = \langle\psi|\psi'\rangle$. With this substitution,

$$\begin{aligned} H(Z_A | E) &\geq 1 + \frac{1}{2}S((1-\eta)|\psi\rangle\langle\psi| + \eta|\psi'\rangle\langle\psi'|) \\ &\quad + \frac{1}{2}S(\eta|\psi\rangle\langle\psi| + (1-\eta)|\psi'\rangle\langle\psi'|) \\ &\quad - S(\frac{1}{2}(|\psi\rangle\langle\psi| + |\psi'\rangle\langle\psi'|)). \end{aligned} \quad (3.21)$$

The eigenvalues of the operator $(1-\eta)|\psi\rangle\langle\psi| + \eta|\psi'\rangle\langle\psi'|$ are easily found to be $\frac{1}{2} \pm \frac{1}{2}\sqrt{1-4\eta(1-\eta)(1-|\langle\psi|\psi'\rangle|^2)}$. Consequently,

$$\begin{aligned} H(Z_A | E) &\geq 1 - h(\frac{1}{2} + \frac{1}{2}F(\rho_E, \rho'_E)) \\ &\quad + h(\frac{1}{2} + \frac{1}{2}\sqrt{1-4\eta(1-\eta)(1-F(\rho_E, \rho'_E)^2)}). \end{aligned} \quad (3.22)$$

In appendix 3.A, the right-hand side of (3.22) is shown to be an increasing function of the fidelity. Given a lower bound $F(\rho_E, \rho'_E) \geq F_z$ on the fidelity, then, we obtain the analytic lower bound

$$\begin{aligned} r &\geq 1 + h(\frac{1}{2} + \frac{1}{2}\sqrt{1-4\eta(1-\eta)(1-F_z^2)}) \\ &\quad - h(\frac{1}{2} + \frac{1}{2}F_z) - h((1-\eta)\delta_z + \eta(1-\delta_z)) \end{aligned} \quad (3.23)$$

for the key rate.

The best result for the key rate is obtained by maximising the right-hand side of (3.23) over η , which, if necessary, is readily done numerically. Typically, as the channel noise approaches the maximal threshold, the optimal fraction η approaches 1/2. In this regime, the behaviour of the key-rate bound can be studied by substituting $\eta = (1-\varepsilon)/2$ and expanding the resulting expression in powers of ε . The result, to the first non-trivial order in ε , is

$$r \gtrsim \left(-\frac{1-F_z^2}{4F_z} \log\left(\frac{1+F_z}{1-F_z}\right) + \frac{(1-2\delta_z)^2}{2\ln(2)} \right) \varepsilon^2, \quad (3.24)$$

where $\ln$ is the natural logarithm. Consequently, threshold error rates can be obtained by identifying corresponding roots of the expression

$$(1-F_z^2) \ln\left(\frac{1+F_z}{1-F_z}\right) - 2F_z(1-2\delta_z)^2. \quad (3.25)$$

3.2.3 Bounding the min-entropy

As alluded to above, the Devetak-Winter bound, and consequently the bound (3.4) on the key rate that we will use in the remainder of this thesis, is only guaranteed to hold in the asymptotic limit of an infinitely long key. In the realistic finite-key case, a finite-key result can be derived based on a “one shot” version of the Devetak-Winter rate [60] where the main problem in obtaining a security proof is reduced to lower bounding the conditional min-entropy $H_{\min}(Z_A | E)$ [77], defined by

$$H_{\min}(Z_A | E) = \max_{\sigma_E} \sup \{ \lambda \in \mathbb{R} : 2^{-\lambda} \mathbb{1}_Z \otimes \sigma_E \geq \tau_{ZE} \}, \quad (3.26)$$

rather than the von Neumann entropy $H(Z_A | E)$. To obtain a result that converges asymptotically to the key rate given by the Devetak-Winter bound, the min-entropy should ideally be evaluated with a smoothing parameter, which can roughly be interpreted as a small additional probability of failure traded in exchange for a higher key rate, and which can be made arbitrarily small in the asymptotic limit. We refer to [49, 78] for a discussion of the details.

Whether and how the smooth min-entropy can be bounded in such a way as to recover (3.14) asymptotically will be left as a problem for future work and will not be further addressed. Here, we show that the min-entropy (without smoothing), which in any case may be useful as a lower bound on the von Neumann entropy, has an exact expression in terms of the trace distance. To obtain this, we insert the expression (3.12) for τ_{ZE} into the condition $2^{-\lambda} \mathbb{1}_Z \otimes \sigma_E - \tau_{ZE} \geq 0$ appearing in (3.26). This expands to

$$|0\rangle\langle 0|_Z \otimes (2^{-\lambda} \sigma_E - \tfrac{1}{2} \rho_E) + |1\rangle\langle 1|_Z \otimes (2^{-\lambda} \sigma_E - \tfrac{1}{2} \rho'_E) \geq 0, \quad (3.27)$$

from which we extract the conditions $2^{1-\lambda} \sigma_E \geq \rho_E$ and $2^{1-\lambda} \sigma_E \geq \rho'_E$. The tightest fit is obtained with

$$2^{1-\lambda} \sigma_E = \tfrac{1}{2}(\rho_E + \rho'_E) + \tfrac{1}{2}|\rho_E - \rho'_E|. \quad (3.28)$$

We determine $\lambda = H_{\min}(Z_A | E)$ by taking the trace of both sides:

$$\begin{aligned} 2^{1-\lambda} &= 1 + \tfrac{1}{2} \text{Tr}[|\rho_E - \rho'_E|] \\ &= 1 + D(\rho_E, \rho'_E), \end{aligned} \quad (3.29)$$

which rearranges to

$$H_{\min}(Z_A | E) = 1 - \log(1 + D(\rho_E, \rho'_E)). \quad (3.30)$$

The end result (3.30) is a special case of the min-entropy’s expression in terms of the guessing probability [77]. Applying the bound $D(\rho_E, \rho'_E) \leq$

$\sqrt{1 - F(\rho_E, \rho'_E)^2}$, we can alternatively bound the min-entropy in terms of the fidelity. While the result would be less than optimal and still limited to collective attacks, it is worth noting that using (3.30) in place of (3.14) would already allow the obtention of finite-key bounds using the general approach explored in this chapter.

3.3 BB84 with ideal source

In the special case of an ideal BB84 source, characterised by $\theta = \pi/2$, the general cloning bound (3.8) reduces to

$$F(\rho_E, \rho'_E) \geq D(\sigma_B, \sigma'_B), \quad (3.31)$$

and we recover the famous key rate $r \geq 1 - h(\delta_x) - h(\delta_z)$ due to Shor and Preskill [53]. While the proof of (3.8) is not complicated, a direct proof of (3.31) is especially simple and will serve to introduce our techniques.

In an ideal BB84 implementation, the source states form two mutually unbiased orthogonal bases, i.e., satisfy

$$\langle \alpha | \alpha' \rangle = \langle \beta | \beta' \rangle = 0 \quad (3.32)$$

and, in an appropriate phase convention,

$$\begin{aligned} |\alpha\rangle &= \frac{1}{\sqrt{2}}(|\beta\rangle + |\beta'\rangle), \\ |\alpha'\rangle &= \frac{1}{\sqrt{2}}(|\beta\rangle - |\beta'\rangle). \end{aligned} \quad (3.33)$$

Equivalently, one can identify $Z = \rho - \rho'$ and $X = \sigma - \sigma'$ with the Pauli z and x operators. With this notation, $D(\rho_B, \rho'_B) = \frac{1}{2}\|Z_B\|_1$ and $D(\sigma_B, \sigma'_B) = \frac{1}{2}\|X_B\|_1$.

We obtain (3.31) as follows. Let U_B be the (Hermitian) unitary operator acting on $\mathcal{H}_B$ such that $\frac{1}{2}\|X_B\|_1 = \frac{1}{2}\text{Tr}_B[U_B X_B]$. Then,

$$\begin{aligned} D(\sigma_B, \sigma'_B) &= \frac{1}{2}\text{Tr}_B[U_B X_B] \\ &= \frac{1}{2}\text{Tr}[(U_B \otimes \mathbb{1}_E)] \\ &= \frac{1}{2}(\langle \alpha | U_B \otimes \mathbb{1}_E | \alpha' \rangle + \langle \alpha' | U_B \otimes \mathbb{1}_E | \alpha \rangle) \\ &= \text{Re}[\langle \alpha | U_B \otimes \mathbb{1}_E | \alpha' \rangle] \\ &\leq |\langle \alpha | U_B \otimes \mathbb{1}_E | \alpha' \rangle|. \end{aligned} \quad (3.34)$$

Now, $|\alpha\rangle$ and $|\alpha'\rangle$ are by definition purifications of ρ_E and ρ'_E , and since $U_B \otimes \mathbb{1}_E$ acts nontrivially only on $\mathcal{H}_B$, the state $U_B \otimes \mathbb{1}_E |\alpha'\rangle$ is still a purification

of ρ'_E . Thus, by Uhlmann's theorem, the last line of (3.34) provides a lower bound on the fidelity between ρ_E and ρ'_E , which concludes our proof. Note that, similar to the derivation in section 1.4.4, we never actually used the orthogonality relations in the proof above, meaning that the proof also holds for the entanglement-based BB84 variant.

We conclude this discussion of the ideal BB84 scenario by noting that (3.31) can be seen as a strengthened version the tradeoff relation,

$$D(\rho_E, \rho'_E)^2 + D(\sigma_B, \sigma'_B)^2 \leq 1. \quad (3.35)$$

cited as a corollary of the derivation of the FGGNP bound summarised in section 1.4.4. Specifically, (3.35) follows by applying the upper bound $D(\rho_E, \rho'_E) \leq \sqrt{1 - F(\rho_E, \rho'_E)^2}$ to (3.31).

3.4 BB84 with arbitrary source states

3.4.1 Derivation of fidelity bound

We now derive (3.8). Our starting point is the quantity

$$\Delta = \frac{1}{2\sqrt{2}} (\langle \alpha | \beta \rangle + \langle \alpha' | \beta \rangle + \langle \alpha | \beta' \rangle - \langle \alpha' | \beta' \rangle). \quad (3.36)$$

Note that (3.36) as expressed above is dependent on physically irrelevant phase factors (e.g. $|\alpha\rangle$ and $e^{i\phi}|\alpha\rangle$ denote the same state); to obtain the best result, one should adopt the phase convention that maximises (3.36), which can then always be taken to be real.

Let P and Q be orthogonal projectors such that $P - Q = U$ and $P + Q = \mathbb{1}$, where $U = U_B \otimes \mathbb{1}_E$ is the Hermitian unitary such that $\frac{1}{2} \text{Tr}[U(\sigma - \sigma')] = D(\sigma_B, \sigma'_B)$, as in the proof of (3.31). For convenience, we also define the (generally not normalised) states $|\alpha_{\pm}\rangle = (|\alpha\rangle \pm |\alpha'\rangle)/\sqrt{2}$. Then,

$$\begin{aligned} 2\Delta &= \langle \alpha_+ | (P + Q) | \beta \rangle + \langle \alpha_- | (P + Q) | \beta' \rangle \\ &\leq |\langle \alpha_+ | P | \beta \rangle| + |\langle \alpha_+ | Q | \beta \rangle| \\ &\quad + |\langle \alpha_- | P | \beta' \rangle| + |\langle \alpha_- | Q | \beta' \rangle|. \end{aligned} \quad (3.37)$$

Applying the Cauchy-Schwarz inequality twice on the first and fourth terms on the right-hand side,

$$\begin{aligned} &|\langle \alpha_+ | P | \beta \rangle| + |\langle \alpha_- | Q | \beta' \rangle| \\ &\leq \sqrt{\langle \alpha_+ | P | \alpha_+ \rangle} \sqrt{\langle \beta | P | \beta \rangle} + \sqrt{\langle \alpha_- | Q | \alpha_- \rangle} \sqrt{\langle \beta' | Q | \beta' \rangle} \\ &\leq \sqrt{\langle \alpha_+ | P | \alpha_+ \rangle + \langle \alpha_- | Q | \alpha_- \rangle} \sqrt{\langle \beta | P | \beta \rangle + \langle \beta' | Q | \beta' \rangle} \\ &= \sqrt{1 + \text{Re}[\langle \alpha | U | \alpha' \rangle]} \sqrt{1 + D(\sigma_B, \sigma'_B)}. \end{aligned} \quad (3.38)$$

In a similar manner, $|\langle\alpha_+|Q|\beta\rangle| + |\langle\alpha_-|P|\beta'\rangle|$ provides a lower bound on $\sqrt{1 - \text{Re}[\langle\alpha|U|\alpha'\rangle]}\sqrt{1 - D(\sigma_B, \sigma'_B)}$. We thus obtain

$$\begin{aligned} \Delta \leq & \sqrt{\frac{1}{2} + \frac{1}{2} \text{Re}[\langle\alpha|U|\alpha'\rangle]}\sqrt{\frac{1}{2} + \frac{1}{2} D(\sigma_B, \sigma'_B)} \\ & + \sqrt{\frac{1}{2} - \frac{1}{2} \text{Re}[\langle\alpha|U|\alpha'\rangle]}\sqrt{\frac{1}{2} - \frac{1}{2} D(\sigma_B, \sigma'_B)}, \end{aligned} \quad (3.39)$$

where $\text{Re}[\langle\alpha|U|\alpha'\rangle]$ in turn provides a lower bound on $F(\rho_E, \rho'_E)$, as in the proof of (3.31). Substituting $\Delta = \sqrt{1 + |\sin(\theta)|}/\sqrt{2}$ and rearranging, we arrive at (3.8).

3.4.2 Resulting key rate

Explicitly combining these with the generic key-rate expression (3.23), we obtain the bound

$$\begin{aligned} r \geq & 1 + h\left(\frac{1}{2} + \frac{1}{2}\sqrt{1 - 4\eta(1 - \eta)(1 - f_\theta(|1 - 2\delta_x|)^2)}\right) \\ & - h\left(\frac{1}{2} + \frac{1}{2}f_\theta(|1 - 2\delta_x|)\right) - h(\tilde{\delta}_z), \end{aligned} \quad (3.40)$$

with $\tilde{\delta}_z = (1 - \eta)\delta_z + \eta(1 - \delta_z)$ and f_θ defined by

$$f_\theta(x) = \begin{cases} |\sin(\theta)|x - |\cos(\theta)|\sqrt{1 - x^2} & : x \geq |\cos(\theta)| \\ 0 & : x \leq |\cos(\theta)| \end{cases}, \quad (3.41)$$

for the key rate with local randomisation applied. For $\eta = 0$, we recover the key rate

$$r \geq 1 - h\left(\frac{1}{2} + \frac{1}{2}f_\theta(|1 - 2\delta_x|)\right) - h(\delta_z) \quad (3.42)$$

given earlier, which itself coincides with the rate derived in [20] in the setting under consideration here.

The rates (3.40) and (3.42) (with and without local randomisation, respectively) are illustrated for a few values of θ in figure 3.1, assuming symmetric errors (i.e., $\delta_z = \delta_x = \delta$) for simplicity. The depicted rates with local randomisation were found by numerically maximising (3.40) over η . For $\theta = \pi/2 = 90^\circ$, corresponding to an ideal BB84 source, we recover the Shor-Preskill rate [53] and the improvement with local randomisation depicted in figure 2 of Ref. [32].

The threshold error rates, i.e., the error rates for which the key rates (3.42) without preprocessing and (3.40) with optimal local randomisation become zero, again for $\delta_z = \delta_x = \delta$, are depicted in figure 3.2 as a function of θ . The threshold curve with local randomisation was found by identifying the

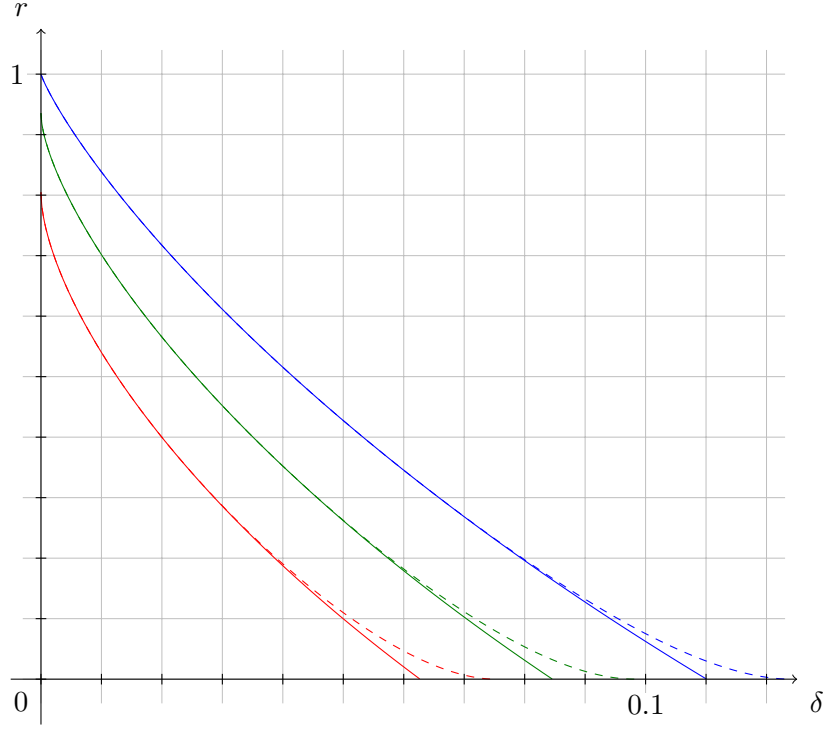


Figure 3.1: Key rate (3.40) secure against collective attacks for a BB84 implementation of source characterisation angle $\theta = 90^\circ$, $\theta = 80^\circ$, and $\theta = 70^\circ$, with (dashed curves) and without (solid curves) local randomisation, for $\delta = \delta_z = \delta_x$.

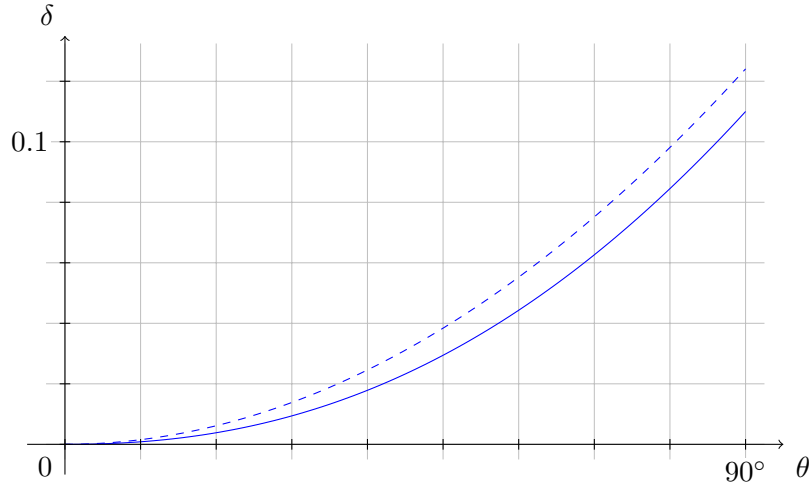


Figure 3.2: Threshold error rate, i.e., the error rate $\delta = \delta_z = \delta_x$ for which the key-rate bound (3.40) becomes zero, with (dashed curve) and without (solid curve) local randomisation, for source characterisation angles $0 \leq \theta \leq 90^\circ$.

corresponding root of (3.25). For $\theta = \pi/2 = 90^\circ$ we recover the threshold error rates of $\delta \approx 12.4120\%$ and $\delta \approx 11.0028\%$ originally found in Refs. [21] and [53], respectively. For an ideal BB84 implementation, this corresponds to a relative increase of around 12.81% to the provably tolerable channel noise. This difference becomes more significant as θ decreases: for instance the relative improvement becomes around 20.00% ($\delta \approx 7.5191\%$ compared with $\delta \approx 6.2660\%$) for $\theta = 70^\circ$, around 33.84% ($\delta \approx 3.1120\%$ vs $\delta \approx 2.3251\%$) for $\theta = 45^\circ$, and around 83.38% ($\delta \approx 0.1538\%$ vs $\delta \approx 0.08390\%$) if θ is as low as 10° , indicating that the benefit of additional preprocessing becomes more pronounced for a realistic BB84 implementation expected to suffer from device imprecisions.

Finally, if $\dim \mathcal{H}_A = 2$ and the source states form two orthogonal bases, then θ as defined by (3.7) coincides with the angle separating the bases on the Bloch sphere. In this particular case, the source is basis independent (i.e., $\frac{1}{2}\rho + \frac{1}{2}\rho' = \frac{1}{2}\sigma + \frac{1}{2}\sigma'$), and we note that the key rate (3.42) is an improvement over the key rate predicted by the uncertainty relation [59, 60], which for comparison is

$$r \geq 1 - \log(1 + |\cos(\theta)|) - h(\delta_x) - h(\delta_z). \quad (3.43)$$

3.4.3 Optimality

For the nonideal BB84 implementation considered in the preceding section, the key-rate bound (3.40) is tight in the sense that the Devetak-Winter rate (3.11) can be attained for all values of the independent variables θ , δ_z , δ_x , and η . This is demonstrated here by the explicit construction of a family of source states and optimal unitary attacks. Equality between the right-hand sides of (3.11) and (3.40) requires the conditional von Neumann entropy bound (3.22) and the fidelity bound (3.8) to hold with equality simultaneously, which helps in the determination of an optimal attack. First, note that in the case of an equality, (3.8) rearranges to

$$|\sin(\theta)| = F_z D_x + \sqrt{1 - F_z^2} \sqrt{1 - D_x^2}, \quad (3.44)$$

with $F_z = F(\rho_E, \rho'_E)$ and $D_x = D(\sigma_B, \sigma'_B)$ and the condition $F_z \leq D_x$. Equation (3.44) can equivalently be reexpressed as

$$\sqrt{1 + |\sin(\theta)|} = \frac{1}{\sqrt{2}} \left(\sqrt{1 + F_z} \sqrt{1 + D_x} + \sqrt{1 - F_z} \sqrt{1 - D_x} \right). \quad (3.45)$$

Consequently, our goal will be to construct source states such that the definition of the source characterisation (3.7) equals the right-hand side of (3.45).

Requiring $D(\rho_B, \rho'_B) = D_z$ suggests setting the z-basis states to the form

$$|\alpha\rangle = \sqrt{\frac{1+D_z}{2}}|0\rangle_B|\psi_0\rangle_E + \sqrt{\frac{1-D_z}{2}}|1\rangle_B|\psi'_1\rangle_E, \quad (3.46)$$

$$|\alpha'\rangle = \sqrt{\frac{1-D_z}{2}}|0\rangle_B|\psi_1\rangle_E + \sqrt{\frac{1+D_z}{2}}|1\rangle_B|\psi'_0\rangle_E, \quad (3.47)$$

with $|0\rangle_B$ and $|1\rangle_B$ orthonormal. The trace distance $D(\rho_B, \rho'_B)$ will equal D_z if $\langle\psi_0|\psi'_1\rangle = \langle\psi_1|\psi'_0\rangle = 0$. In order for the fidelity $F(\rho_E, \rho'_E)$ to equal F_z , and in such a way that the von Neumann entropy bound (3.22) becomes an equality, we additionally require $\langle\psi_0|\psi_1\rangle = \langle\psi'_0|\psi'_1\rangle = 0$ and $\langle\psi_0|\psi'_0\rangle = \langle\psi_1|\psi'_1\rangle = F_z \in \mathbb{R}^+$, such that $\{|\psi_0\rangle, |\psi'_0\rangle\}$ and $\{|\psi_1\rangle, |\psi'_1\rangle\}$ span two mutually orthogonal subspaces. Note that, with these definitions, $|\alpha\rangle$ and $|\alpha'\rangle$ are normalised and orthogonal.

The right-hand side of the source characterisation (3.7) can be reexpressed as $\frac{1}{\sqrt{2}}|\langle\alpha_+|\beta\rangle + \langle\alpha_-|\beta'\rangle|$ with $|\alpha_\pm\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle \pm |\alpha'\rangle)$. Introducing, for convenience, the states

$$|\alpha_k\rangle = |0\rangle_B|\psi_k\rangle_E, \quad |\alpha'_k\rangle = |1\rangle_B|\psi'_k\rangle_E, \quad (3.48)$$

and $|\alpha_k^\pm\rangle = \frac{1}{\sqrt{2}}(|\alpha_k\rangle \pm |\alpha'_k\rangle)$, $k \in \{0, 1\}$, we find

$$|\alpha_k^+\rangle = \sqrt{\frac{1+F_z}{2}}|+\rangle_B|\psi_k^+\rangle_E + \sqrt{\frac{1-F_z}{2}}|-\rangle_B|\psi_k^-\rangle_E, \quad (3.49)$$

$$|\alpha_k^-\rangle = \sqrt{\frac{1-F_z}{2}}|+\rangle_B|\psi_k^-\rangle_E + \sqrt{\frac{1+F_z}{2}}|-\rangle_B|\psi_k^+\rangle_E, \quad (3.50)$$

where $|\pm\rangle_B = \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B)$ and the states

$$|\psi_k^\pm\rangle_E = \frac{|\psi_k\rangle_E \pm |\psi'_k\rangle_E}{\sqrt{2 \pm 2F_z}} \quad (3.51)$$

are orthonormal. In terms of $|\alpha_k^\pm\rangle$,

$$|\alpha_+\rangle = \sqrt{\frac{1+D_z}{2}}|\alpha_0^+\rangle + \sqrt{\frac{1-D_z}{2}}|\alpha_1^+\rangle, \quad (3.52)$$

$$|\alpha_-\rangle = \sqrt{\frac{1+D_z}{2}}|\alpha_0^-\rangle - \sqrt{\frac{1-D_z}{2}}|\alpha_1^-\rangle. \quad (3.53)$$

It is then fairly straightforward to construct x-basis states for which the right-hand side of the source characterisation (3.7) will take the form of the right-hand side of (3.45). We set

$$|\beta_k\rangle = \sqrt{\frac{1+D_x}{2}}|+\rangle_B|\psi_k^+\rangle_E + \sqrt{\frac{1-D_x}{2}}|-\rangle_B|\psi_k^-\rangle_E, \quad (3.54)$$

$$|\beta'_k\rangle = \sqrt{\frac{1-D_x}{2}}|+\rangle_B|\psi_k^-\rangle_E + \sqrt{\frac{1+D_x}{2}}|-\rangle_B|\psi_k^+\rangle_E, \quad (3.55)$$

and

$$|\beta\rangle = \sqrt{\frac{1+D_z}{2}}|\beta_0\rangle + \sqrt{\frac{1-D_z}{2}}|\beta_1\rangle, \quad (3.56)$$

$$|\beta'\rangle = \sqrt{\frac{1+D_z}{2}}|\beta'_0\rangle - \sqrt{\frac{1-D_z}{2}}|\beta'_1\rangle. \quad (3.57)$$

With these definitions we find

$$\langle\alpha_+|\beta\rangle = \langle\alpha_-|\beta'\rangle = \frac{1}{2}\left(\sqrt{1+F_z}\sqrt{1+D_x} + \sqrt{1-F_z}\sqrt{1-D_x}\right), \quad (3.58)$$

independently of D_z , from which we recover the right-hand side of the rearrangement (3.45) of the fidelity bound (3.8).

Explicitly, from the expressions (3.46), (3.47), (3.56), and (3.57) for the z- and x-basis states, Bob's marginals are given by

$$\rho_B = \frac{1+D_z}{2}|0\rangle\langle 0|_B + \frac{1-D_z}{2}|1\rangle\langle 1|_B, \quad (3.59)$$

$$\rho'_B = \frac{1-D_z}{2}|0\rangle\langle 0|_B + \frac{1+D_z}{2}|1\rangle\langle 1|_B, \quad (3.60)$$

$$\sigma_B = \frac{1+D_x}{2}|+\rangle\langle +|_B + \frac{1-D_x}{2}|-\rangle\langle -|_B, \quad (3.61)$$

$$\sigma'_B = \frac{1-D_x}{2}|+\rangle\langle +|_B + \frac{1+D_x}{2}|-\rangle\langle -|_B. \quad (3.62)$$

Consequently, Alice and Bob detect errors at the rates $\delta_z = \frac{1}{2} - \frac{1}{2}D_z$ and $\delta_x = \frac{1}{2} - \frac{1}{2}D_x$ if Bob measures (optimally) in the σ_z and σ_x bases. Likewise, Eve's marginals of the z states are given by

$$\rho_E = \frac{1+D_z}{2}|\psi_0\rangle\langle\psi_0|_E + \frac{1-D_z}{2}|\psi'_1\rangle\langle\psi'_1|_E, \quad (3.63)$$

$$\rho'_E = \frac{1-D_z}{2}|\psi_1\rangle\langle\psi_1|_E + \frac{1+D_z}{2}|\psi'_0\rangle\langle\psi'_0|_E, \quad (3.64)$$

for which one can readily verify that $F(\rho_E, \rho'_E) = \|\sqrt{\rho_E}\sqrt{\rho'_E}\|_1 = F_z$ and, for any $p, q \geq 0$ and $p+q=1$,

$$\begin{aligned} S(p\rho_E + q\rho'_E) &= h\left(\frac{1}{2} + \frac{1}{2}D_z\right) \\ &\quad + \frac{1+D_z}{2}S(p|\psi_0\rangle\langle\psi_0|_E + q|\psi'_0\rangle\langle\psi'_0|_E) \\ &\quad + \frac{1-D_z}{2}S(p|\psi'_1\rangle\langle\psi'_1|_E + q|\psi_1\rangle\langle\psi_1|_E) \\ &= h\left(\frac{1}{2} + \frac{1}{2}D_z\right) \\ &\quad + h\left(\frac{1}{2} + \frac{1}{2}\sqrt{1-4pq(1-F_z^2)}\right). \end{aligned} \quad (3.65)$$

Using (3.65) to directly evaluate the expression (3.20) for the conditional von Neumann entropy $H(Z_A | E)$, we find that its bound (3.22) in terms of fidelity $F(\rho_E, \rho'_E)$ is attained with equality for the entire family of sources and attacks just constructed.

Equations (3.46), (3.47), (3.56), and (3.57) give the optimal attack for a family of sources identified by the relations

$$\langle\alpha|\alpha'\rangle = \langle\beta|\beta'\rangle = 0 \quad (3.66)$$

and

$$\langle \alpha | \beta \rangle = \langle \alpha' | \beta \rangle = \langle \alpha | \beta' \rangle = -\langle \alpha' | \beta' \rangle = \frac{\sqrt{1+|\sin(\theta)|}}{2}, \quad (3.67)$$

for which the bound (3.40) on the Devetak-Winter rate is attained with equality independently of the fraction η of bits flipped by Alice in the local randomisation preprocessing step. The family of optimal attacks given here generalises the optimal individual attack derived for an ideal BB84 source in [14], which is recovered for $|\sin(\theta)| = 1$ or, equivalently, by setting $F_z = D_x$. Another extreme worth noting is the case $F_z = 0$ and $D_z = D_x = 1$, in which case $|\sin(\theta)| = 0$ and

$$|\alpha\rangle = |0\rangle_B |0\rangle_E, \quad |\alpha'\rangle = |1\rangle_B |1\rangle_E, \quad (3.68)$$

$$|\beta\rangle = |+\rangle_B |+\rangle_E, \quad |\beta'\rangle = |-\rangle_B |+\rangle_E, \quad (3.69)$$

i.e., the adversary acquires perfect copies of Bob's z-basis states without introducing any errors.

3.5 BB84 with arbitrary qubit states

3.5.1 Arbitrary measurements

We now turn to a more detailed study of qubit sources. It will be convenient to characterise such a source in terms of three parameters: an angle φ representing the angle between the two bases on the Bloch sphere, and angles α and β measuring the nonorthogonality of the states constituting each basis and defined by $|\sin(\alpha)| = |\langle \alpha | \alpha' \rangle|$ and $|\sin(\beta)| = |\langle \beta | \beta' \rangle|$ respectively. With this characterisation, the differences between the z- and x-basis density operators can be expressed as

$$\rho - \rho' = \cos(\alpha)Z, \quad (3.70)$$

$$\sigma - \sigma' = \cos(\beta)X, \quad (3.71)$$

where Z and X are Pauli-type operators (i.e., two dimensional and of eigenvalues $+1$ and -1) and separated by an angle φ on the Bloch sphere such that their anticommutator satisfies $\frac{1}{2}\{Z, X\} = \cos(\varphi)\mathbb{1}_A$.

In order to bound the distinguishability between ρ_E and ρ'_E , we first introduce another Pauli-type operator W orthogonal to Z on the Bloch sphere and in the same plane as Z and X in such a way that X can be expressed as

$$X = \cos(\varphi)Z + \sin(\varphi)W. \quad (3.72)$$

With a suitable choice of basis, Z and W can be identified with σ_z and σ_x , respectively. The trace norm $\frac{1}{2}\|W_B\|_1$ is a priori unknown, since it is not the difference between two states that are actually prepared by Alice or measured by Bob. It can, however, be bounded in terms of $\frac{1}{2}\|X_B\|_1$, which is in turn lower bounded in terms of the x-basis error rate. To see this, similar to the preceding subsections, let $U = U_B \otimes \mathbf{1}_E$ be the Hermitian unitary operator such that $\frac{1}{2}\|X_B\|_1 = \frac{1}{2} \text{Tr}[UX]$. Then,

$$\begin{aligned} \frac{1}{2}\|X_B\|_1 &= \frac{1}{2}\|\cos(\varphi)Z_B + \sin(\varphi)W_B\|_1 \\ &\leq |\cos(\varphi)|\frac{1}{2}|\text{Tr}[UZ]| + |\sin(\varphi)|\frac{1}{2}|\text{Tr}[UW]| \\ &\leq |\cos(\varphi)|\sqrt{1 - \frac{1}{4}\text{Tr}[UW]^2} + |\sin(\varphi)|\frac{1}{2}|\text{Tr}[UW]|, \end{aligned} \quad (3.73)$$

where we used that $\frac{1}{4}\text{Tr}[UZ]^2 + \frac{1}{4}\text{Tr}[UW]^2 \leq 1$. Rearranging the final line and using $\frac{1}{2}|\text{Tr}[UW]|$ as a lower bound on $\frac{1}{2}\|W_B\|_1$, we obtain

$$\frac{1}{2}\|W_B\|_1 \geq |\sin(\varphi)|\frac{1}{2}\|X_B\|_1 - |\cos(\varphi)|\sqrt{1 - \frac{1}{4}\|X_B\|_1^2}. \quad (3.74)$$

Since $\frac{1}{2}\|W_B\|_1$ is also necessarily positive, the lower can be expressed as

$$\frac{1}{2}\|W_B\|_1 \geq f_\varphi\left(\frac{1}{2}\|X_B\|_1\right), \quad (3.75)$$

where f is the same function earlier defined in (3.41), except that we use the angle φ between the bases in place of the basis overlap angle θ defined in (3.7).

If we are content with obtaining a lower bound on the min-entropy, which depends on the trace distance $D(\rho_E, \rho'_E)$ we simply use that $D(\rho_E, \rho'_E) = |\cos(\alpha)|\frac{1}{2}\|Z_E\|_1$. Using that $\frac{1}{4}\|Z_E\|_1^2 + \frac{1}{4}\|W_B\|_1^2 \leq 1$, which was first derived in section 1.4.4 and recovered as (3.35) in section 3.3, $\frac{1}{2}\|Z_E\|_1$ can be upper bounded by

$$\frac{1}{2}\|Z_E\|_1 \leq |\cos(\varphi)|\|X_B\|_1 + |\sin(\varphi)|\sqrt{1 - \frac{1}{4}\|X_B\|_1^2}, \quad (3.76)$$

provided that $\frac{1}{2}\|X_B\|_1 \geq |\cos(\varphi)|$.

Deriving a lower bound on the conditional von Neumann entropy via its lower bound on the fidelity $F(\rho_E, \rho'_E)$ requires additional work. If $|\alpha\rangle$ and $|\alpha'\rangle$ happen to be orthogonal, i.e., if $\sin(\alpha) = 0$, then according to (3.31) Eve's fidelity is simply bounded by $F(\rho_E, \rho'_E) \geq \frac{1}{2}\|W_B\|_1$. This, however, does not hold more generally: a numerical minimisation of $F(\rho_E, \rho'_E) - \frac{1}{2}\|W_B\|_1$ will easily find counterexamples. Nor is this an artefact of bounding the fidelity as an intermediate step in bounding the key rate, as one can likewise find cases where $H(Z_A | E) \not\geq 1 - h\left(\frac{1}{2} + \frac{1}{4}\|W_B\|_1\right)$ when minimising the conditional von Neumann entropy directly.

It follows that a generally valid bound will depend explicitly on the non-orthogonality measured by the angle α . In the following, we will show that

$$F(\rho_E, \rho'_E) \geq g_\alpha\left(\frac{1}{2}\|W_B\|_1\right), \quad (3.77)$$

with the function g_α defined by

$$g_\alpha(x) = \begin{cases} (1 + |\sin(\alpha)|)x - |\sin(\alpha)| & : x \geq \frac{2|\sin(\alpha)|}{1+|\sin(\alpha)|} \\ |\sin(\alpha)| & : x \leq \frac{2|\sin(\alpha)|}{1+|\sin(\alpha)|} \end{cases}. \quad (3.78)$$

Combining (3.75) and (3.77), we obtain

$$F(\rho_E, \rho'_E) \geq g_\alpha \circ f_\varphi\left(\frac{D(\sigma_B, \sigma'_B)}{|\cos(\beta)|}\right), \quad (3.79)$$

where we have used that $\sigma - \sigma' = \cos(\beta)X$. We see here that explicitly introducing the non-orthogonality of the x-basis states can only improve the key rate bound. This is not surprising since the x-basis states are only used for the purpose of testing the channel. Explicitly writing the resulting key rate,

$$r \geq 1 - h\left[\frac{1}{2} + \frac{1}{2}g_\alpha \circ f_\varphi\left(\frac{|1-2\delta_x|}{|\cos(\beta)|}\right)\right] - h(\delta_z). \quad (3.80)$$

The remainder of this section will be devoted to the determination of the function g_α . We first express $|\alpha\rangle$ and $|\alpha'\rangle$ as

$$|\alpha\rangle = \cos\left(\frac{\alpha}{2}\right)|z\rangle + e^{i\phi} \sin\left(\frac{\alpha}{2}\right)|z'\rangle, \quad (3.81)$$

$$|\alpha'\rangle = \sin\left(\frac{\alpha}{2}\right)|z\rangle + e^{i\phi} \cos\left(\frac{\alpha}{2}\right)|z'\rangle \quad (3.82)$$

in terms of the (orthogonal) eigenstates $|z\rangle$ and $|z'\rangle$ of the operator Z defined at the beginning of this subsection, such that $\langle\alpha|\alpha'\rangle = \sin(\alpha)$ and $\rho - \rho' = \cos(\alpha)(|z\rangle\langle z| - |z'\rangle\langle z'|)$. We define the Hermitian unitary $U = U_B \otimes \mathbb{1}_E$ such that $\frac{1}{2}\|W_B\|_1 = \frac{1}{2}\text{Tr}[UW] = \text{Re}[\langle z|U|z'\rangle]$. Without loss of generality, we can take the quantity $\Gamma = \langle z|U|z'\rangle$ to be real (if necessary, this can be achieved by absorbing its phase into the phase ϕ already present in (3.81)). A lower bound is already given by

$$F(\rho_E, \rho'_E) \geq |\langle\alpha|U|\alpha'\rangle|, \quad (3.83)$$

however the result this would produce is not optimal. In order to obtain a better result, note that any unitary of the form $\tilde{U}_B \otimes \mathbb{1}_E$ can be used in place of U in (3.83). In particular, since U is Hermitian, the family of operators

$$U(\gamma) = i \sin(\gamma)\mathbb{1} + \cos(\gamma)U \quad (3.84)$$

are unitary and satisfy this requirement. Evaluating $\langle \alpha | U(\gamma) | \alpha' \rangle$, we find

$$\begin{aligned} \langle \alpha | U(\gamma) | \alpha' \rangle &= \frac{1}{2} \sin(\alpha) (\langle z | U(\gamma) | z \rangle + \langle z' | U(\gamma) | z' \rangle) \\ &\quad + e^{i\phi} \cos(\frac{\alpha}{2})^2 \langle z | U(\gamma) | z' \rangle + e^{-i\phi} \sin(\frac{\alpha}{2})^2 \langle z' | U(\gamma) | z \rangle \\ &= i \sin(\gamma) \sin(\alpha) + \cos(\gamma) \sin(\alpha) K \\ &\quad + \cos(\gamma) (\cos(\phi) + i \cos(\alpha) \sin(\phi)) \Gamma, \end{aligned} \quad (3.85)$$

where we have set $K = \frac{1}{2}(\langle z | U | z \rangle + \langle z' | U | z' \rangle)$. Squaring the last line,

$$\begin{aligned} |\langle \alpha | U(\gamma) | \alpha' \rangle|^2 &= \cos(\gamma)^2 (\sin(\alpha) K + \cos(\phi) \Gamma)^2 \\ &\quad + (\sin(\gamma) \sin(\alpha) + \cos(\gamma) \cos(\alpha) \sin(\phi) \Gamma)^2. \end{aligned} \quad (3.86)$$

Our task now is to maximise this quantity over γ and minimise over possible values of K and ϕ . We begin by minimising over K for fixed γ and ϕ . First, note that $|K| + \Gamma \leq 1$ (this can be inferred by evaluating (3.86) for $\gamma = \phi = 0$ and $\alpha = \pm\pi/2$ and noting that the result should never exceed 1). With this constraint, we wish to minimise $Q = |\sin(\alpha) K + \cos(\phi) \Gamma|$. If Γ is sufficiently small, we are able to choose K such that this quantity is zero. Otherwise, we simply set $|K| = 1 - \Gamma$. Calling Q_* the minimum possible value of $|\sin(\alpha) K + \cos(\phi) \Gamma|$, we have

$$Q_* = \begin{cases} (|\sin(\alpha)| + |\cos(\phi)|) \Gamma - |\sin(\alpha)| & : \Gamma \geq \frac{|\sin(\alpha)|}{|\sin(\alpha)| + |\cos(\phi)|} \\ 0 & : \Gamma \leq \frac{|\sin(\alpha)|}{|\sin(\alpha)| + |\cos(\phi)|} \end{cases}, \quad (3.87)$$

The next step is the maximisation over γ . Since Q_* has no dependence on γ , this is straightforward. For the optimal value γ_* of γ , we find

$$\begin{aligned} |\langle \alpha | U(\gamma_*) | \alpha' \rangle|^2 &= \frac{1}{2} (Q_*^2 + \cos(\alpha)^2 \sin(\phi)^2 \Gamma^2 + \sin(\alpha)^2) \\ &\quad + \frac{1}{2} \left\{ (Q_*^2 + \cos(\alpha)^2 \sin(\phi)^2 \Gamma^2 - \sin(\alpha)^2)^2 \right. \\ &\quad \left. + 4 \sin(\alpha)^2 \cos(\alpha)^2 \sin(\phi)^2 \Gamma^2 \right\}^{1/2}. \end{aligned} \quad (3.88)$$

Finally, we minimise over ϕ . For values of ϕ such that $|\cos(\phi)| \leq |\sin(\alpha)| \frac{1-\Gamma}{\Gamma}$, we set $Q_* = 0$, and (3.88) simplifies to

$$|\langle \alpha | U(\gamma_*) | \alpha' \rangle|^2 = \sin(\alpha)^2 + \cos(\alpha)^2 \sin(\phi)^2 \Gamma^2. \quad (3.89)$$

If $\Gamma \leq \frac{|\sin(\alpha)|}{1+|\sin(\alpha)|}$, then $Q_* = 0$ regardless of ϕ , and the minimum of (3.89) is

$$|\langle \alpha | U(\gamma_*) | \alpha' \rangle| = |\sin(\alpha)|. \quad (3.90)$$

Otherwise, the minimum of (3.86) for values of ϕ where $Q_* = 0$ is found by using the minimum allowed value of $|\sin(\phi)|$ in (3.89), which simplifies to

$$|\langle \alpha | U(\gamma_*) | \alpha' \rangle| = \sin(\alpha)^2 + \cos(\alpha)^2 \Gamma. \quad (3.91)$$

If $\Gamma > \frac{|\sin(\alpha)|}{1+|\sin(\alpha)|}$, then we must separately consider the range of possible values of ϕ for which $|\cos(\phi)| > |\sin(\alpha)| \frac{1-\Gamma}{\Gamma}$, where $Q_* = (|\sin(\alpha)| + |\cos(\phi)|) - |\sin(\alpha)|$. We have

$$Q_*^2 + \cos(\alpha)^2 \sin(\phi)^2 \Gamma^2 = Y\Gamma(Y\Gamma - 2) + 2\cos(\alpha)^2 \Gamma + \sin(\alpha)^2, \quad (3.92)$$

where we have set $Y = 1 + |\sin(\alpha)||\cos(\phi)|$. Then,

$$\begin{aligned} |\langle \alpha | U(\gamma_*) | \alpha' \rangle|^2 &= \sin(\alpha)^2 + \cos(\alpha)^2 \Gamma + \frac{1}{2} Y\Gamma(Y\Gamma - 2) \\ &\quad + \frac{1}{2} \Gamma \left((Y(Y\Gamma - 2) + 2\cos(\alpha)^2)^2 \right. \\ &\quad \left. + 4\sin(\alpha)^2 \cos(\alpha)^2 \sin(\phi)^2 \right)^{1/2}, \end{aligned} \quad (3.93)$$

which we can simplify down to

$$\begin{aligned} |\langle \alpha | U(\gamma_*) | \alpha' \rangle|^2 &= \sin(\alpha)^2 + \cos(\alpha)^2 \Gamma + \frac{1}{2} Y\Gamma(Y\Gamma - 2) \\ &\quad + \frac{1}{2} Y\Gamma \sqrt{(Y\Gamma - 2)^2 + 4\cos(\alpha)^2(\Gamma - 1)}. \end{aligned} \quad (3.94)$$

This is a decreasing function in Y and is therefore minimised by taking $|\cos(\phi)| = 1$, or $Y_* = 1 + |\sin(\alpha)|$. We find

$$\begin{aligned} |\langle \alpha | U(\gamma_*) | \alpha' \rangle|^2 &= \sin(\alpha)^2 + \cos(\alpha)^2 \Gamma \\ &\quad + \frac{1}{2} Y_* \Gamma [Y_* \Gamma - 2 + |Y_* \Gamma - 2| \sin(\alpha)|]. \end{aligned} \quad (3.95)$$

If $\Gamma \geq \frac{2|\sin(\alpha)|}{1+|\sin(\alpha)|}$, then the minimum we find is

$$|\langle \alpha | U(\gamma_*) | \alpha' \rangle| = (1 + |\sin(\alpha)|)\Gamma - |\sin(\alpha)|. \quad (3.96)$$

Otherwise, the minimum is simply $|\sin(\alpha)|$. Since (3.96) is always less than (3.91), our final result is

$$|\langle \alpha | U(\gamma_*) | \alpha' \rangle| \geq \begin{cases} (1 + |\sin(\alpha)|)\Gamma - |\sin(\alpha)| & : \Gamma \geq \frac{2|\sin(\alpha)|}{1+|\sin(\alpha)|} \\ |\sin(\alpha)| & : \Gamma \leq \frac{2|\sin(\alpha)|}{1+|\sin(\alpha)|} \end{cases}. \quad (3.97)$$

Recalling that $F(\rho_E, \rho'_E) \geq |\langle \alpha | U(\gamma_*) | \alpha' \rangle|$ and that we defined $\Gamma = \frac{1}{2} \|W_B\|_1$, this concludes the proof of (3.77).

3.5.2 Qubit source and detector

In most security analyses of the BB84 protocol accounting for device alignment errors, including the results derived in sections 3.4 and 3.5.1, only the x-basis error rate is used to bound the information an eavesdropper could have about the key. This leaves open the possibility that we could derive better results if the z-basis error rate were also used for this purpose. Here, we will illustrate how such a result can be derived in the case where Alice's source emits qubit states and Bob's detector is assumed to perform two-dimensional measurements.

As in section 3.5.1, we take $\rho - \rho' \propto Z$ and $\sigma - \sigma' \propto X$, where $X = \cos(\varphi)Z + \sin(\varphi)W$, and Z and W can be identified with the Pauli z and x operators. The error rates provide lower bounds on $D(\rho_B, \rho'_B) = |\cos(\alpha)|\frac{1}{2}\|Z_B\|_1$ and $D(\sigma_B, \sigma'_B) = |\cos(\beta)|\frac{1}{2}\|X_B\|_1$. The main intuition for deriving an improved key rate is that, if $\dim \mathcal{H}_B = 2$, the quantities $\frac{1}{2}\|Z_B\|_1$, $\frac{1}{2}\|X_B\|_1$, and $\frac{1}{2}\|W_B\|_1$ are constrained in the values they can take. Specifically, we will be able to show that

$$\sqrt{1 - \frac{1}{4}\|X_B\|_1^2} \geq |\sin(\varphi)|\sqrt{1 - \frac{1}{4}\|W_B\|_1^2} - |\cos(\varphi)|\sqrt{1 - \frac{1}{4}\|Z_B\|_1^2}. \quad (3.98)$$

Since we have $F(\rho_E, \rho'_E) \geq g_\alpha(\frac{1}{2}\|W_B\|_1)$ from (3.77), (3.98) provides a second bound on the fidelity to complement (3.79). Our end result is similar in form to (3.79):

$$F(\rho_E, \rho'_E) \geq g_\alpha \circ f_\varphi^{(2)}\left(\frac{D(\rho_B, \rho'_B)}{|\cos(\alpha)|}, \frac{D(\sigma_B, \sigma'_B)}{|\cos(\beta)|}\right), \quad (3.99)$$

with the function $f_\varphi^{(2)}$ defined piecewise by

$$f_\varphi^{(2)}(z, x) = \begin{cases} h_\varphi(z, x) & : q(z, x) \geq |\cos(\varphi)| \\ f_\varphi(x) & : q(z, x) \leq |\cos(\varphi)| \end{cases}, \quad (3.100)$$

in turn with h_φ defined such that

$$|\sin(\varphi)|\sqrt{1 - h_\varphi(z, x)^2} = \sqrt{1 - x^2} + |\cos(\varphi)|\sqrt{1 - z^2}, \quad (3.101)$$

f_φ as in (3.41), and q by

$$q(z, x) = zx - \sqrt{(1 - z^2)(1 - x^2)}. \quad (3.102)$$

The resulting key rate is

$$r \geq 1 - h\left[\frac{1}{2} + \frac{1}{2}g_\alpha \circ f_\varphi^{(2)}\left(\frac{|1 - 2\delta_z|}{|\cos(\alpha)|}, \frac{|1 - 2\delta_x|}{|\cos(\beta)|}\right)\right] - h(\delta_z). \quad (3.103)$$

We note that, provided $\sin(\varphi) \neq 0$, if $\frac{1}{2}\|Z_B\|_1 = 1$ and $\frac{1}{2}\|X_B\|_1 = 1$ then (3.98) implies $\frac{1}{2}\|W_B\|_1 = 1$. Thus, except in the pathological case where the z and x bases coincide, if $\delta_z = (1 - |\cos(\alpha)|)/2$ and $\delta_x = (1 - |\cos(\beta)|)/2$ (the minimum possible error rates) we certify that Bob is in full control of Alice's source space and we find $H(Z_A | E) = 1$ and $r \geq 1 - h(\delta_z)$ (i.e., the only reduction in the key rate is due to error correction).

We now prove (3.98). Our strategy will be to derive an upper bound on $\frac{1}{2}\|X_B\|_1$ given $\frac{1}{2}\|Z_B\|_1$ and $\frac{1}{2}\|W_B\|_1$, and then invert the resulting bound for $\frac{1}{2}\|W_B\|_1$. Where $\mathcal{H}_B$ is two dimensional, this is especially easy as all three operators are expressible as combinations of Pauli operators. We set $Z_B = \mathbf{z} \cdot \boldsymbol{\sigma}$, $X_B = \mathbf{x} \cdot \boldsymbol{\sigma}$, and $W_B = \mathbf{w} \cdot \boldsymbol{\sigma}$, with $\|\mathbf{z}\|, \|\mathbf{x}\|, \|\mathbf{w}\| \leq 1$. For simplicity of notation we will generally denote by e.g. a the norm $\|\mathbf{a}\|$ of a vector $\mathbf{a}$. Because $a = \|\mathbf{a}\| = \frac{1}{2}\|\mathbf{a} \cdot \boldsymbol{\sigma}\|_1$, deriving the desired bound is reduced to bounding x in terms of z and w . Using that $X_B = \cos(\varphi)Z_B + \sin(\varphi)W_B$, we have

$$\begin{aligned} x &= \frac{1}{2}\|X_B\|_1 \\ &= \frac{1}{2}\|\cos(\varphi)Z_B + \sin(\varphi)W_B\|_1 \\ &= \|\cos(\varphi)\mathbf{z} + \sin(\varphi)\mathbf{w}\|. \end{aligned} \quad (3.104)$$

Squaring this and developing,

$$x^2 = \frac{1}{2}(z^2 + w^2) + \frac{1}{2}\cos(2\varphi)(z^2 - w^2) + \sin(2\varphi)\mathbf{w} \cdot \mathbf{z}. \quad (3.105)$$

Now, while we are working with a *given* value of φ , we can arrive at a nontrivial bound on x by noting that $\frac{1}{2}\|\cos(\varphi)Z_B + \sin(\varphi)W_B\|_1$ must be less than 1 for *all* values of φ . The maximum value of (3.105), if φ is allowed to take any value, is given by

$$x_{\max}^2 = \frac{1}{2}(z^2 + w^2) + \sqrt{\frac{1}{4}(z^2 - w^2)^2 + (\mathbf{w} \cdot \mathbf{z})^2}. \quad (3.106)$$

Requiring that the right-hand side is less than 1, we find

$$(2 - z^2 - w^2)^2 \geq (z^2 - w^2)^2 + 4(\mathbf{w} \cdot \mathbf{z})^2, \quad (3.107)$$

which simplifies to

$$|\mathbf{w} \cdot \mathbf{z}| \leq \sqrt{(1 - z^2)(1 - w^2)}. \quad (3.108)$$

From this, we deduce that

$$x^2 \leq \frac{1}{2}(z^2 + w^2) + \frac{1}{2}\cos(2\varphi)(z^2 - w^2) + |\sin(2\varphi)|\sqrt{(1 - z^2)(1 - w^2)}, \quad (3.109)$$

or equivalently,

$$\sqrt{1 - x^2} \geq |\cos(\varphi)|\sqrt{1 - z^2} - |\sin(\varphi)|\sqrt{1 - w^2}. \quad (3.110)$$

Invoking the triangle inequality on (3.104), we also trivially have

$$x \leq |\cos(\varphi)|z + |\sin(\varphi)|w \quad (3.111)$$

and

$$x \geq \left| |\cos(\varphi)|z - |\sin(\varphi)|w \right|. \quad (3.112)$$

Inverting these, we find two lower bounds on w :

$$|\sin(\varphi)|\sqrt{1-w^2} \leq \sqrt{1-x^2} + |\cos(\varphi)|\sqrt{1-z^2} \quad (3.113)$$

and

$$|\sin(\varphi)|w \geq \left| x - |\cos(\varphi)|z \right|, \quad (3.114)$$

and the bound on w we are seeking is simply whichever is the stronger of the two, which depends on z and x . Specifically, we should use (3.113) if $zx - \sqrt{(1-z^2)(1-x^2)} \geq |\cos(\varphi)|$, and (3.114) otherwise.

Equation (3.113) is just a rearrangement of (3.98), and for sufficiently good bounds on z and x provides a better bound on $F(\rho_E, \rho'_E)$ than (3.77). Equation (3.114), by contrast, proves to be of little interest as in practice we have only lower bounds of the form $z \geq z_0$ and $x \geq x_0$ on z and x . Because (3.114) is always a decreasing function of either z or x , we cannot safely substitute lower bounds z_0 and x_0 in their place.

For the sake of completeness, we show here that (3.113) and (3.114) together are no more useful than (3.113) and (3.77) in this respect. First, note that we certify nothing if $x_0 \leq |\cos(\varphi)|$, as this permits $x = |\cos(\varphi)|$ and $z = 1$, for which both (3.113) and (3.114) reduce to $x \geq 0$. If $x_0 > |\cos(\varphi)|$, then (3.114) is minimised by maximising z (i.e. setting $z = 1$) and minimising x (i.e., setting $x = x_0$). The lower bound on w implied by (3.113) is minimised by minimising both z and x . In both cases we can safely substitute $x = x_0$. If z_0 is such that $z_0 x_0 - \sqrt{(1-z_0^2)(1-x_0^2)} \geq |\cos(\varphi)|$, then we simply use $z = z_0$ in (3.113). Otherwise, we use $z = |\cos(\varphi)|x_0 + |\sin(\varphi)|\sqrt{1-x_0^2}$ in either (3.113) or (3.114), which both reduce to $w \geq |\sin(\varphi)|x_0 - |\cos(\varphi)|\sqrt{1-x_0^2}$.

To summarise, given lower bounds z_0 and x_0 on $z = \frac{1}{2}\|Z_B\|_1$ and $x = \|X_B\|_1$, respectively, we have shown that

$$w = \frac{1}{2}\|W_B\|_1 \geq f_\varphi^{(2)}(z_0, x_0), \quad (3.115)$$

with the function $f_\varphi^{(2)}$ defined in (3.100) above. This concludes the derivation of (3.99).

3.A Convexity of entropy bound

The right-hand side of (3.22) has the form

$$H(F) = 1 + \phi(R) - \phi(F), \quad (3.116)$$

where, for convenience, we have set

$$R = \sqrt{\lambda + \mu F^2}, \quad (3.117)$$

$\lambda = (1 - 2\eta)^2$ and $\mu = 4\eta(1 - \eta)$ (such that $0 \leq \lambda, \mu \leq 1$ and $\lambda + \mu = 1$), and the function ϕ is defined by

$$\begin{aligned} \phi(x) &= h\left(\frac{1}{2} + \frac{1}{2}x\right) \\ &= 1 - \frac{1}{2}(1+x) \log(1+x) \\ &\quad - \frac{1}{2}(1-x) \log(1-x) \end{aligned} \quad (3.118)$$

for $-1 < x < 1$ and $\phi(1) = \phi(-1) = 0$.

If $\mu = 1$ (and $\lambda = 0$), (3.116) reduces to $H(F) = 1$. In the following we show that, for $\mu < 1$, H is a convex function by showing that its second derivative in F is nonnegative. Since its global minimum is $H(0) = 0$, it will follow that H is an increasing function over the range $0 \leq F \leq 1$.

We first evaluate the first and second derivatives of ϕ ; respectively, they are

$$\phi'(x) = -\frac{1}{2} \log\left(\frac{1+x}{1-x}\right) \quad (3.119)$$

and

$$\phi''(x) = -\frac{1}{\ln(2)} \frac{1}{1-x^2}. \quad (3.120)$$

For the first and second derivatives of R (viewed as a function of F), we obtain $R' = \mu F/R$ and $R'' = \lambda \mu/R^3$. In terms of ϕ and its derivatives and R , the first and second derivatives of H are

$$H'(F) = \phi'(R) \frac{\mu F}{R} - \phi'(F), \quad (3.121)$$

and

$$H''(F) = \phi''(R) \frac{\mu^2 F^2}{R^2} + \phi'(R) \frac{\lambda \mu}{R^3} - \phi''(F). \quad (3.122)$$

Using that $\phi''(F) = \mu \phi''(R)$ and that $\mu F^2 - R^2 = \mu - 1 = -\lambda$, (3.122) can be rearranged to

$$\begin{aligned} H''(F) &= \frac{\lambda \mu}{R^3} \left(-R \phi''(R) + \phi'(R) \right) \\ &= \frac{1}{\ln(2)} \frac{\lambda \mu}{R^3} \left[\frac{R}{1-R^2} - \frac{1}{2} \ln\left(\frac{1+R}{1-R}\right) \right] \\ &= \frac{1}{4 \ln(2)} \frac{\lambda \mu}{R^3} \left(Z - \frac{1}{Z} - 2 \ln(Z) \right), \end{aligned} \quad (3.123)$$

where we set $Z = (1 + R)/(1 - R)$ and we used that

$$\frac{4R}{1 - R^2} = \frac{(1 + R)^2 - (1 - R)^2}{(1 + R)(1 - R)} = Z - \frac{1}{Z}. \quad (3.124)$$

Finally, we note that $(\lambda\mu)/(4\ln(2)R^3) \geq 0$ and that, for $Z \geq 1$,

$$\begin{aligned} Z - \frac{1}{Z} - 2\ln(Z) &= \int_1^Z dz \left(1 + \frac{1}{z^2}\right) - 2 \int_1^Z dz \frac{1}{z} \\ &= \int_1^Z dz \left(1 - \frac{1}{z}\right)^2 \\ &\geq 0, \end{aligned} \quad (3.125)$$

which together imply $H''(F) \geq 0$.

From (3.121), we see that $H'(0) = 0$, confirming that $F = 0$ is at least a local extremum. Since H is convex, the only possibility is that $F = 0$ is, in fact, the global minimum, in turn implying that H is an increasing function of F over the range $0 \leq F \leq 1$.

Chapter 4

Semi-device-independent QKD

This chapter presents two semi-device-independent BB84-like protocols in which security is based on the estimation of a CHSH-like correlator and the assumption that Alice's source emits (a priori unknown) qubit states. The protocols differ in whether Alice or Bob perform the additional measurements needed to estimate the correlator. The first protocol, mostly reported in [27], differs only from the BB84 protocol in that Alice performs additional CHSH-type measurements. The estimated CHSH-type correlator value is used to bound the source characterisation angles that were introduced in section 3.5, which in the semi-device-independent protocol no longer need to be assumed a priori. In the second, the subject of an article in preparation [28], the CHSH-type correlator is estimated by measurements performed by Bob and serves both the purposes of testing the channel and (implicitly) characterising the source. Both protocols are proved secure against collective attacks.

4.1 Introduction

So-called *device-independent* cryptographic protocols aim to establish security or correct functioning based on limited or no assumptions on the detailed functioning of the devices used. The first proposed and most well known example is device-independent quantum key distribution (DIQKD) [25], in which an entanglement-based QKD protocol includes a CHSH Bell test to certify the absence of an adversary. A principal interest in device-independence is that it adds a level of self-certification to a protocol, in that

the security no longer depends on assumptions about the functioning of the devices (such as the source characterisation angles θ or φ , α , and β of the previous chapter). Faulty construction, for instance, or simply natural wear and degradation in performance over time is automatically detected during the execution of the protocol. This advantage comes at the cost of added complexity (DIQKD specifically requires entanglement as well as additional measurements compared with the BB84 protocol) and reduced tolerance to channel noise and losses.

Fully trusted and fully device-independent protocols represent two extremes on a range of possible approaches to cryptography. One may also consider intermediate, partially device-independent schemes. We have already seen an example with the BB84 protocol itself in the previous chapter, in that the main security bounds that were derived hold with minimal assumptions made about Bob's measurements. In this chapter, we will consider the extent to which a prepare-and-measure BB84-like protocol can be made device-independent.

It is clear that a prepare-and-measure protocol cannot be made fully device-independent, as this would in principle allow a source that could simply transmit any necessary classical information about Alice's choice of state preparation which could easily be intercepted and copied by an adversary. We will thus need to make an a priori assumption about the source. Specifically, we will consider how security can be certified in a QKD protocol if Alice's source is assumed to be two dimensional, as was first proposed by Pawłowski and Brunner [26].

We will outline two ways in which the BB84 protocol can be modified in such a way that its security (at least against collective attacks) can be proved if the source is assumed two-dimensional. Both are based on the estimation of a correlator, which we define by

$$S = \frac{1}{2} \sum_{axy} (-1)^{a+b+xy} P(b \mid axy), \quad (4.1)$$

in which $a, x \in \{0, 1\}$ are binary variables chosen by Alice corresponding to a choice of one of four possible source states, $y \in \{0, 1\}$ corresponds to a choice between one of two possible measurements, and $b \in \{0, 1\}$ is a binary measurement outcome. The correlator defined in (4.1) was presented as a dimension witness in [26] and can be viewed as a prepare-and-measure version of the CHSH Bell correlator [79].

The protocols differ in where the measurements are made and how the estimation of S is used. In section 4.3, we consider a modified version of the

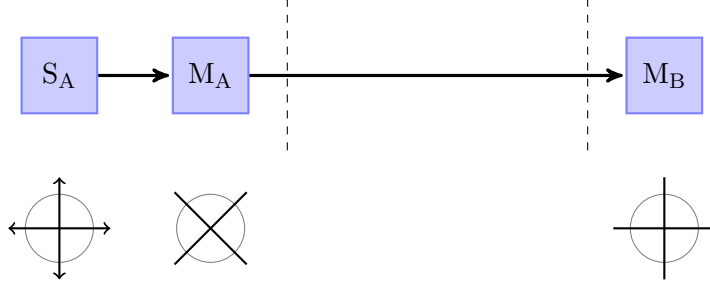


Figure 4.1: Semi-device-independent protocol with local CHSH-type test depicted with the *ideal* states and measurements. Alice’s source (S_A) can emit one of four (a priori unknown) qubit states. Some of the emitted states are intercepted by Alice’s measurement device (M_A) which performs the two measurements required for the local CHSH-type test. The remaining states are detected at Bob’s measurement device (M_B), which performs measurements which serve the same purpose as in the BB84 protocol.

BB84 protocol in which Alice estimates the value of the correlator (4.1) locally, measuring a fraction of the emitted source states rather than transmitting them to Bob, in order to obtain information about the source, as illustrated in figure 4.1. (The idea is similar to the entanglement-based DIQKD protocol based on a local Bell test proposed in [80].) Specifically, we will show that the expectation value of S is related to the angles φ , α , and β defined in section 3.5.1 by

$$|\cos(\alpha)||\cos(\beta)||\sin(\varphi)| \geq S^2/4 - 1. \quad (4.2)$$

This can then be used to complement the results obtained in section 3.5 for the BB84 protocol with the assumption of qubit source states, which depended explicitly on the angles φ , α , and β that are now constrained by the estimation of S according to (4.2).

In section 4.4, we consider a protocol more closely analogous to the DIQKD protocol considered in [25], in which Bob performs the measurements necessary for the estimation of S , as illustrated in figure 4.2. In this case, the CHSH-type test fulfills both the purposes of testing the channel and (indirectly) characterising the source, and the two measurements performed by Bob replace the x-basis measurement used in the BB84 protocol, which is no longer used. The main result we will derive in this case is an upper bound on the adversary’s ability to distinguish the z-basis states as measured by the trace distance:

$$D(\rho_E, \rho'_E) \leq \sqrt{2 - S^2/4}. \quad (4.3)$$

Applying this result to the expression (3.30) for the min-entropy derived in



Figure 4.2: Semi-device-independent QKD protocol with the CHSH-type correlator used as a channel test depicted with the *ideal* states and measurements. Alice's source (S_A) can emit one of four different qubit states. Bob's measurement device (M_B) performs three measurements: one measurement intended for key generation and two measurements intended for the CHSH-type estimation.

the previous chapter, one obtains a lower bound on the min-entropy,

$$H_{\min}(Z_A | E) \geq 1 - \log(1 + \sqrt{2 - S^2/4}), \quad (4.4)$$

that depends only on S and does not explicitly depend on any assumptions about the source states beyond that they are constrained to a qubit Hilbert space. This result is the subject of an article still in preparation at the time of writing [28].

In both cases, we will mainly concentrate on obtaining a lower bound on the conditional min-entropy for simplicity. We recall that the min-entropy provides a lower bound on the conditional von Neumann entropy which appears in the Devetak-Winter rate.

Before discussing the semi-device-independent protocols in more detail, it is worth stressing that the assumption of a two-dimensional source is necessary in order to prove security. For instance, the protocols are completely broken if the source emits the states

$$|\alpha\rangle = |0\rangle_B |0\rangle_E, \quad |\alpha'\rangle = |1\rangle_B |1\rangle_E, \quad (4.5)$$

$$|\beta\rangle = |+\rangle_B |+\rangle_E, \quad |\beta'\rangle = |-\rangle_B |-\rangle_E, \quad (4.6)$$

in which $|0\rangle$ and $|1\rangle$ are orthonormal and $|\pm\rangle = \frac{1}{\sqrt{2}}[|0\rangle \pm |1\rangle]$. These states span a three-dimensional space (one can readily verify that $|\alpha\rangle + |\alpha'\rangle = |\beta\rangle + |\beta'\rangle$) and always result in the adversary holding a perfect copy of whichever ideal BB84 state Bob received. The protocols are thus completely insecure if the source is allowed to be three dimensional.

4.2 Notation

We use the same notations as in section 3.5. The z-basis and x-basis states and corresponding density operators are noted $\rho = |\alpha\rangle\langle\alpha|$, $\rho' = |\alpha'\rangle\langle\alpha'|$, $\sigma = |\beta\rangle\langle\beta|$, and $\sigma' = |\beta'\rangle\langle\beta'|$. The Pauli operators Z and X are defined by

$$\begin{aligned}\rho - \rho' &= \cos(\alpha)Z, \\ \sigma - \sigma' &= \cos(\beta)X,\end{aligned}\tag{4.7}$$

with angles α and β , defined by $|\sin(\alpha)| = \langle\alpha|\alpha'\rangle$ and $|\sin(\beta)| = \langle\beta|\beta'\rangle$, measuring the nonorthogonality between the z and x states, respectively. Note that Z and X are traceless and have eigenvalues $+1$ and -1 . For convenience we also introduce the states $|z\rangle$ and $|z'\rangle$, defined such that

$$Z = |z\rangle\langle z| - |z'\rangle\langle z'|\tag{4.8}$$

with $\langle z|z'\rangle = 0$, as in section 3.5.

The measurements needed for the CHSH-type correlator estimation are represented by operators U and V given as the difference between to POVM elements

$$U = M - M',\tag{4.9}$$

$$V = N - N',\tag{4.10}$$

In section 4.4, where Bob performs the additional measurements, we add a subscript ‘B’ so that the operators are noted U_B and V_B .

4.3 Correlator as source characterisation

We now proceed to derive the source characterisation bound (4.2). First, we expand the sum over the indices a and b in order to insert the operators Z and X :

$$\begin{aligned}S &= \frac{1}{2} \sum_{xy} (-1)^{xy} (P(0 | 0xy) - P(1 | 0xy) - P(0 | 1xy) + P(1 | 1xy)) \\ &= \frac{1}{2} \cos(\alpha) \text{Tr}[UZ + VZ] + \frac{1}{2} \sin(\alpha) \text{Tr}[UX - VX], \\ &= \frac{1}{2} \text{Tr}[UZ_\alpha + VZ_\alpha + UX_\beta - VX_\beta],\end{aligned}\tag{4.11}$$

where, in the last line, we set $Z_\alpha = \cos(\alpha)Z$ and $X_\beta = \cos(\beta)X$ for convenience. Using the identity (1.126) described in section 1.B.1, which for a qubit Hilbert space is

$$\text{Tr}[AB] = 2\langle\Phi^+|A \otimes B^\top|\Phi^+\rangle, \quad (4.12)$$

we can rewrite S as the expectation value of an operator $\mathcal{S}$ in a $|\Phi^+\rangle$ state:

$$S = \langle\mathcal{S}\rangle_{\Phi^+} = \langle Z_\alpha \otimes U^\top + Z_\alpha \otimes V^\top + X_\beta \otimes U^\top - X_\beta \otimes V^\top \rangle_{\Phi^+}. \quad (4.13)$$

In order to obtain the desired source characterisation, we proceed in a manner analogous to a simple derivation of the CHSH inequality's Tsirelson bound [81] (its maximal quantum violation of $2\sqrt{2}$) given in [82, 83]. Specifically, we use that

$$S^2 = \langle\mathcal{S}\rangle_{\Phi^+}^2 \leq \langle\mathcal{S}^2\rangle_{\Phi^+}. \quad (4.14)$$

Inserting the expression for $\mathcal{S}$ in (4.13) and expanding $\mathcal{S}^2$, we find

$$\mathcal{S}^2 = \mathcal{O} - [Z_\alpha, X_\beta] \otimes [U^\top, V^\top], \quad (4.15)$$

where we have collected most of the terms into an operator $\mathcal{O}$ given by

$$\begin{aligned} \mathcal{O} = & (Z_\alpha^2 + X_\beta^2) \otimes ((U^\top)^2 + (V^\top)^2) + (Z_\alpha^2 - X_\beta^2) \otimes \{U, V\}^\top \\ & + \{Z_\alpha, X_\beta\} \otimes ((U^\top)^2 - (V^\top)^2), \end{aligned} \quad (4.16)$$

and we used the notation $[A, B] = AB - BA$, and $\{A, B\} = AB + BA$ for the commutator and anticommutator, respectively. Rearranging the resulting inequality for S and using that $-[U^\top, V^\top] = [U, V]^\top$,

$$\langle[Z_\alpha, X_\beta] \otimes [U, V]^\top\rangle_{\Phi^+} \geq S^2 - \langle\mathcal{O}\rangle_{\Phi^+}, \quad (4.17)$$

or, reapplying (4.12) to the left-hand side,

$$\frac{1}{2} \text{Tr}[[Z_\alpha, X_\beta][U, V]] \geq S^2 - \langle\mathcal{O}\rangle_{\Phi^+}. \quad (4.18)$$

In order to make use of (4.18), we need to determine an upper bound on $\langle\mathcal{O}\rangle_{\Phi^+}$. Specifically, we find that $\langle\mathcal{O}\rangle_{\Phi^+} \leq 4$. In order to show this, it is sufficient to show that $\mathcal{O}$ satisfies the operator inequality $\mathcal{O} \leq 4\mathbb{1}$. Rearranging the terms in (4.16),

$$\begin{aligned} \mathcal{O} = & (Z_\alpha + X_\beta)^2 \otimes (U^\top)^2 + (Z_\alpha - X_\beta)^2 \otimes (V^\top)^2 \\ & + (Z_\alpha^2 - X_\beta^2) \otimes \{U, V\}^\top \\ \leq & (Z_\alpha + X_\beta)^2 \otimes \mathbb{1} + (Z_\alpha - X_\beta)^2 \otimes \mathbb{1} + |Z_\alpha^2 - X_\beta^2| \otimes 2\mathbb{1} \\ = & 2(Z_\alpha^2 + X_\beta^2 + |Z_\alpha^2 - X_\beta^2|) \otimes \mathbb{1}. \end{aligned} \quad (4.19)$$

Because the operators Z_α and X_β are traceless, we can express them as linear combinations

$$Z_\alpha = \mathbf{z} \cdot \boldsymbol{\sigma}, \quad X_\beta = \mathbf{x} \cdot \boldsymbol{\sigma} \quad (4.20)$$

of Pauli operators, for vectors $\mathbf{z}$ and $\mathbf{x}$ with $\|\mathbf{z}\|, \|\mathbf{x}\| \leq 1$. Using this,

$$\begin{aligned} Z_\alpha^2 + X_\beta^2 + |Z_\alpha^2 - X_\beta^2| &= (\|\mathbf{z}\|^2 + \|\mathbf{x}\|^2 + \|\mathbf{z}\|^2 - \|\mathbf{x}\|^2) \mathbb{1} \\ &\leq 2\mathbb{1}, \end{aligned} \quad (4.21)$$

which, applied to (4.19), implies $\mathcal{O} \leq 4\mathbb{1}$.

We have thus arrived at

$$\frac{1}{2} \text{Tr}[[Z, X][U, V]] \geq S^2 - 4. \quad (4.22)$$

Using now that $\|[U, V]\| \leq 2\mathbb{1}$, and using that for arbitrary operators A and B , $\text{Tr}[AB] \leq \|A\|_1 \|B\|_\infty$, we obtain a constraint involving only $[Z_\alpha, X_\beta]$:

$$\|[Z_\alpha, X_\beta]\|_1 \geq S^2 - 4. \quad (4.23)$$

Finally, using that $[Z_\alpha, X_\beta] = 2i(\mathbf{z} \times \mathbf{x}) \cdot \boldsymbol{\sigma}$, we arrive at

$$\|\mathbf{z} \times \mathbf{x}\| \geq S^2/4 - 1. \quad (4.24)$$

Recalling that $\|\mathbf{z}\| = |\cos(\alpha)|$ and $\|\mathbf{x}\| = |\cos(\beta)|$ and that $\|\mathbf{z} \times \mathbf{x}\| = \|\mathbf{z}\| \|\mathbf{x}\| |\sin(\varphi)|$, we obtain the alternative expression

$$|\cos(\alpha)| |\cos(\beta)| |\sin(\varphi)| \geq S^2/4 - 1 \quad (4.25)$$

already given above.

Note that, if S attains its maximal value of $2\sqrt{2}$, this implies $|\cos(\alpha)| = |\cos(\beta)| = |\sin(\varphi)| = 1$, which would certify that Alice was emitting states satisfying the ideal BB84 relations. More generally, any value $S > 2$ will provide some nontrivial characterisation of the source states.

The source characterisation derived here can be combined with an analysis of the sort already done in sections 3.5.1 or 3.5.2 in order to obtain an asymptotic security result. The simplest approach is to derive a lower bound on the min-entropy, as this depends on a trace-distance bound which does not require the “penalty” function g_α derived in section 3.5.1 for the fidelity. In particular, the distinguishability of Eve’s marginal z-basis states can be bounded by

$$D(\rho_E, \rho'_E) \leq |\cos(\varphi)| D(\sigma_B, \sigma'_B) + |\sin(\varphi)| \sqrt{1 - D(\sigma_B, \sigma'_B)^2}, \quad (4.26)$$

which can be used in the expression $H_{\min}(Z_A | E) = 1 - \log(1 - D(\rho_E, \rho'_E))$ for the min-entropy and with $|\sin(\varphi)| \geq S^2/4 - 1$ substituted in as a bound on the angle φ . Note that the use of (4.26) produces a better result than was originally reported in [27].

The fidelity, and thus the conditional von Neumann entropy, could likewise be lower bounded by, for instance, minimising the right-hand side of the inequality

$$F(\rho_E, \rho'_E) \geq g_\alpha \circ f_\varphi(D(\sigma_B, \sigma'_B)) \quad (4.27)$$

subject to $|\sin(\alpha)||\cos(\varphi)| \geq S^2/4 - 1$. This proved difficult to solve analytically, however the minimisation can easily be solved numerically if desired.

4.4 Correlator as channel test

4.4.1 Outline

In this section we describe how to obtain the bound

$$D(\rho_E, \rho'_E) \leq \sqrt{2 - S^2/4} \quad (4.28)$$

on Eve's ability to distinguish the z-basis states, as measured by the trace distance, mentioned in this chapter's introduction. Combined with the expression given in section 3.2.3, this implies the lower bound

$$H_{\min}(Z_A | E) \geq 1 - \log(1 + \sqrt{2 - S^2/4}). \quad (4.29)$$

for the min-entropy. We note that a relation with the same form as (4.29) was first derived in the context of DIQKD in [84] (and earlier for device-independent randomness generation in the supplementary information to [85]). The main result of this section can be viewed as the analogous result for the prepare-and-measure scenario.

Because the derivation is somewhat lengthy, we begin by briefly outlining the main ideas and approach followed. The starting point is the CHSH-type correlator, whose expression,

$$S = \frac{1}{2} \text{Tr}[(\cos(\alpha)Z_B + \cos(\beta)X_B)U_B + (\cos(\alpha)Z_B - \cos(\beta)X_B)V_B], \quad (4.30)$$

is similar to (4.11). The only difference is that, because the measurements are performed by Bob, it is the partial traces Z_B and X_B of Z and X that appear in (4.30). The main idea in the approach is to reexpress (4.30) in terms of a new orthonormal basis (the “y basis”) $\{|y\rangle, |y'\rangle\}$ defined such that

the operator $Y = |y\rangle\langle y| - |y'\rangle\langle y'|$ is orthogonal to both Z and X on the Bloch sphere. In the special case where Bob is restricted to qubit measurements, and if we consider the case $|\cos(\alpha)| = |\cos(\beta)| = 1$ for simplicity, it is not too difficult to show that the bound

$$\frac{1}{2}\|Y_B\|_1 \geq \sqrt{S^2/4 - 1} \quad (4.31)$$

holds by constructing a Hermitian unitary operator W_B with the property that $\frac{1}{2}\text{Tr}[W_B Y_B] \geq \sqrt{S^2/4 - 1}$ (recalling that, for any such operator, $\frac{1}{2}\|Y_B\|_1 \geq \frac{1}{2}\text{Tr}[W_B Y_B]$). Recalling that the trace distance can alternatively be expressed as $D(\rho_E, \rho'_E) = |\cos(\alpha)|\frac{1}{2}\|Z_E\|_1 \leq \|Z_E\|_1$, the upper bound (4.28) follows by applying a version of the bound (3.35) derived in section 3.3 for ideal BB84-type source states, which we express here as

$$\frac{1}{4}\|Y_B\|_1^2 + \frac{1}{4}\|Z_E\|_1^2 \leq 1. \quad (4.32)$$

The lower bound on $\frac{1}{2}\|Y_B\|_1$ also lends itself to lower bounding the conditional von Neumann entropy. For instance, if the z-basis states are assumed to be orthogonal, the relation $F(\rho_E, \rho'_E) \geq \frac{1}{2}\|Y_B\|_1$ holds and can be used to obtain the lower bound

$$H(Z_A | E) \geq 1 - h\left(\frac{1}{2} + \frac{1}{2}\sqrt{S^2/4 - 1}\right), \quad (4.33)$$

which has the same form as the security bound obtained for the DIQKD protocol against collective attacks in [25].

In order to show that the trace-distance bound (4.28) still holds if Bob's measurements are *not* restricted to two dimensions, we will use the Jordan lemma [86] to break the problem into considering orthogonal two-dimensional subspaces on Bob's side. The derivation we will follow will be more complicated than that outlined above in two ways. First, while we will obtain an analogue of (4.31) in each qubit subspace, the function $\sqrt{S^2/4 - 1}$ is unfortunately concave in S (it would need to be convex in order for (4.31) – a lower bound – to hold in general). Because of this, (4.31) does not hold if Bob's measurements are more than two dimensional, and in general the best bound that can be obtained is the linear interpolation $\frac{1}{2}\|Y_B\|_1 \geq (S - 2)/(2\sqrt{2} - 2)$. Second, while our main goal is the trace-distance bound, in order to keep open the possibility of deriving a useful lower bound on the fidelity we will also explicitly keep track of the factors $|\cos(\alpha)|$ and $|\cos(\beta)|$. Rather than deriving (4.31), we will show that, for each two-dimensional subspace (indexed by a variable k), there is a Hermitian unitary operator $W_k = W_B^k \otimes \mathbb{1}_E$ such that

$$\frac{1}{2p_k} \text{Tr}[W_B^k Y_B] \geq \frac{\sqrt{(S_k/p_k)^2/4 - 1}}{|\cos(\alpha)|}, \quad (4.34)$$

where S_k is the contribution to S from the k th subspace (with $\sum_k S_k = S$) and $p_k = \frac{1}{2} \text{Tr}[\mathbb{1}_B^k \mathcal{I}_B]$ is a probabilistic weight associated with the k th subspace (with $\sum_k p_k = 1$), defined in terms of the identity $\mathbb{1}_B^k$ in the k th subspace and $\mathcal{I}_B = \text{Tr}_E[\mathcal{I}]$, where $\mathcal{I} = |z\rangle\langle z| + |z'\rangle\langle z'|$ is the identity in Alice's source space. Finally, we will use concavity to get from (4.34) to the trace-distance bound (4.28).

If we restrict ourselves to the case of qubit measurements on Bob's side, (4.34) simplifies to $\frac{1}{2}\|Y_B\|_1 \geq \sqrt{S^2/4 - 1}|\cos(\alpha)|$. Applying the results of section 3.5.1, it is possible to obtain a lower bound on the fidelity, which we express as

$$F(\rho_E, \rho'_E) \geq g_*(\sqrt{S^2/4 - 1}). \quad (4.35)$$

The function g_* is defined in terms of the function g_α derived in section 3.5.1 by $g_*(x) = \min_\alpha g_\alpha(x/|\cos(\alpha)|)$, the minimisation over α reflecting the fact that α is treated as a priori unknown in the semi-device-independent protocol we are considering. A precise characterisation of g_* is given in appendix 4.C.

It is probably the case that the resulting bound on the conditional von Neumann entropy,

$$H(Z_A | E) \geq 1 - h\left[\frac{1}{2} + \frac{1}{2}g_*(\sqrt{S^2/4 - 1})\right], \quad (4.36)$$

or a bound resembling it still holds in general (i.e., if the restriction to qubit measurements is removed). This proved difficult to show in general, however as a partial result in this direction, in appendix 4.A we show that the bound $H(Z_A | E) \geq 1 - h\left(\frac{1}{2} + \frac{1}{2}\sqrt{S^2/4 - 1}\right)$ holds in the special case where the z -basis states are assumed to be orthogonal.

4.4.2 Derivation of qubit y -basis bound

In this section we derive the intermediate result (4.34) from the outline above. After breaking the problem into qubit subspaces, the approach followed will be to determine an upper bound on the contribution S_k to S from each qubit subspace k in terms of the quantities appearing in (4.34), and then to rearrange the final result.

Anticipating that the final bound (4.28) we will obtain is a decreasing function of S , and because Eve's information about the z -basis states does not depend on Bob's measurements, we will consider the maximal value of S for fixed source states and a fixed unitary attack. This can be expressed as

$$S = \frac{1}{2}\|\cos(\alpha)Z_B + \cos(\beta)X_B\|_1 + \frac{1}{2}\|\cos(\alpha)Z_B - \cos(\beta)X_B\|_1, \quad (4.37)$$

which is attained if U_B and V_B in (4.30) are Hermitian unitary operators corresponding to the projective measurements which maximise (4.30).

Since U_B and V_B are Hermitian unitaries, they admit a common block diagonalisation

$$U_B = \bigoplus_k U_B^k, \quad V_B = \bigoplus_k V_B^k, \quad (4.38)$$

in which U_B^k and V_B^k are still Hermitian and unitary and of dimension at most 2, $\forall k$ (the Jordan lemma, see Lemma 2 of [87] for a short proof). For each subspace k , we define the corresponding contribution to S by

$$\begin{aligned} S_k &= \frac{1}{2} \cos(\alpha) \text{Tr}[(U_B^k + V_B^k)Z_B] + \frac{1}{2} \cos(\beta) \text{Tr}[(U_B^k - V_B^k)X_B] \\ &= \frac{1}{2} \cos(\alpha) \text{Tr}[(U_k + V_k)Z] + \frac{1}{2} \cos(\beta) \text{Tr}[(U_k - V_k)X], \end{aligned} \quad (4.39)$$

such that $S = \sum_k S_k$, where we have set $U_k = U_B^k \otimes \mathbb{1}_E$ and $V_k = V_B^k \otimes \mathbb{1}_E$ in the second line.

If Z and X are separated by an angle φ on the Bloch sphere, then there is an orthonormal basis $\{|y\rangle, |y'\rangle\}$ (the “ y basis” alluded to in the outline) in which Z and X take the expressions

$$Z = e^{i\frac{\varphi}{2}} |y\rangle\langle y'| + e^{-i\frac{\varphi}{2}} |y'\rangle\langle y|, \quad (4.40)$$

$$X = e^{-i\frac{\varphi}{2}} |y\rangle\langle y'| + e^{i\frac{\varphi}{2}} |y'\rangle\langle y|. \quad (4.41)$$

Note that Z and X are orthogonal to the operator $Y = |y\rangle\langle y| - |y'\rangle\langle y'|$. The correct relations between these operators can be verified by checking that $\{Z, Y\} = \{X, Y\} = 0$, $[Z, X] = 2i \sin(\varphi)Y$, and $\{Z, X\} = \cos(\varphi)\mathcal{I}$ where $\mathcal{I} = |y\rangle\langle y| + |y'\rangle\langle y'|$ is the identity in the qubit source space. In the basis of y states, S_k becomes

$$\begin{aligned} S_k &= \cos(\alpha) \text{Re}[\langle y|e^{-i\frac{\varphi}{2}}(U_k + V_k)|y'\rangle] \\ &\quad + \cos(\beta) \text{Re}[\langle y|e^{i\frac{\varphi}{2}}(U_k - V_k)|y'\rangle]. \end{aligned} \quad (4.42)$$

In each subspace k , the only interesting case is where both U_B^k and V_B^k are two-dimensional and of eigenvalues $+1$ and -1 . In this case, if U_B^k and V_B^k are separated by an angle γ_k on the Bloch sphere, one can choose an orthonormal basis $\{|w_k\rangle_B, |w'_k\rangle_B\}$ in which

$$U_B^k + V_B^k = 2 \cos\left(\frac{\gamma_k}{2}\right) (|w_k\rangle\langle w'_k|_B + |w'_k\rangle\langle w_k|_B), \quad (4.43)$$

$$U_B^k - V_B^k = -2 \sin\left(\frac{\gamma_k}{2}\right) (-i|w_k\rangle\langle w'_k|_B + i|w'_k\rangle\langle w_k|_B). \quad (4.44)$$

Inserting this into (4.42), we obtain

$$S_k/2 = \text{Re}[\langle y|(\lambda_k |w_k\rangle\langle w'_k|_B + \mu_k |w'_k\rangle\langle w_k|_B) \otimes \mathbb{1}_E |y'\rangle], \quad (4.45)$$

where we have collected the various angles into

$$\begin{aligned}\lambda_k &= \cos(\alpha) \cos\left(\frac{\gamma_k}{2}\right) e^{-i\frac{\varphi}{2}} + i \cos(\beta) \sin\left(\frac{\gamma_k}{2}\right) e^{i\frac{\varphi}{2}}, \\ \mu_k &= \cos(\alpha) \cos\left(\frac{\gamma_k}{2}\right) e^{-i\frac{\varphi}{2}} - i \cos(\beta) \sin\left(\frac{\gamma_k}{2}\right) e^{i\frac{\varphi}{2}}\end{aligned}\quad (4.46)$$

for convenience. Introducing now the vectors

$$|A_k\rangle = (\langle w_k|_B \otimes \mathbb{1}_E)|y\rangle, \quad |A'_k\rangle = (\langle w'_k|_B \otimes \mathbb{1}_E)|y'\rangle, \quad (4.47)$$

$$|B_k\rangle = (\langle w'_k|_B \otimes \mathbb{1}_E)|y\rangle, \quad |B'_k\rangle = (\langle w_k|_B \otimes \mathbb{1}_E)|y'\rangle \quad (4.48)$$

(in $\mathcal{H}_E$), the expression for S_k can be simplified to

$$S_k/2 = \text{Re}[\lambda_k \langle A_k|A'_k\rangle + \mu_k \langle B_k|B'_k\rangle]. \quad (4.49)$$

Finally, in order to reexpress S_k in a form better suited for determining an upper bound, we introduce new coefficients

$$\xi_k = \cos\left(\frac{\gamma_k + \varphi}{2}\right) + i \sin\left(\frac{\gamma_k - \varphi}{2}\right), \quad (4.50)$$

$$\nu_k = \cos\left(\frac{\gamma_k - \varphi}{2}\right) - i \sin\left(\frac{\gamma_k + \varphi}{2}\right), \quad (4.51)$$

and

$$c_{\pm} = \frac{1}{2}(\cos(\alpha) \pm \cos(\beta)), \quad (4.52)$$

such that

$$\lambda_k = c_+ \xi_k + c_- \nu_k, \quad (4.53)$$

$$\mu_k = c_+ \nu_k + c_- \xi_k. \quad (4.54)$$

Inserting these into the expression for S_k ,

$$\begin{aligned}S_k/2 &= \text{Re}[\xi_k (c_+ \langle A_k|A'_k\rangle + c_- \langle B_k|B'_k\rangle) \\ &\quad + \nu_k (c_- \langle A_k|A'_k\rangle + c_+ \langle B_k|B'_k\rangle)].\end{aligned}\quad (4.55)$$

In order to obtain a useful upper bound on S_k , we begin by taking the absolute value of the various terms in (4.55), obtaining

$$\begin{aligned}S_k/2 &\leq |\xi_k| (|c_+| |\langle A_k|A'_k\rangle| + |c_-| |\langle B_k|B'_k\rangle|) \\ &\quad + |\nu_k| (|c_-| |\langle A_k|A'_k\rangle| + |c_+| |\langle B_k|B'_k\rangle|).\end{aligned}\quad (4.56)$$

Applying the Cauchy-Schwarz inequality, using that $|\xi_k|^2 + |\nu_k|^2 = 2$, and developing,

$$\begin{aligned}S_k^2/4 &\leq 2 \left(|c_+| |\langle A_k|A'_k\rangle| + |c_-| |\langle B_k|B'_k\rangle| \right)^2 \\ &\quad + 2 \left(|c_-| |\langle A_k|A'_k\rangle| + |c_+| |\langle B_k|B'_k\rangle| \right)^2 \\ &= 2(c_+^2 + c_-^2) \left(|\langle A_k|A'_k\rangle|^2 + |\langle B_k|B'_k\rangle|^2 \right) \\ &\quad + 8|c_+ c_-| |\langle A_k|A'_k\rangle| |\langle B_k|B'_k\rangle| \\ &\leq 2(c_+^2 + c_-^2) \left(\|A_k\|^2 \|A'_k\|^2 + \|B_k\|^2 \|B'_k\|^2 \right) \\ &\quad + 8|c_+ c_-| \|A_k\| \|A'_k\| \|B_k\| \|B'_k\|,\end{aligned}\quad (4.57)$$

where we used the Cauchy-Schwarz inequality again to substitute $|\langle A_k | A'_k \rangle| \leq \|A_k\| \|A'_k\|$ and $|\langle B_k | B'_k \rangle| \leq \|B_k\| \|B'_k\|$. Using now that, e.g., $2\|A_k\| \|A'_k\| \leq \|A_k\|^2 + \|A'_k\|^2$,

$$S_k^2/4 \leq \frac{1}{2}(c_+^2 + c_-^2) [(\|A_k\|^2 + \|A'_k\|^2)^2 + (\|B_k\|^2 + \|B'_k\|^2)^2] + 2|c_+c_-|(\|A_k\|^2 + \|A'_k\|^2)(\|B_k\|^2 + \|B'_k\|^2). \quad (4.58)$$

We now reintroduce the definitions of the vectors $|A_k\rangle$, $|A'_k\rangle$, $|B_k\rangle$, and $|B'_k\rangle$. We first note that

$$\|A_k\|^2 + \|A'_k\|^2 = \frac{1}{2} \text{Tr}[\mathbb{1}_B^k \mathcal{I}_B] + \frac{1}{2} \text{Tr}[W_B^k Y_B], \quad (4.59)$$

$$\|B_k\|^2 + \|B'_k\|^2 = \frac{1}{2} \text{Tr}[\mathbb{1}_B^k \mathcal{I}_B] - \frac{1}{2} \text{Tr}[W_B^k Y_B], \quad (4.60)$$

where we recall that we defined

$$\mathcal{I} = |y\rangle\langle y| + |y'\rangle\langle y'|, \quad (4.61)$$

$$Y = |y\rangle\langle y| - |y'\rangle\langle y'|, \quad (4.62)$$

$\mathcal{I}_B = \text{Tr}_E[\mathcal{I}]$, $Y_B = \text{Tr}_E[Y]$, and we have introduced

$$\mathbb{1}_B^k = |w_k\rangle\langle w_k|_B + |w'_k\rangle\langle w'_k|_B, \quad (4.63)$$

$$W_B^k = |w_k\rangle\langle w_k|_B - |w'_k\rangle\langle w'_k|_B. \quad (4.64)$$

Inserting (4.59) and (4.60) into (4.58) gets us

$$S_k^2/4 \leq (c_+^2 + c_-^2) \left(\frac{1}{4} \text{Tr}[\mathbb{1}_B^k \mathcal{I}_B]^2 + \frac{1}{4} \text{Tr}[W_B^k Y_B]^2 \right) + 2|c_+c_-| \left(\frac{1}{4} \text{Tr}[\mathbb{1}_B^k \mathcal{I}_B]^2 - \frac{1}{4} \text{Tr}[W_B^k Y_B]^2 \right), \quad (4.65)$$

and, simplifying, we arrive at

$$S_k^2/4 \leq \min(\cos(\alpha)^2, \cos(\beta)^2) \frac{1}{4} \text{Tr}[W_B^k Y_B]^2 + \max(\cos(\alpha)^2, \cos(\beta)^2) \frac{1}{4} \text{Tr}[\mathbb{1}_B^k \mathcal{I}_B]^2. \quad (4.66)$$

Since the x basis is only used for the purpose of testing the channel, we use that $\min(\cos(\alpha)^2, \cos(\beta)^2) \leq \cos(\alpha)^2$ and $\max(\cos(\alpha)^2, \cos(\beta)^2) \leq 1$ to eliminate the factor $\cos(\beta)$ from the above upper bound, obtaining

$$S_k^2/4 \leq \cos(\alpha)^2 \frac{1}{4} \text{Tr}[W_B^k Y_B]^2 + p_k^2, \quad (4.67)$$

where we have set $p_k = \frac{1}{2} \text{Tr}[\mathbb{1}_B^k \mathcal{I}_B]$. Rearranging (4.67), we arrive at the intermediate result (4.34) asserted in the outline. (If $\frac{1}{2} \text{Tr}[W_B^k Y_B]$ is negative, we simply redefine $W_B^k \mapsto -W_B^k$.)

In the special case where Bob's measurements are two dimensional, the index k is redundant, $\frac{1}{2} \text{Tr}[\mathbb{1}_B \mathcal{I}_B] = 1$, $\frac{1}{2} \text{Tr}[W_B^k Y_B] \leq \frac{1}{2} \|Y_B\|_1$, and we arrive at the lower bound

$$\frac{1}{2} \|Y_B\|_1 \geq \frac{\sqrt{S^2/4 - 1}}{|\cos(\alpha)|} \quad (4.68)$$

for $\frac{1}{2}\|Y_B\|_1$, from which the lower bound (4.35) given for the fidelity in the outline follows after minimisation over α .

Finally, before deriving the trace-distance bound, we note that, because $\frac{1}{2}\text{Tr}[W_B^k Y_B] \leq p_k$, S_k as given in (4.66) is upper bounded by

$$S_k \leq 2\sqrt{\cos(\alpha)^2 + \cos(\beta)^2} p_k. \quad (4.69)$$

Summing over k , we obtain the correct tight upper bound for S ,

$$S \leq 2\sqrt{\cos(\alpha)^2 + \cos(\beta)^2}, \quad (4.70)$$

which indicates that the bound (4.34) is tight.

4.4.3 Trace-distance bound

The derivation of the trace-distance bound (4.28) is done in a manner similar to the derivation of the “information gain” for the BB84 protocol given in section 1.4.4 in the introduction. We set U_E (not to be confused with U_B from the previous subsection) to be the Hermitian unitary such that $\frac{1}{2}\|Z_E\|_1 = \frac{1}{2}\text{Tr}[U_E Z_E]$, which we decompose into its positive and negative projections P_E and Q_E such that $P_E - Q_E = U_E$ and $P_E + Q_E = \mathbb{1}_E$. Using that Y can be expressed as $Y = e^{i\frac{\varphi}{2}}|z\rangle\langle z'| + e^{-i\frac{\varphi}{2}}|z'\rangle\langle z|$ in terms of the z basis states and developing,

$$\begin{aligned} \frac{1}{2}\text{Tr}[W_B^k Y_B] &= \text{Re}[e^{-i\frac{\varphi}{2}}\langle z|W_B^k \otimes \mathbb{1}_E|z'\rangle] \\ &= \text{Re}[e^{-i\frac{\varphi}{2}}\langle z|W_B^k \otimes P_E|z'\rangle] + \text{Re}[e^{-i\varphi}\langle z|W_B^k \otimes Q_E|z'\rangle] \\ &\leq |\langle z|W_B^k \otimes P_E|z'\rangle| + |\langle z|W_B^k \otimes Q_E|z'\rangle| \\ &\leq \sqrt{\langle z|\mathbb{1}_B^k \otimes P_E|z\rangle}\sqrt{\langle z'|\mathbb{1}_B^k \otimes P_E|z'\rangle} \\ &\quad + \sqrt{\langle z|\mathbb{1}_B^k \otimes Q_E|z\rangle}\sqrt{\langle z'|\mathbb{1}_B^k \otimes Q_E|z'\rangle} \\ &\leq \sqrt{\langle z|\mathbb{1}_B^k \otimes P_E|z\rangle + \langle z'|\mathbb{1}_B^k \otimes Q_E|z'\rangle} \\ &\quad \times \sqrt{\langle z'|\mathbb{1}_B^k \otimes P_E|z'\rangle + \langle z|\mathbb{1}_B^k \otimes Q_E|z\rangle} \\ &= \sqrt{p_k + \frac{1}{2}\text{Tr}[(\mathbb{1}_B^k \otimes U_E)Z]}\sqrt{p_k - \frac{1}{2}\text{Tr}[(\mathbb{1}_B^k \otimes U_E)Z]} \\ &= \sqrt{p_k^2 - \frac{1}{4}\text{Tr}[(\mathbb{1}_B^k \otimes U_E)Z]^2}, \end{aligned} \quad (4.71)$$

which rearranges to a per-qubit-subspace version of the bound (4.32) cited in the outline for BB84-type states:

$$\frac{1}{4}\text{Tr}[W_B^k Y_B]^2 + \frac{1}{4}\text{Tr}[(\mathbb{1}_B^k \otimes U_E)Z]^2 \leq p_k^2. \quad (4.72)$$

Rearranging again and inserting the lower bound (4.67) (with the replacement $\cos(\alpha)^2 \leq 1$),

$$\frac{1}{4} \text{Tr}[(\mathbf{1}_B^k \otimes U_E)Z]^2 \leq 2p_k^2 - S_k^2/4, \quad (4.73)$$

or

$$\frac{1}{2} \text{Tr}[(\mathbf{1}_B^k \otimes U_E)Z] \leq p_k \sqrt{2 - (S_k/p_k)^2/4}. \quad (4.74)$$

Finally, summing over k and using that the function $x \mapsto \sqrt{2 - x^2/4}$ is concave,

$$\begin{aligned} \frac{1}{2} \|Z_B\|_1 &= \sum_k \frac{1}{2} \text{Tr}[(\mathbf{1}_B^k \otimes U_E)Z] \\ &\leq \sum_k p_k \sqrt{2 - (S_k/p_k)^2/4} \\ &\leq \sqrt{2 - (\sum_k S_k)^2/4} \\ &= \sqrt{2 - S^2/4}. \end{aligned} \quad (4.75)$$

The trace-distance bound $D(\rho_E, \rho'_E) \leq \sqrt{2 - S^2/4}$ then follows from the fact that $D(\rho_E, \rho'_E) = |\cos(\alpha)| \frac{1}{2} \|Z_B\|_1 \leq \frac{1}{2} \|Z_B\|_1$.

4.4.4 Optimal collective attack

The trace-distance bound (4.28) is tight and is attained with a prepare-and-measure version of the optimal collective attack described in [25], which we describe here. We set the z states to

$$|\alpha\rangle = |0\rangle_B |\psi\rangle_E, \quad |\alpha'\rangle = |1\rangle_B |\psi'\rangle_E, \quad (4.76)$$

in which $|0\rangle_B$ and $|1\rangle_B$ are orthonormal and $|\psi\rangle_E$ and $|\psi'\rangle_E$ are normalised and we set $\langle\psi|\psi'\rangle = F_z$ for some real constant $0 \leq F_z \leq 1$. We define the x states in terms of the z states by

$$|\beta\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle + |\alpha'\rangle), \quad (4.77)$$

$$|\beta'\rangle = \frac{1}{\sqrt{2}}(|\alpha\rangle - |\alpha'\rangle). \quad (4.78)$$

With these definitions, $|\alpha\rangle$, $|\alpha'\rangle$, $|\beta\rangle$, and $|\beta'\rangle$ span a qubit subspace. Note that $\langle\alpha|\alpha'\rangle = \langle\beta|\beta'\rangle = 0$, such that $|\cos(\alpha)| = |\cos(\beta)| = 1$.

From the expression we gave for the z states, $\rho'_E = |\psi\rangle\langle\psi|_E$ and $\rho_E = |\psi'\rangle\langle\psi'|_E$, and

$$D(\rho_E, \rho'_E) = \sqrt{1 - F_z^2}. \quad (4.79)$$

For the operators $Z = |\alpha\rangle\langle\alpha| - |\alpha'\rangle\langle\alpha'|$ and $X = |\beta\rangle\langle\beta| - |\beta'\rangle\langle\beta'| = |\alpha\rangle\langle\alpha'| + |\alpha'\rangle\langle\alpha|$, we find the partial traces

$$Z_B = |0\rangle\langle 0|_B - |1\rangle\langle 1|_B = \sigma_z \quad (4.80)$$

and

$$X_B = F_z(|0\rangle\langle 1|_B + |1\rangle\langle 0|_B) = F_z\sigma_x. \quad (4.81)$$

With the optimal measurements on Bob's side,

$$\begin{aligned} S &= \frac{1}{2}\|Z_B + X_B\|_1 + \frac{1}{2}\|Z_B - X_B\|_1 \\ &= \frac{1}{2}\|\sigma_z + F_z\sigma_x\|_1 + \frac{1}{2}\|\sigma_z - F_z\sigma_x\|_1 \\ &= 2\sqrt{1 + F_z^2}, \end{aligned} \quad (4.82)$$

which rearranges to

$$F_z = \sqrt{S^2/4 - 1}. \quad (4.83)$$

Substituting this into (4.79), we find that $D(\rho_E, \rho'_E) = \sqrt{2 - S^2/4}$.

4.5 Comparison for depolarising channel

The two protocols studied here have different advantages. Specifically, the version in which Alice performs the CHSH-type correlator estimation requires a more complicated implementation, since both Alice and Bob must be able to perform measurements and Alice must be able to select between measuring her states and transmitting them to Bob, but may better tolerate channel noise. This is illustrated here in the case of a depolarising channel in which any state $|\psi\rangle$ transmitted by Alice to Bob is mixed with white noise according to

$$|\psi\rangle\langle\psi| \mapsto (1 - p)|\psi\rangle\langle\psi| + p\mathbb{1} \quad (4.84)$$

for some parameter p . For simplicity, we assume the implementation is otherwise perfect, i.e., the source states perfectly satisfy the BB84 relations and Bob and Alice perform any measurements ideally, and we compare the key-rate bounds obtained via the min-entropy.

In the first protocol, in which Alice performs the CHSH measurements, the ideal result $S = 2\sqrt{2}$ implies $|\sin(\varphi)| = 1$, in which case the upper bound (4.26) on the trace distance reduces to

$$D(\rho_E, \rho'_E) \leq \sqrt{1 - D(\sigma_B, \sigma'_B)^2}. \quad (4.85)$$

Following the action of the depolarising channel, $D(\sigma_B, \sigma'_B) = 1 - p$ and the z-basis error rate is given by $\delta_z = p/2$. The resulting lower bound as a function of p is then

$$r \geq 1 - \log(1 - \sqrt{2p - p^2}) - h(p/2), \quad (4.86)$$

which becomes zero for $p \approx 0.1516$, or the equivalent of an error rate of about $\delta \approx 7.58\%$.

In the version where Bob performs the CHSH-type test, the expectation value of the correlator is reduced to $S = 2\sqrt{2}(1 - p)$, in which case

$$r \geq 1 - \log(1 - \sqrt{4p - 2p^2}) - h(p/2), \quad (4.87)$$

which is worse than (4.86), with a threshold channel noise parameter of $p \approx 0.1045$.

As is generally the case with protocols featuring partial or full device independence, both protocols are vulnerable to the detection loophole and thus impractical with current technology, with similar relative performance.

4.A Orthogonal source states

In this appendix, we show that the tight bound $H(Z_A | E) \geq 1 - h(\frac{1}{2} + \frac{1}{2}\sqrt{S^2/4 - 1})$ for the conditional von Neumann entropy holds if Bob's measurements are no longer two dimensional, provided that the z-basis states are orthogonal. The approach is similar to the approach used to bound the trace distance, though it is complicated by the fact that the fidelity bound $F(\rho_E, \rho'_E) \geq \sqrt{S^2/4 - 1}$ is not convex in S . The main idea is to derive a per-qubit-subspace bound for the conditional entropy which will turn out to be convex in S_k .

In the case we are considering, $|\alpha\rangle = |z\rangle$ and $|\alpha'\rangle = |z'\rangle$, where $|z\rangle$ and $|z'\rangle$ are the eigenstates of the operator Z used throughout this chapter. We recall that the operator Y can be expressed in terms of these states as $Y = e^{i\frac{\varphi}{2}}|z\rangle\langle z'| + e^{-i\frac{\varphi}{2}}|z'\rangle\langle z|$. We insert this into the quantity $\frac{1}{2}\text{Tr}[W_B^k Y_B]$, for which we derived a lower bound on in section 4.4.2, obtaining

$$\begin{aligned} \frac{1}{2}\text{Tr}[W_B^k Y_B] &= \text{Re}[e^{-i\frac{\varphi}{2}}\langle z|W_B^k \otimes \mathbb{1}_E|z'\rangle] \\ &\leq |\langle z|W_B^k \otimes \mathbb{1}_E|z'\rangle| \\ &= |\langle z|(\mathbb{1}_B^k \otimes \mathbb{1}_E)(W_B^k \otimes \mathbb{1}_E)|z'\rangle|, \end{aligned} \quad (4.88)$$

where W_B^k and $\mathbb{1}_B^k$ were defined in (4.64) and (4.63). By Uhlmann's theorem, the last line of (4.88) provides a lower bound on the fidelity between two states $\rho_E^{(k)}$ and $\rho_E'^{(k)}$ obtained as partial traces of $(\mathbb{1}_B^k \otimes \mathbb{1}_E)|z\rangle$ and $(W_B^k \otimes \mathbb{1}_E)|z'\rangle$. With the correct normalisation factored out, these are

$$p_k(1 + \varepsilon_k)\rho_E^{(k)} = \text{Tr}_B[(\mathbb{1}_B^k \otimes \mathbb{1}_E)|z\rangle\langle z|], \quad (4.89)$$

$$p_k(1 - \varepsilon_k)\rho_E'^{(k)} = \text{Tr}_B[(\mathbb{1}_B^k \otimes \mathbb{1}_E)|z'\rangle\langle z'|], \quad (4.90)$$

where $p_k = \frac{1}{2} \text{Tr}[\mathbb{1}_B^k \mathcal{I}_B]$ as in section 4.4.2 and

$$p_k(1 + \varepsilon_k) = \langle z | \mathbb{1}_B^k \otimes \mathbb{1}_E | z \rangle, \quad (4.91)$$

$$p_k(1 - \varepsilon_k) = \langle z' | \mathbb{1}_B^k \otimes \mathbb{1}_E | z' \rangle. \quad (4.92)$$

Note that $\sum_k p_k = \sum_k p_k(1 + \varepsilon_k) = p_k(1 - \varepsilon_k) = 1$ and that

$$\sum_k p_k(1 + \varepsilon_k)\rho_E^{(k)} = \rho_E, \quad (4.93)$$

$$\sum_k p_k(1 - \varepsilon_k)\rho_E'^{(k)} = \rho_E'. \quad (4.94)$$

For these states, (4.88) and Uhlmann's theorem together imply that

$$F(\rho_E^{(k)}, \rho_E'^{(k)}) \geq \frac{1}{\sqrt{1 - \varepsilon_k^2}} \frac{1}{2p_k} \text{Tr}[W_B^k Y_B]. \quad (4.95)$$

Note that, because the fidelity can never exceed 1 for normalised states, the asymmetry ε_k is limited by $\varepsilon_k^2 \leq 1 - \frac{1}{4p_k^2} \text{Tr}[W_B^k Y_B]^2 \leq 2 - (S_k/p_k)^2/4$.

Our goal is to find a lower bound for the conditional von Neumann entropy $H(Z_A | E)$, which we recall should be evaluated on the classical-quantum state

$$\tau_{ZE} = \frac{1}{2}(|0\rangle\langle 0|_Z \otimes \rho_E + |1\rangle\langle 1|_Z \otimes \rho_E'). \quad (4.96)$$

Because the conditional von Neumann entropy is concave in the state it is evaluated on,

$$H(Z_A | E) \geq \sum_k p_k H(Z_A | E)_{\tau_{ZE}^{(k)}}, \quad (4.97)$$

where $H(Z_A | E)_{\tau_{ZE}^{(k)}}$ is the conditional entropy evaluated on

$$\tau_{ZE}^{(k)} = \frac{1 + \varepsilon_k}{2} |0\rangle\langle 0|_Z \otimes \rho_E^{(k)} + \frac{1 - \varepsilon_k}{2} |1\rangle\langle 1|_Z \otimes \rho_E'^{(k)}, \quad (4.98)$$

and ε_k , $\rho_E^{(k)}$, and $\rho_E'^{(k)}$ are as defined in (4.91) and (4.92) above, such that $\sum_k p_k \tau_{ZE}^{(k)} = \tau_{ZE}$. In order to use (4.95), we use the lower bound

$$H(Z_A | E)_{\tau_{ZE}^{(k)}} \geq h\left(\frac{1 + \varepsilon_k}{2}\right) - h\left(\frac{1}{2} + \frac{1}{2}\sqrt{\varepsilon_k^2 + (1 - \varepsilon_k^2)F(\rho_E^{(k)}, \rho_E'^{(k)})^2}\right) \quad (4.99)$$

for the conditional entropy that was obtained for a classical-quantum state of the type (4.98) in section 3.2.1. Using the lower bound (4.95) on the fidelity,

$$H(Z_A | E)_{\tau_{ZE}^{(k)}} \geq h\left(\frac{1+\varepsilon_k}{2}\right) - h\left(\frac{1}{2} + \frac{1}{2}\sqrt{\varepsilon_k^2 + \frac{1}{4p_k^2} \text{Tr}[W_B^k Y_B]^2}\right). \quad (4.100)$$

In appendix 4.B, we show that (4.100) is convex in ε_k with its global minimum at $\varepsilon_k = 0$. Applying this simplifies the lower bound to

$$H(Z_A | E)_{\tau_{ZE}^{(k)}} \geq 1 - h\left(\frac{1}{2} + \frac{1}{4p_k} \text{Tr}[W_B^k Y_B]\right). \quad (4.101)$$

Finally, summing over k again,

$$\begin{aligned} H(Z_A | E) &\geq \sum_k p_k \left(1 - h\left(\frac{1}{2} + \frac{1}{4p_k} \text{Tr}[W_k Y]\right)\right) \\ &\geq \sum_k p_k \left(1 - h\left(\frac{1}{2} + \frac{1}{2}\sqrt{(S_k/p_k)^2/4 - 1}\right)\right) \\ &\geq 1 - h\left(\frac{1}{2} + \frac{1}{2}\sqrt{S^2/4 - 1}\right), \end{aligned} \quad (4.102)$$

where the end result follows from convexity of the function $x \mapsto 1 - h\left(\frac{1}{2} + \frac{1}{2}\sqrt{x^2/4 - 1}\right)$.

4.B Convexity of asymmetric entropy bound

We consider the function

$$f(\varepsilon) = h\left(\frac{1}{2} + \frac{\varepsilon}{2}\right) - h\left(\frac{1}{2} + \frac{1}{2}\sqrt{\varepsilon^2 + \mu^2}\right) \quad (4.103)$$

with $\varepsilon^2 + \mu^2 \leq 1$. In order to avoid carrying an extra factor of $\ln(2)$ around, we define the binary entropy in nats rather than bits:

$$h(x) = -x \ln(x) - (1-x) \ln(1-x). \quad (4.104)$$

The derivative of h is

$$h'(x) = -\ln(x) + \ln(1-x) = -\ln\left(\frac{x}{1-x}\right), \quad (4.105)$$

Using this, the first derivative of f is

$$f'(\varepsilon) = -\frac{1}{2} \ln\left(\frac{1+\varepsilon}{1-\varepsilon}\right) + \frac{1}{2} \ln\left(\frac{1+\sqrt{\varepsilon^2 + \mu^2}}{1-\sqrt{\varepsilon^2 + \mu^2}}\right) \frac{\varepsilon}{\sqrt{\varepsilon^2 + \mu^2}}. \quad (4.106)$$

We see that $f'(0) = 0$, confirming already that $\varepsilon = 0$ is at least a local extremum. Proceeding with the evaluation of the second derivative, we obtain

$$f''(\varepsilon) = -\frac{1}{1-\varepsilon^2} + \frac{1}{1-\varepsilon^2-\mu^2} \frac{\varepsilon^2}{\varepsilon^2+\mu^2} + \frac{1}{2} \ln\left(\frac{1+\sqrt{\varepsilon^2+\mu^2}}{1-\sqrt{\varepsilon^2+\mu^2}}\right) \frac{\mu^2}{(\varepsilon^2+\mu^2)^{3/2}}. \quad (4.107)$$

Using now that $\ln\left(\frac{1+|x|}{1-|x|}\right) \geq 2|x|$ and simplifying,

$$\begin{aligned} f''(\varepsilon) &\geq -\frac{1}{1-\varepsilon^2} + \frac{1}{1-\varepsilon^2-\mu^2} \frac{\varepsilon^2}{\varepsilon^2+\mu^2} + \frac{\mu^2}{\varepsilon^2+\mu^2} \\ &= -\frac{1}{1-\varepsilon^2} + \frac{1}{\varepsilon^2+\mu^2} \frac{\varepsilon^2+\mu^2-\mu^2(\varepsilon^2+\mu^2)}{1-\varepsilon^2-\mu^2} \\ &= -\frac{1}{1-\varepsilon^2} + \frac{1-\mu^2}{1-\varepsilon^2-\mu^2} \\ &= \frac{\varepsilon^2\mu^2}{(1-\varepsilon^2)(1-\varepsilon^2-\mu^2)} \\ &\geq 0, \end{aligned} \quad (4.108)$$

with equality if and only if $\mu = 0$. The function f is thus strictly convex unless $\mu = 0$, in which case $f(\varepsilon) = 0, \forall \varepsilon$, which implies that the local extremum at $\varepsilon = 0$ is in fact the global minimum. We have thus shown that

$$f(\varepsilon) \geq f(0) = \ln(2) - h\left(\frac{1}{2} + \frac{1}{2}|\mu|\right). \quad (4.109)$$

4.C Characterisation of g_*

The bound (4.35) on the fidelity is given in terms of a “penalty” function

$$g_*(x) = \min_{\alpha} g_{\alpha}(x/|\cos(\alpha)|), \quad (4.110)$$

accounting for the possibility that the z-basis states may not be orthogonal, with g_{α} defined by (3.78) in section 3.5.1. The angle α can be restricted to the range $[0, \pi/2]$, in which case both $\sin(\alpha)$ and $\cos(\alpha)$ are nonnegative. Explicitly,

$$g_*(x) = \min_{\alpha \in [0, \pi/2]} \begin{cases} \sin(\alpha) & : x \leq \frac{2\sin(\alpha)\cos(\alpha)}{1+\sin(\alpha)} \\ \frac{1+\sin(\alpha)}{\cos(\alpha)}x - \sin(\alpha) & : x \geq \frac{2\sin(\alpha)\cos(\alpha)}{1+\sin(\alpha)} \end{cases}. \quad (4.111)$$

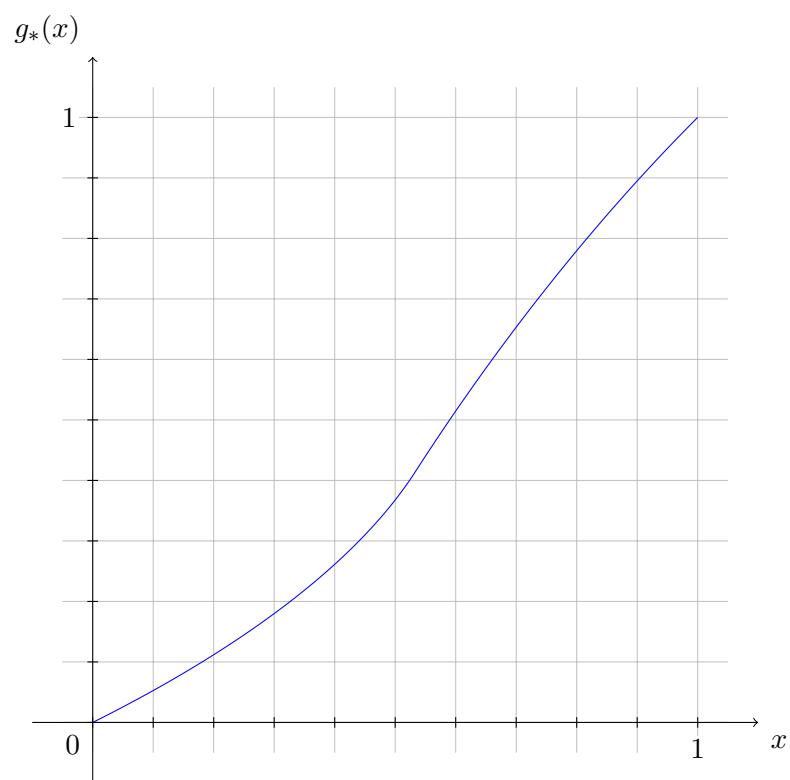


Figure 4.3: Graph of $g_*(x)$.

For some values of x , the minimum is simply the minimum of the function $h_x(\alpha) = \frac{1+\sin(\alpha)}{\cos(\alpha)}x - \sin(\alpha)$, whose derivative is

$$h'_x(\alpha) = \frac{x}{1 - \sin(\alpha)} - \cos(\alpha). \quad (4.112)$$

In these cases, the minimum is obtained with the angle α such that

$$\cos(\alpha)(1 - \sin(\alpha)) = x. \quad (4.113)$$

The values of x for which this is the minimum are those for which α in (4.113) satisfies

$$1 - \sin(\alpha) \geq \frac{2 \sin(\alpha)}{1 + \sin(\alpha)}, \quad (4.114)$$

or $\sin(\alpha) \leq \sin(\alpha_0) = \sqrt{2} - 1$, for which $x \geq x_0 = 2\sqrt{5\sqrt{2} - 7}$ and $g_*(x) \geq g_0 = \sqrt{2} - 1$. Otherwise, $g_*(x)$ is simply obtained with the angle α such that

$$x = \frac{2 \sin(\alpha) \cos(\alpha)}{1 + \sin(\alpha)} = 2 \sin(\alpha) \sqrt{\frac{1 - \sin(\alpha)}{1 + \sin(\alpha)}}. \quad (4.115)$$

In this way, we obtain a profile of the graph $(x, g_*(x))$. The part for which $x \leq x_0$ and $g_*(x) \leq g_0$ is given by

$$\begin{aligned} x &= 2 \sin(\alpha) \sqrt{\frac{1 - \sin(\alpha)}{1 + \sin(\alpha)}} \\ g_*(x) &= \sin(\alpha), \end{aligned} \quad (4.116)$$

while the part for which $x \geq x_0$ and $g_*(x) \geq g_0$ is given by

$$\begin{aligned} x &= \cos(\alpha)(1 - \sin(\alpha)) \\ g_*(x) &= \cos(\alpha)^2 - \sin(\alpha), \end{aligned} \quad (4.117)$$

in both cases for $0 \leq \alpha \leq \alpha_0$. The graph obtained from (4.115) and (4.116) is depicted in Fig. 4.3.

From (4.115) and (4.116), we have $\sin(\alpha) = g_*(x)$ and $\sin(\alpha) = \frac{1}{2}(-1 + \sqrt{5 - 4g_*(x)})$, respectively, which allows x to be expressed in terms of $g_*(x)$. With some further work, $g_*(x)$ can be identified with the roots of a pair of polynomials. Specifically, if $0 \leq x \leq x_0$, $g_*(x)$ is the (unique) real root $P_x(g_*(x)) = 0$ in the range $[0, g_0]$ of the cubic polynomial

$$P_x(g) = 4g^3 - 4g^2 + x^2g + x^2, \quad (4.118)$$

while for $x_0 \leq x \leq 1$, $g_*(x)$ is the real root $Q_x(g_*(x)) = 0$ in the range $[g_0, 1]$ of the quartic polynomial

$$Q_x(g) = g^4 + 2g^3 + 2x^2g^2 - 2(7x^2 + 1)g + x^4 + 11x^2 - 1. \quad (4.119)$$

Chapter 5

Conclusion

The main theme of the work presented in chapters 2, 3, and 4 was the security of prepare-and-measure BB84-like protocols with source and detector alignment imprecisions against individual and collective attacks. Chapter 2 demonstrated the advantage an adversary could gain by exploiting source and detector measurement imprecisions, emphasising the necessity of accounting for such imprecisions in security analyses of QKD protocols. The two subsequent chapters gave security results for the BB84 protocol with source and detector imprecisions and for two semi-device-independent protocols, both against collective attacks.

Table 5.1 summarises the conditions for which the main results obtained in chapter 3 apply. Section 3.4 presented a key rate for a BB84 variant, for which the four possible (pure) source states may be arbitrary provided a single source characterisation angle θ was known, and gave a proof of its optimality. For the optimal attack derived in section 3.4.3, it is worth remarking that the key-rate bound is attained if Bob performs the ideal σ_z and σ_x measurements and that there are no “cross correlations” between the two bases, i.e., Bob obtains a completely random result if he measures (e.g.) one of the z-basis source states in the x basis. This suggests there is little potential to improve the result by introducing a characterisation of Bob’s detector or using estimated parameters, such as correlations from mismatched basis choices, in addition to the x- and z-basis error rates. Section 3.5 developed tools for the more specific case of a qubit source. In this special case, the result of section 3.5.1 can slightly outperform the more general key rate for arbitrary source states and the result of 3.5.2 gives a significantly better result if Bob’s measurements are additionally assumed constrained to a two-dimensional Hilbert space. In the case of a characterised qubit source,

Sec.	Source states	Measurements	Key rate
3.4	arbitrary (θ known)	uncharacterised	$r = r_\theta(\delta_x, \delta_z)$
3.5.1	qubit (φ, α, β known)	uncharacterised	$r = r_{\varphi, \alpha, \beta}(\delta_x, \delta_z)$
3.5.2	qubit (φ, α, β known)	uncharacterised qubit	$r = r_{\varphi, \alpha, \beta}(\delta_x, \delta_z)$

Table 5.1: Summary of conditions for which the security results for the BB84 protocol derived in chapter 3 hold. For the source states, “arbitrary” means that the set $\text{Src} = \{|\alpha\rangle, |\alpha'\rangle, |\beta\rangle, |\beta'\rangle\}$ of source states may span a Hilbert space of dimension up to four; “qubit” means that Src is constrained to a Hilbert space of dimension two; θ is the angular source characterisation parameter defined in (3.7); φ, α , and β are the qubit source characterisation angles defined at the beginning of section 3.5.1. For Bob’s measurements, “uncharacterised” means that Bob performs either of two binary-outcome measurements separately on each received state that are otherwise uncharacterised; “uncharacterised qubit” means that Bob’s measurements are, additionally, assumed to be restricted to a two-dimensional Hilbert space. The “key rate” column summarises the form of the resulting key-rate bound: the subscripted indices ($\theta, \varphi, \alpha, \beta$) are assumed a priori while the parameters in parentheses (the error rates δ_x and δ_z) are estimated during the execution of the protocol.

Sec.	Alice's meas.	Bob's meas.	Key rate
4.3	CHSH	x, z	$r = r(S, \delta_x, \delta_z)$
4.4	/	CHSH, z	$r = r(S, \delta_z)$

Table 5.2: Overview of semi-device-independent protocols considered in chapter 4. In both protocols, Alice’s source is assumed to emit four a priori unknown qubit states, Alice’s (if any) and Bob’s measurements are left uncharacterised, and the “z-basis” key bits are used to generate the final key. The two protocols differ in the measurements performed by Alice and by Bob: “/” indicates that a party performs no measurements, “CHSH” indicates that a party performs the measurements required for the estimation of the CHSH correlator S , and “x” and “z” indicate that a party should perform the “x-basis” and “z-basis” measurements, respectively. The “key rate” column summarises the dependence of the key-rate bound on the CHSH correlator value S and the x- and z-basis error rates δ_x and δ_z that are estimated in the course of the protocol.

however, there will typically be correlations in the results for mismatched basis choices. No attempt was made to exploit this – the key rates were bounded only in terms of the usual x- and z-basis error rates – and there is likely significant room for improvement in both results.

Table 5.2 gives an overview of the two semi-device-independent protocols considered in chapter 4, which differ in implementation mainly in which measurements are performed by each party. In both protocols, Alice’s source is assumed to emit (a priori unknown) qubit states, and their security is tested based on the estimation of a CHSH-like correlator. (For the main results which give a lower bound on the min-entropy, the qubit source assumption can be slightly relaxed to requiring only that the two operators $\rho - \rho'$ and $\sigma - \sigma'$ share their support on a common two-dimensional Hilbert space.) In both cases, Alice’s (if any) and Bob’s measurements are uncharacterised and may be of unbounded dimension. The protocol considered in section 4.3 uses the CHSH value, estimated entirely by Alice, as a partial source characterisation. The resulting key rate is obtained simply by substituting the resulting source characterisation into a lower bound for the min-entropy that was derived in section 3.5.1. Because of this, like the main result of section 3.5.1, there is no guarantee of optimality for the resulting key rate. In the second protocol, studied in section 4.4, the CHSH-like correlator is estimated by Alice and Bob working in cooperation and serves the purpose of testing the channel between the two parties. Compared with the BB84 protocol, the CHSH estimation replaces the x-basis measurement for this purpose, which is no longer used. Based on the example comparison in section 4.5, one may expect a possible tradeoff between the two approaches to

semi-device-independent QKD considered in this chapter: in particular, the source characterisation protocol has the potential to better tolerate channel noise and possibly detector losses, but requires a more complicated implementation in that Alice must have the ability to switch at will between measuring her states and transmitting them to Bob. In practice, both protocols are likely to be impractical in the near future due to the requirement of high detection efficiencies. The security analysis should strictly also be made more robust with regard to the qubit source assumption: even a single-photon state is infinite-dimensional and one will never be able to guarantee that four differently-prepared photon states are prepared in *exactly* the same two-dimensional subspace.

Aside from the key rates themselves, the approach followed in chapter 3 showed that the security of the prepare-and-measure BB84 protocol can be related in a particularly simple way to a particular characterisation of the no-cloning principle introduced for this purpose. In this respect, the approach is similar to that explored in [59, 60] in that security was related as closely as possible to a simple generic characterisation of the structure of quantum physics, with the tradeoff bounds such as (3.8) playing a role analogous to the entropic tradeoff relations of [59, 60]. Notably, in a departure from recent security analyses, we saw that a much more direct treatment of the prepare-and-measure scenario is possible, without the need to recast the protocol into its entanglement-based form as part of the analysis. There is a possibility that some of the techniques introduced in chapters 3 and 4 may be useful for prepare-and-measure randomness protocols, such as [88].

The main security results in chapters 3 and 4 hold with Bob’s device left largely uncharacterised. Specifically, the key rates hold wherever the Helstrom bound does, which is at least wherever Bob’s measurement operators are separable (i.e., Bob measures each state sent by Alice individually). This is in line with a trend in recent security analyses featuring automatic “one-sided device independence” [89]. (See also [90] for a recent security proof of BB84 that is completely device-independent on Bob’s side.) Since this makes the results susceptible to the detection loophole, a more refined consideration of losses, such as the approach in [20] in which nondetection was treated as a third possible measurement outcome, may be necessary.

The main limitation of these security results, which use the Devetak-Winter bound as a starting point, is the restriction to collective attacks, in which the eavesdropper is assumed to attack each qubit unitarily individually and identically. The extension to general attacks is left as a problem for future work. One way this may possibly be addressed is by adapting the quantum de Finetti theorem [66] or the related post-selection technique of [67], which is already used in entanglement-based QKD. The more recent security proof

of the entanglement-based BB84 protocol in [49], based on a “one shot” version of the entropic inequality for the smooth min- and max-entropies in [60], however gives some hope that an analogously simple approach may be possible in the prepare-and-measure scenario. In this case the question arises as to how tradeoff relations such as (3.8) should be generalised and how the smooth min-entropy should be bounded in such a way as to recover lower bounds for the conditional von Neumann entropy such as (3.14) asymptotically.

Appendix A

Partially deterministic polytopes

This appendix summarises research work concerning classes of probabilistic correlations – the *partially deterministic polytopes* – intermediate between the local and no-signalling polytopes of the theory of Bell nonlocality. They have a possible application as relaxations of the local polytope for which membership testing – an NP complete problem for the local polytope – is more easily accomplished. The work presented in this appendix is the subject of an article currently in preparation [29] and should, to some degree, be read as work in progress. In particular, some formulations and proofs may not be as streamlined as they could be, and the connection between the partially deterministic polytopes and the problem of device-independent randomness certification is currently only explicitly discussed in the simplest cases where the local and global guessing probabilities – corresponding to two of the most relaxed polytopes with one or two deterministic measurements – are used as the figure of merit.

A.1 Introduction

Two sets of *behaviours* widely studied in the field of Bell nonlocality (see [91] for a review) are the sets of *local* and *no-signalling* behaviours, both of which are contrasted from the set of quantum behaviours. In the two-party case, the local set is comprised of probability distributions $\mathbf{P} = (P(ab | xy))$ that admit a factorisation in the form of a locally causal model

$$P(ab | xy) = \int d\lambda \rho(\lambda) P_A(a | x; \lambda) P_B(b | y; \lambda) \quad \Leftrightarrow \quad \mathbf{P} \in \mathbb{L}. \quad (\text{A.1})$$

The no-signalling set $\mathbb{NS}$ is the (generally larger) set of behaviours satisfying the no-signalling constraints

$$\sum_b P(ab | xy) = \sum_b P(ab | x0) = P_A(a | x), \quad (\text{A.2})$$

$$\sum_a P(ab | xy) = \sum_a P(ab | 0y) = P_B(b | y) \quad \Leftrightarrow \quad \mathbf{P} \in \mathbb{NS}, \quad (\text{A.3})$$

such that the marginal distributions $P_A(a | x)$ and $P_B(b | y)$ do not depend on y and x respectively. Though we will not generally be concerned with the quantum set $\mathbb{Q}$, it can be defined as the set of behaviours compatible with the Born rule, i.e.,

$$P(ab | xy) = \text{Tr}[M_a^{(x)} \otimes N_b^{(y)} \rho_{AB}] \quad \Leftrightarrow \quad \mathbf{P} \in \mathbb{Q}, \quad (\text{A.4})$$

in which ρ_{AB} is a density operator and $\{M_a^{(x)}\}_a$ and $\{N_b^{(y)}\}_b$ are POVMs. In all but the most trivial Bell scenarios, a core theorem is the strict hierarchy $\mathbb{L} \subset \mathbb{Q} \subset \mathbb{NS}$. That in general $\mathbb{L} \neq \mathbb{Q}$ was effectively first proved by Bell [92], while the inequivalence $\mathbb{Q} \neq \mathbb{NS}$ was illustrated by a counterexample in [82, 93, 94]. In the simplest nontrivial scenario (two parties with binary inputs and dichotomic outcomes), these distinctions can be established by comparing the respective maximal values of the CHSH correlator,

$$S = \mathbf{I} \cdot \mathbf{P} = \sum_{abxy} I_{abxy} P(ab | xy), \quad (\text{A.5})$$

with $I_{abxy} = (-1)^{a+b+xy}$, $a, b, x, y \in \{0, 1\}$. The well known results are that

$$\max_{\mathbf{P} \in \mathbb{L}} \mathbf{I} \cdot \mathbf{P} = 2, \quad (\text{A.6})$$

$$\max_{\mathbf{P} \in \mathbb{Q}} \mathbf{I} \cdot \mathbf{P} = 2\sqrt{2}, \quad (\text{A.7})$$

$$\max_{\mathbf{P} \in \mathbb{NS}} \mathbf{I} \cdot \mathbf{P} = 4, \quad (\text{A.8})$$

which were determined in [79, 81, 82, 93, 94].

Aside from its foundational interest (Bell's concept of locality was motivated by relativistic causality [95, 96] and originated in a response [92] to an argument against the completeness of quantum mechanics by Einstein, Podolsky, and Rosen [97]), Bell nonlocality has a more pragmatic interest as a resource for device-independent cryptography [25, 98, 99] and the related problem of randomness generation [85]. In the case of an adversary limited by quantum mechanics, the amount of randomness that can be certified device-independently for a given behaviour $\mathbf{P}$, as measured by the adversary's worst-case guessing probability, can be determined by solving a family of semidefinite programs [100, 101] corresponding to a hierarchy

of relaxations that converge to the quantum set [102–104]. For an adversary restricted only by the no-signalling principle, evaluating the device-independent guessing probability reduces to solving a linear program.

This work introduces a hierarchy of sets of behaviours – the *partially deterministic* polytopes – intermediate between the local and no-signalling sets and proves some simple relations between them. Their characterisation can be considered the complementary problem to device-independent randomness certification in the no-signalling scenario in that the partially deterministic polytopes are precisely the sets of behaviours against which randomness cannot be certified on a given set of measurements against a no-signalling adversary. For small numbers of measurements, membership of a behaviour in a partially deterministic polytope can be tested by determining whether the corresponding guessing probability – the solution to a comparatively small linear program – is 1. Membership testing becomes progressively more difficult as the number of deterministic measurements is increased and the corresponding partially deterministic polytope converges to the local set (for which membership testing is an NP-complete problem [105]).

A.2 Preliminaries

A.2.1 Scenarios and behaviours

Bell scenarios

In general, a (two-party or bipartite) *Bell scenario* is characterised by sets of *inputs* $\mathcal{X} = \{0, \dots, m_A - 1\}$ and $\mathcal{Y} = \{0, \dots, m_B - 1\}$ that the respective parties (“Alice” and “Bob”) can select, and corresponding sets of *outputs*. It will be convenient to associate to each of Alice’s and Bob’s inputs $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ the symbols A_x and B_y respectively, which we also take to be the names of the corresponding sets of outputs

$$A_x = \{0, \dots, d_A - 1\}, \quad x \in \mathcal{X}, \quad (\text{A.9})$$

$$B_y = \{0, \dots, d_B - 1\}, \quad y \in \mathcal{Y}. \quad (\text{A.10})$$

For a given Bell scenario, we denote $\mathcal{A} = \{A_x\}_{x \in \mathcal{X}}$ and $\mathcal{B} = \{B_y\}_{y \in \mathcal{Y}}$ the sets of all Alice’s and Bob’s input symbols and $\mathcal{S} = \mathcal{A} \cup \mathcal{B} = \{A_0, A_1, \dots, B_0, B_1, \dots\}$ the set of all inputs.

Note that, in general, the numbers $d_A = d_A(x)$ and $d_B = d_B(y)$ of possible outputs may be allowed to depend on the inputs $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. We will refer any two-party Bell scenario in which d_A and d_B are *independent* of x and y as a “ $m_A m_B d_A d_B$ scenario”. For instance, the 3322 scenario is the Bell scenario where Alice and Bob can each select from one of three different inputs with corresponding dichotomic outputs, in which case $\mathcal{S} = \{A_0, A_1, A_2, B_0, B_1, B_2\}$ and $A_x = B_y = \{0, 1\}$, $\forall x, y \in \{0, 1, 2\}$.

Behaviours

In a Bell scenario of inputs $\mathcal{S}$, A *behaviour* is a probability distribution $\mathbf{P} = (P(ab | xy))$ of elements

$$P(ab | xy), \quad a \in A_x, b \in B_y, A_x, B_y \in \mathcal{S}. \quad (\text{A.11})$$

It will be implicit that anything called a “probability distribution” satisfies the necessary defining normalisation ($\sum_{ab} P(ab | xy) = 1$) and positivity ($P(ab | xy) \geq 0$) constraints.

Marginal and conditional behaviours will be denoted with subscripts A, B, etc., in the usual way, such that $P_A(a | xy) = \sum_b P(ab | xy)$ and $P(ab | xy) = P_{A|B}(a | bxy)P_B(b | xy)$. For no-signalling probability distributions, the inputs of “traced out” parties becomes redundant and will be excluded, for instance $P_A(a | xy) = P_A(a | x)$. Conversely, a behaviour compatible with a given marginal will be called an *extension* of that marginal. For instance, if a bipartite behaviour $\mathbf{P}_{AB}$ is a marginal of a tripartite behaviour $\mathbf{P}_{ABE}$, then $\mathbf{P}_{ABE}$ is an extension of $\mathbf{P}_{AB}$.

Where necessary, behaviours can conveniently be expressed in the table notation

$$\mathbf{P} = \left[\begin{array}{ccc|cc} P(00 | 00) & P(01 | 00) & \cdots & P(00 | 01) & \cdots \\ P(10 | 00) & P(11 | 00) & & P(10 | 01) & \\ \vdots & & \ddots & \vdots & \ddots \\ \hline P(00 | 10) & P(01 | 10) & \cdots & P(00 | 11) & \cdots \\ \vdots & & \ddots & \vdots & \ddots \end{array} \right]. \quad (\text{A.12})$$

For instance, the Popescu-Rohrlich (PR) box [82, 93, 94], defined in the 2222 scenario by

$$P_{\text{PR}}(ab | xy) = \begin{cases} 1/2 & \text{if } a + b = xy \pmod{2} \\ 0 & \text{otherwise} \end{cases}, \quad (\text{A.13})$$

is represented by the table

$$\mathbf{P}_{\text{PR}} = \left[\begin{array}{cc|cc} 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ \hline 1/2 & 0 & 0 & 1/2 \\ 0 & 1/2 & 1/2 & 0 \end{array} \right]. \quad (\text{A.14})$$

The no-signalling polytope

The largest sets of behaviours that we will be concerned with are the no-signalling polytopes, whose elements satisfy no-signalling constraints of the type given in section A.1. For a Bell scenario $\mathcal{S} = \mathcal{A} \cup \mathcal{B}$, we call the corresponding no-signalling polytope $\text{NS}(\mathcal{S})$, though the input set may be omitted where the scenario is clear from context. Note that, if a behaviour $\mathbf{P}$ is contained in a given no-signalling polytope, its marginal after a party is removed is in the corresponding no-signalling polytope. For instance, if $\mathbf{P}_{\text{ABE}} \in \text{NS}(\mathcal{A} \cup \mathcal{B} \cup \mathcal{E})$ in a tripartite Bell scenario, then the marginal $\mathbf{P}_{\text{AB}} \in \text{NS}(\mathcal{A} \cup \mathcal{B})$.

That NS is a polytope is mostly evident from its definition in terms of a finite number of linear equality (normalisation and no-signalling) constraints and a finite number of inequality (positivity) constraints (which constitute its facets). (That NS is closed is evident from the fact that any behaviour is contained in the box $0 \leq P(ab | xy) \leq 1$, $\forall a, b, x, y$.) In a $m_A m_B d_A d_B$ scenario, the inequality constraints mean that the no-signalling polytope is constrained to a plane of dimension $m_A(d_A - 1) + m_B(d_B - 1) + m_A m_B(d_A - 1)(d_B - 1)$. This can be seen from the fact that a behaviour in this case can be reconstructed from its elements $P(ab | xy)$ and marginals $P_A(a | x)$ and $P_B(b | y)$ for all but one of the outputs on each side (the Collins-Gisin projection [106]).

For a single party (e.g., Alice), the no-signalling polytope $\text{NS}(\mathcal{A})$ reduces to the set of marginal behaviours $\{\mathbf{P}_A = (P_A(a | x))\}$. For the empty input set, we adopt the convention that $\text{NS}(\emptyset) = \{1\}$.

The local polytope

As was first shown by Fine [107], the response functions $P_A(a | x; \lambda)$ and $P_B(b | y; \lambda)$ appearing in the definition of a local model cited in section A.1 can be taken to be deterministic without any loss of generality, i.e., a (bipartite) local set $\mathbb{L}(\mathcal{A} \cup \mathcal{B})$ can equivalently be defined as the set of behaviours

$\mathbf{P}$ which admit a decomposition of the form

$$P(ab \mid xy) = \sum_{\mathbf{a}, \mathbf{b}} p_{\mathbf{a}\mathbf{b}} D_A(a \mid x; \mathbf{a}) D_B(b \mid y; \mathbf{b}), \quad (\text{A.15})$$

with the sum taken over all vectors

$$\mathbf{a} = (a_x) \in A_\Pi = A_0 \times A_1 \times \cdots, \quad (\text{A.16})$$

$$\mathbf{b} = (b_y) \in B_\Pi = B_0 \times B_1 \times \cdots \quad (\text{A.17})$$

describing deterministic mappings $x \mapsto a_x$ and $y \mapsto b_y$ of inputs to outputs and where the corresponding deterministic response functions are defined by

$$D_A(a \mid x; \mathbf{a}) = \delta_{a, a_x}, \quad (\text{A.18})$$

$$D_B(b \mid y; \mathbf{b}) = \delta_{b, b_y}, \quad (\text{A.19})$$

δ is the Kronecker delta, and the coefficients $p_{\mathbf{a}\mathbf{b}}$ satisfy $p_{\mathbf{a}\mathbf{b}} \geq 0$ and $\sum_{\mathbf{a}\mathbf{b}} p_{\mathbf{a}\mathbf{b}} = 1$. The local set is thus a polytope, being the convex hull of a finite number of local deterministic behaviours which constitute its vertices. Its (non positivity) facets correspond to tight Bell inequalities.

In general, one can restrict to just summing over (say) Alice's deterministic strategies. Setting $p_{\mathbf{a}\mathbf{b}} = p_{\mathbf{b}|\mathbf{a}} p_{\mathbf{a}}$ with $p_{\mathbf{a}} = \sum_{\mathbf{b}} p_{\mathbf{a}\mathbf{b}}$,

$$P(ab \mid xy) = \sum_{\mathbf{a}} p_{\mathbf{a}} D_A(a \mid x; \mathbf{a}) \sum_{\mathbf{b}} p_{\mathbf{b}|\mathbf{a}} D_B(b \mid y; \mathbf{b}). \quad (\text{A.20})$$

Recognising that $P_B(b \mid y; \mathbf{a}) = \sum_{\mathbf{b}} p_{\mathbf{b}|\mathbf{a}} D_B(b \mid y; \mathbf{b})$ is a probability distribution, we find that all local points admit an expression of the form

$$P(ab \mid xy) = \sum_{\mathbf{a}} p_{\mathbf{a}} D_A(a \mid x; \mathbf{a}) P_B(b \mid y; \mathbf{a}). \quad (\text{A.21})$$

Similar to the no-signalling polytope, the local polytope for a single party reduces to the set of marginal behaviours, such that $\mathbb{L}(\mathcal{A}) = \mathbb{NS}(\mathcal{A})$ and $\mathbb{L}(\mathcal{B}) = \mathbb{NS}(\mathcal{B})$. We also set $\mathbb{L}(\emptyset) = \{1\}$.

Facets for dichotomic outputs

Because any polytope containing only no-signalling behaviours is constrained to the no-signalling plane, there is no unique format for expressing its facets. For instance, the CHSH and CH74 [108] inequalities correspond to the same

facet of the local polytope in the 2222 scenario and can be derived from each other by substituting the normalisation and no-signalling constraints. The simplest way to constrain the expression of a facet is to work in a projection of the set of behaviours with the same dimension as the no-signalling plane. In the case where all the outputs are dichotomic – the only case we will explicitly derive facets for – we will use the projection of expectation values defined by

$$\langle A_x \rangle = P_A(0 | x) - P_A(1 | x), \quad (\text{A.22})$$

$$\langle B_y \rangle = P_B(0 | y) - P_B(1 | y), \quad (\text{A.23})$$

$$\begin{aligned} \langle A_x B_y \rangle &= P(00 | xy) - P(01 | xy) \\ &\quad - P(10 | xy) + P(11 | xy). \end{aligned} \quad (\text{A.24})$$

Given the expectation values, the full probability distribution can be reconstructed from the inverse relations

$$P(00 | xy) = \frac{1}{4}(1 + \langle A_x \rangle + \langle B_y \rangle + \langle A_x B_y \rangle), \quad (\text{A.25})$$

$$P(01 | xy) = \frac{1}{4}(1 + \langle A_x \rangle - \langle B_y \rangle - \langle A_x B_y \rangle), \quad (\text{A.26})$$

$$P(10 | xy) = \frac{1}{4}(1 - \langle A_x \rangle + \langle B_y \rangle - \langle A_x B_y \rangle), \quad (\text{A.27})$$

$$P(11 | xy) = \frac{1}{4}(1 - \langle A_x \rangle - \langle B_y \rangle + \langle A_x B_y \rangle). \quad (\text{A.28})$$

With this projection, linear inequalities can be expressed in the form

$$\sum_x I_x^A \langle A_x \rangle + \sum_y I_y^B \langle B_y \rangle + \sum_{xy} I_{xy} \langle A_x B_y \rangle \leq I_0. \quad (\text{A.29})$$

We will express inequalities in the table form

$$N \times \begin{array}{c|ccc} & I_0^B & I_1^B & \dots \\ I_0^A & I_{00} & I_{01} & \dots \\ I_1^A & I_{10} & I_{11} & \\ \vdots & \vdots & & \ddots \end{array} \leq I_0, \quad (\text{A.30})$$

similar to the table notation for behaviours introduced earlier. N , if indicated, is the multiplicity, i.e., the number of facets that are equivalent to the one indicated up to relabelling of inputs and outputs. (Since we will consider examples where Alice and Bob have different numbers of inputs, we will count facets related by interchanging Alice and Bob separately.) For instance, the CHSH inequality, which is

$$\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle \leq 2 \quad (\text{A.31})$$

in the expectation value projection, could be expressed as

$$8 \times \begin{array}{c|cc} & 0 & 0 \\ \hline 0 & 1 & 1 \\ 0 & 1 & -1 \end{array} \leq 2, \quad (\text{A.32})$$

for the 2222 scenario. For convenience, full lists of facets for the 3322 and 4322 local polytopes have been given in [appendix A.A](#)

A.2.2 Device-independent randomness

In a bipartite Bell scenario $\mathcal{A} \cup \mathcal{B}$, the device-independent guessing probability is the worst-case probability with which a third party (Eve) can correctly guess an output. There are two main guessing probabilities – the local and global guessing probabilities – that we will be concerned with.

Local guessing probability

For a given behaviour $\mathbf{P} \in \mathcal{NS}(\mathcal{A} \cup \mathcal{B})$ and for (say) one of Alice's inputs $A_x \in \mathcal{A}$, we define the *local* guessing probability as

$$G_{A_x}(\mathbf{P}) = \max_{\mathbf{P}_{ABE}} \sum_a P_{AE}(aa | x), \quad (\text{A.33})$$

where the minimisation is taken over all tripartite no-signalling extensions $\mathbf{P}_{ABE} \in \mathcal{NS}(\mathcal{A} \cup \mathcal{B} \cup \{E\})$ of $\mathbf{P}_{AB}$ for which Eve's (only) input E is a copy of Alice's input, i.e., $a \in A_x \Leftrightarrow a \in E$. Using that $P_{AE}(ae | x) = P_E(e)P_{A|E}(a | x; e)$, the local guessing probability can alternatively and equivalently be defined by

$$G_{A_x}(\mathbf{P}) = \max_{\{p_e, \mathbf{P}_{AB}^{(e)}\}} \sum_a p_a P_A^{(a)}(a | x), \quad (\text{A.34})$$

with the minimisation taken over decompositions $\{p_e, \mathbf{P}_{AB}^{(e)}\}_{e \in A_x}$ satisfying $\sum_e p_e P_{AB}^{(e)}(ab | xy) = P(ab | xy)$, $\forall a, b, x, y$.

Global guessing probability

For a pair of inputs $\{A_x, B_y\}$, we define the *global* guessing probability by

$$G_{A_x B_y}(\mathbf{P}) = \max_{\mathbf{P}_{ABE}} \sum_{ab} P_{ABE}(ab(ab) | xy), \quad (\text{A.35})$$

with the minimisation taken over all extensions $\mathbf{P}_{\text{ABE}} \in \mathbb{NS}(\mathcal{A} \cup \mathcal{B} \cup \{\text{E}\})$ in which $\text{E} \simeq \text{A}_x \times \text{B}_y$, i.e., $a \in \text{A}_x, b \in \text{B}_y \Leftrightarrow (ab) \in \text{E}$. Similar to the case with the local guessing probability, $G_{\text{A}_x \text{B}_y}$ can equivalently be expressed as

$$G_{\text{A}_x \text{B}_y}(\mathbf{P}) = \max_{\{p_e, \mathbf{P}_{\text{AB}}^{(e)}\}} \sum_{ab} p_{ab} P^{(ab)}(ab | xy), \quad (\text{A.36})$$

with $e \in \text{A}_x \times \text{B}_y$, subject to the constraint $\sum_e p_e P_{\text{AB}}^{(e)}(ab | xy) = P(ab | xy)$.

Convex guessing set

For $\mathbf{P}_k \in \mathbb{NS}(\mathcal{A} \cup \mathcal{B})$ and convex coefficients p_k , it is not difficult to see that

$$G_{\mathcal{I}}\left(\sum_k p_k \mathbf{P}_k\right) \geq \sum_k p_k G_{\mathcal{I}}(\mathbf{P}_k) \quad (\text{A.37})$$

for both the local ($\mathcal{I} = \{\text{A}_x\}$) and global ($\mathcal{I} = \{\text{A}_x, \text{B}_y\}$) guessing probabilities. A consequence is that, for a probability $0 \leq p \leq 1$, the set

$$\mathbb{G}_{\mathcal{I}}(\mathcal{S})_p = \{\mathbf{P} \in \mathbb{NS}(\mathcal{S}) \mid G_{\mathcal{I}}(\mathbf{P}) \geq p\} \quad (\text{A.38})$$

is convex.

A.2.3 Operations for behaviours

Products of behaviours

For two bipartite Bell scenarios of input sets $\mathcal{I}$ and $\mathcal{J}$ which are disjoint (i.e., $\mathcal{I} \cap \mathcal{J} = \emptyset$), we introduce a product

$$*: \mathbb{NS}(\mathcal{I}) \times \mathbb{NS}(\mathcal{J}) \rightarrow \mathbb{NS}(\mathcal{I} \cup \mathcal{J}), \quad (\text{A.39})$$

with the product behaviour $\mathbf{P} * \mathbf{Q}$, for $\mathbf{P} \in \mathbb{NS}(\mathcal{I})$ and $\mathbf{Q} \in \mathbb{NS}(\mathcal{J})$, defined by the elements

$$\begin{aligned} (P * Q)(ab | xy) &= P(ab | xy), & \text{A}_x, \text{B}_y \in \mathcal{I}, \\ (P * Q)(ab | xv) &= P_{\text{A}}(a | x) Q_{\text{B}}(b | v), & \text{A}_x \in \mathcal{I}, \text{B}_v \in \mathcal{J}, \\ (P * Q)(ab | \xi y) &= Q_{\text{A}}(a | \xi) P_{\text{B}}(b | y), & \text{A}_{\xi} \in \mathcal{J}, \text{B}_y \in \mathcal{I}, \\ (P * Q)(ab | \xi v) &= Q(ab | \xi v) & \text{A}_{\xi}, \text{B}_v \in \mathcal{J}. \end{aligned} \quad (\text{A.40})$$

Note that the definition still applies if either $\mathcal{I}$ or $\mathcal{J}$ or both contain inputs by only one party. For instance, if $\mathcal{I} = \{\text{A}_0, \dots\}$ contains only inputs available

to Alice (in which case, $\mathbf{P}_A \in \mathbb{NS}(\mathcal{I})$ is a marginal behaviour of elements $P_A(a | x)$, $A_x \in \mathcal{I}$), the product definition simply reduces to the second and fourth lines of (A.40). If $\mathcal{I}$ does not contain any of Bob's inputs and $\mathcal{J}$ does not contain any of Alice's inputs, then the product $\mathbf{P}_A * \mathbf{P}_B$ of $\mathbf{P}_A \in \mathbb{NS}(\mathcal{I})$ and $\mathbf{P}_B \in \mathbb{NS}(\mathcal{J})$ is the uncorrelated behaviour of elements

$$(P_A * P_B)(ab | xy) = P_A(a | x)P_B(b | y) \quad A_x \in \mathcal{I}, B_y \in \mathcal{J}. \quad (\text{A.41})$$

If $\mathcal{I}$ and $\mathcal{J}$ both only contain (say) Alice's inputs, we take the combination $\mathbf{P}_A * \mathbf{Q}_A$ to be defined such that

$$\begin{aligned} (P_A * Q_A)(a | x) &= P_A(a | x), & A_x \in \mathcal{I}, \\ (P_A * Q_A)(a | \xi) &= Q_A(a | \xi), & A_\xi \in \mathcal{J}. \end{aligned} \quad (\text{A.42})$$

If one of the input sets is empty, e.g., $\mathcal{J} = \emptyset$ in which case $\mathbb{NS}(\emptyset) = 1$, $\mathbf{P} * 1 = \mathbf{P}$. Note that, in all cases, the product is bilinear in the sense that

$$\left(\sum_j p_j \mathbf{P}_j \right) * \left(\sum_k q_k \mathbf{Q}_k \right) = \sum_{jk} p_j q_k \mathbf{P}_j * \mathbf{Q}_k \quad (\text{A.43})$$

for convex coefficients satisfying $\sum_j p_j = \sum_k q_k = 1$. We also take $*$ to be commutative, i.e., we do not make a distinction between $\mathbf{P} * \mathbf{Q}$ and $\mathbf{Q} * \mathbf{P}$.

For sets $\mathbb{P} \subseteq \mathbb{NS}(\mathcal{I})$ and $\mathbb{Q} \subseteq \mathbb{NS}(\mathcal{J})$ of behaviours, the product $\mathbb{P} * \mathbb{Q}$ can be defined in the obvious way by

$$\mathbb{P} * \mathbb{Q} = \{ \mathbf{P} * \mathbf{Q} \mid \mathbf{P} \in \mathbb{P}, \mathbf{Q} \in \mathbb{Q} \}. \quad (\text{A.44})$$

If $\mathbb{P}$ and $\mathbb{Q}$ are convex, it does not generally follow that $\mathbb{P} * \mathbb{Q}$ is convex. Accordingly, we introduce a convex product $*_c$ obtained simply by taking the convex hull

$$\mathbb{P} *_c \mathbb{Q} = \text{Conv}(\mathbb{P} * \mathbb{Q}). \quad (\text{A.45})$$

Products of polytopes are polytopes

If $\mathbb{P} \subseteq \mathbb{NS}(\mathcal{I})$ and $\mathbb{Q} \subseteq \mathbb{NS}(\mathcal{J})$ are polytopes, then $\mathbb{P} *_c \mathbb{Q} \subseteq \mathbb{NS}(\mathcal{I} \cup \mathcal{J})$ is likewise a polytope. Furthermore, if we call $\text{Vert } \mathbb{P}$ the set of vertices of a polytope $\mathbb{P}$, the vertices of the convex product $\mathbb{P} *_c \mathbb{Q}$ are given by

$$\text{Vert}(\mathbb{P} *_c \mathbb{Q}) = \text{Vert } \mathbb{P} * \text{Vert } \mathbb{Q}. \quad (\text{A.46})$$

To see this, note that since $\mathbb{P} *_c \mathbb{Q}$ is by construction convex, any behaviour $\mathbf{R} \in \mathbb{P} *_c \mathbb{Q}$ can be expressed as a convex combination

$$\mathbf{R} = \sum_{\lambda} r_{\lambda} \mathbf{P}_{\lambda} * \mathbf{Q}_{\lambda} \quad (\text{A.47})$$

of elements $\mathbf{P}_\lambda * \mathbf{Q}_\lambda \in \mathbb{P} * \mathbb{Q}$. Since $\mathbb{P}$ and $\mathbb{Q}$ are polytopes, $\mathbf{P}_\lambda$ and $\mathbf{Q}_\lambda$ admit convex decompositions $\mathbf{P}_\lambda = \sum_j p_j^\lambda \mathbf{U}_j$ and $\mathbf{Q}_\lambda = \sum_k q_k^\lambda \mathbf{V}_k$ in terms of vertices $\mathbf{U}_j \in \text{Vert } \mathbb{P}$ and $\mathbf{V}_k \in \text{Vert } \mathbb{Q}$. $\mathbf{R}$ can thus be expressed as

$$\begin{aligned} \mathbf{R} &= \sum_\lambda \sum_{jk} r_\lambda p_j^\lambda q_k^\lambda \mathbf{U}_j * \mathbf{V}_k \\ &= \sum_{jk} s_{jk} \mathbf{U}_j * \mathbf{V}_k, \end{aligned} \tag{A.48}$$

in terms of the convex coefficients $s_{jk} = \sum_\lambda r_\lambda p_j^\lambda q_k^\lambda$.

Local polytope construction

The local polytope admits a simple construction in terms of the behaviour product we have just defined. Specifically, if $\mathcal{A} = \{A_x\}_x$ and $\mathcal{B} = \{B_y\}_y$ are Alice's and Bob's input sets and $\mathbb{L}(\mathcal{A})$ and $\mathbb{L}(\mathcal{B})$ are the respective sets of marginal behaviours $\{\mathbf{P}_A = (P_A(a | x))\}$ and $\{\mathbf{P}_B = (P_B(b | y))\}$, then as we remarked above, $\mathbb{L}(\mathcal{A}) * \mathbb{L}(\mathcal{B})$ is the set

$$\mathbb{L}(\mathcal{A}) * \mathbb{L}(\mathcal{B}) = \{\mathbf{P}_A * \mathbf{P}_B = (P_A(a | x) Q_B(b | y))\} \tag{A.49}$$

of uncorrelated behaviours and the identity

$$\mathbb{L}(\mathcal{A} \cup \mathcal{B}) = \mathbb{L}(\mathcal{A}) *_c \mathbb{L}(\mathcal{B}) \tag{A.50}$$

is a simple restatement of the definition of the local set given in section A.1. From the fact that $\mathbb{L}(\mathcal{A}) = \text{NS}(\mathcal{A})$ and $\mathbb{L}(\mathcal{B}) = \text{NS}(\mathcal{B})$ are polytopes, the construction (A.50) is a simple way to see that the local set $\mathbb{L}(\mathcal{A} \cup \mathcal{B})$ is, as was mentioned earlier, indeed a polytope. $\mathbb{L}(\mathcal{A})$ and $\mathbb{L}(\mathcal{B})$ can be further decomposed as

$$\mathbb{L}(\mathcal{A}) = \mathbb{L}(A_0) *_c \mathbb{L}(A_1) *_c \cdots *_c \mathbb{L}(A_{m_A-1}), \tag{A.51}$$

$$\mathbb{L}(\mathcal{B}) = \mathbb{L}(B_0) *_c \mathbb{L}(B_1) *_c \cdots *_c \mathbb{L}(B_{m_B-1}), \tag{A.52}$$

meaning that $\mathbb{L}(\mathcal{A} \cup \mathcal{B})$ can be fully decomposed as

$$\mathbb{L}(\mathcal{A} \cup \mathcal{B}) = \mathbb{L}(A_0) *_c \cdots *_c \mathbb{L}(A_{m_A-1}) *_c \mathbb{L}(B_0) *_c \cdots *_c \mathbb{L}(B_{m_B-1}). \tag{A.53}$$

It follows that the convex product of any two disjoint local polytopes, of input sets $\mathcal{I}$ and $\mathcal{J}$, is another local polytope:

$$\mathbb{L}(\mathcal{I}) *_c \mathbb{L}(\mathcal{J}) = \mathbb{L}(\mathcal{I} \cup \mathcal{J}). \tag{A.54}$$

Projections of behaviours

For an input subset $\mathcal{I} \subseteq S$ we define a projection operation $\Pi_{\mathcal{I}}$ such that, for a behaviour $\mathbf{P} \in \mathbb{NS}(S)$, $\Pi_{\mathcal{I}}\mathbf{P} \in \mathbb{NS}(\mathcal{I})$ is defined by the elements

$$(\Pi_{\mathcal{I}}P)(ab \mid \xi v) = P(ab \mid \xi v), \quad A_{\xi}, B_v \in \mathcal{I}. \quad (\text{A.55})$$

If $\mathcal{I}$ contains only one party's inputs, we take $\Pi_{\mathcal{I}}\mathbf{P}$ to be the corresponding marginal. For instance, if $\mathcal{I} \subseteq \mathcal{A}$,

$$(\Pi_{\mathcal{I}}P)(a \mid \xi) = P_A(a \mid \xi), \quad A_{\xi} \in \mathcal{I}. \quad (\text{A.56})$$

For the empty set of inputs, we define $\Pi_{\emptyset}\mathbf{P} = 1$. For a set $\mathbb{P} \subseteq \mathbb{NS}(S)$, its projection is the image

$$\Pi_{\mathcal{I}}\mathbb{P} = \{\Pi_{\mathcal{I}}\mathbf{P} \mid \mathbf{P} \in \mathbb{P}\}. \quad (\text{A.57})$$

Projecting product behaviours

The projection plays nicely with the behaviour product defined in the previous subsection. For $\mathbf{P} \in \mathbb{NS}(\mathcal{I})$ and $\mathbf{Q} \in \mathbb{NS}(\mathcal{J})$ with $\mathcal{I} \cap \mathcal{J} = \emptyset$, it is easy to see that

$$\Pi_{\mathcal{I}}(\mathbf{P} * \mathbf{Q}) = \mathbf{P}, \quad \Pi_{\mathcal{J}}(\mathbf{P} * \mathbf{Q}) = \mathbf{Q}. \quad (\text{A.58})$$

More generally, for an input set $\mathcal{K} \subseteq \mathcal{I} \cup \mathcal{J}$,

$$\Pi_{\mathcal{K}}(\mathbf{P} * \mathbf{Q}) = (\Pi_{\mathcal{K} \cap \mathcal{I}}\mathbf{P}) * (\Pi_{\mathcal{K} \cap \mathcal{J}}\mathbf{Q}). \quad (\text{A.59})$$

Using that $\mathbf{R} \in \mathbb{P} *_c \mathbb{Q}$ admits a decomposition $\mathbf{R} = \sum_{\lambda} r_{\lambda} \mathbf{P}_{\lambda} * \mathbf{Q}_{\lambda}$,

$$\Pi_{\mathcal{K}}\mathbf{R} = \sum_{\lambda} r_{\lambda} \Pi_{\mathcal{K}}(\mathbf{P}_{\lambda} * \mathbf{Q}_{\lambda}) = \sum_{\lambda} r_{\lambda} (\Pi_{\mathcal{K} \cap \mathcal{I}}\mathbf{P}_{\lambda}) * (\Pi_{\mathcal{K} \cap \mathcal{J}}\mathbf{Q}_{\lambda}), \quad (\text{A.60})$$

which establishes that

$$\Pi_{\mathcal{K}}(\mathbb{P} *_c \mathbb{Q}) = (\Pi_{\mathcal{K} \cap \mathcal{I}}\mathbb{P}) *_c (\Pi_{\mathcal{K} \cap \mathcal{J}}\mathbb{Q}). \quad (\text{A.61})$$

Projections of no-signalling and local polytopes are always no-signalling and local polytopes, respectively. Specifically,

$$\Pi_{\mathcal{I}}\mathbb{L}(S) = \mathbb{L}(\mathcal{I}), \quad (\text{A.62})$$

$$\Pi_{\mathcal{I}}\mathbb{NS}(S) = \mathbb{NS}(\mathcal{I}). \quad (\text{A.63})$$

A.3 Partial determinism

A.3.1 Definition and basic properties

Deterministic points

In order to define the partially deterministic polytopes for a (bipartite) Bell scenario $\mathcal{S} = \mathcal{A} \cup \mathcal{B}$, we first define a class of deterministic behaviours. For a given behaviour $\mathbf{P} \in \mathbb{NS}(\mathcal{S})$ in (say) a bipartite Bell scenario, we will say that $\mathbf{P}$ is *deterministic on an input* (say) $A_x \in \mathcal{S}$ if the corresponding marginal is deterministic, i.e., if $P_A(a \mid x) \in \{0, 1\}$ or, equivalently, $\Pi_{A_x} \mathbf{P} \in \text{Vert } \mathbb{L}(A_x)$. We call the corresponding set of deterministic points $\text{DetPts}_{A_x}(\mathcal{S})$. More generally, we will say that $\mathbf{P}$ is *deterministic on a set* $\mathcal{I} \subseteq \mathcal{S}$ of inputs if $\mathbf{P}$ is deterministic on each input in $\mathcal{I}$, and we call the corresponding set $\text{DetPts}_{\mathcal{I}}(\mathcal{S})$, such that

$$\mathbf{P} \in \text{DetPts}_{\mathcal{I}} \Leftrightarrow P_A(a \mid \xi), P_B(b \mid v) \in \{0, 1\}, \quad \forall A_\xi, B_v \in \mathcal{I}, \quad (\text{A.64})$$

or equivalently,

$$\begin{aligned} \mathbf{P} \in \text{DetPts}_{\mathcal{I}} \Leftrightarrow & \Pi_{A_\xi} \mathbf{P} \in \text{Vert } \mathbb{L}(A_\xi), \\ & \Pi_{B_v} \mathbf{P} \in \text{Vert } \mathbb{L}(B_v), \quad \forall A_\xi, B_v \in \mathcal{I}. \end{aligned} \quad (\text{A.65})$$

By convention, we will consider every no-signalling behaviour to be deterministic on the empty set $\mathcal{I} = \emptyset$, such that $\text{DetPts}_{\emptyset}(\mathcal{S}) = \mathbb{NS}(\mathcal{S})$. Note that if $\mathcal{I}, \mathcal{J} \subseteq \mathcal{S}$ are two sets of inputs, then

$$\text{DetPts}_{\mathcal{I} \cup \mathcal{J}} = \text{DetPts}_{\mathcal{I}} \cap \text{DetPts}_{\mathcal{J}}. \quad (\text{A.66})$$

In particular, a trivial restatement of the definition of $\text{DetPts}_{\mathcal{I}}$ is that it can be identified as the intersection of the sets of behaviours that are deterministic on each input in $\mathcal{I}$:

$$\text{DetPts}_{\mathcal{I}} = \bigcap_{C \in \mathcal{I}} \text{DetPts}_C. \quad (\text{A.67})$$

The sets of deterministic behaviours can be constructed from the behaviour product operation. To see this, consider a point $\mathbf{P} \in \text{DetPts}_{\mathcal{I}}(\mathcal{S})$ and an input $A_\xi \in \mathcal{I}$, and call $a_\xi \in A_\xi$ the output for which $D_A(a_\xi \mid \xi) = \sum_b P(ab \mid \xi y) = 1$. For the other outputs, normalisation implies

$$D_A(a \mid \xi) = \sum_b P(ab \mid \xi y) = 0, \quad \forall a \neq a_\xi \quad (\text{A.68})$$

from which we extract

$$P(ab \mid \xi y) = 0 \quad \forall a \neq a_\xi. \quad (\text{A.69})$$

The relation

$$P_B(b \mid y) = \sum_a P(ab \mid \xi y) = P(a_\xi b \mid \xi y) \quad (\text{A.70})$$

for Bob's marginal then implies that, for both $a = a_\xi$ and $a \neq a_\xi$,

$$P(ab \mid \xi y) = D_A(a \mid \xi) P_B(b \mid y). \quad (\text{A.71})$$

A similar result holds for any of Bob's inputs in $\mathcal{I}$. Combining these, we find that $\mathbf{P}$ is completely determined by the deterministic marginals $D_B(a \mid \xi)$ and $D_B(b \mid v)$ for $A_\xi, B_v \in \mathcal{I}$ and the projection $\Pi_{\mathcal{S} \setminus \mathcal{I}} \mathbf{P}$ on the remaining inputs $\mathcal{S} \setminus \mathcal{I}$:

$$P(ab \mid \xi v) = D_A(a \mid \xi) D_B(b \mid v), \quad A_\xi, B_v \in \mathcal{I}, \quad (\text{A.72})$$

$$P(ab \mid \xi y) = D_A(a \mid \xi) P_B(b \mid y), \quad A_\xi \in \mathcal{I}, B_y \in \mathcal{S} \setminus \mathcal{I}, \quad (\text{A.73})$$

$$P(ab \mid xv) = P_A(a \mid x) D_B(b \mid v), \quad A_x \in \mathcal{S} \setminus \mathcal{I}, B_v \in \mathcal{I}. \quad (\text{A.74})$$

Put differently, $\mathbf{P} = (\Pi_{\mathcal{I}} \mathbf{P}) * (\Pi_{\mathcal{S} \setminus \mathcal{I}} \mathbf{P})$, with $\Pi_{\mathcal{I}} \mathbf{P} \in \text{Vert } \mathbb{L}(\mathcal{I})$. The sets of deterministic points can thus be expressed as

$$\text{DetPts}_{\mathcal{I}}(\mathcal{S}) = \text{Vert } \mathbb{L}(\mathcal{I}) * \mathbb{NS}(\mathcal{S} \setminus \mathcal{I}). \quad (\text{A.75})$$

The partially deterministic polytopes

We finally define the partially deterministic polytope $\mathbb{D}_{\mathcal{I}}(\mathcal{S})$ for an input set $\mathcal{I} \subseteq \mathcal{S}$ and Bell scenario $\mathcal{S} = \mathcal{A} \cup \mathcal{B}$ simply as the convex hull of the corresponding set of deterministic points $\text{DetPts}_{\mathcal{I}}$:

$$\mathbb{D}_{\mathcal{I}}(\mathcal{S}) = \text{Conv } \text{DetPts}_{\mathcal{I}}(\mathcal{S}). \quad (\text{A.76})$$

Taking the convex hull of (A.75), the partially deterministic polytopes can also be expressed as products of local and no-signalling polytopes:

$$\mathbb{D}_{\mathcal{I}}(\mathcal{S}) = \mathbb{L}(\mathcal{I}) *_c \mathbb{NS}(\mathcal{S} \setminus \mathcal{I}). \quad (\text{A.77})$$

From either expression, we see that we recover both the local and no-signalling polytopes as special cases:

$$\mathbb{D}_{\emptyset}(\mathcal{S}) = \mathbb{NS}(\mathcal{S}), \quad (\text{A.78})$$

$$\mathbb{D}_{\mathcal{S}}(\mathcal{S}) = \mathbb{L}(\mathcal{S}). \quad (\text{A.79})$$

From the relation $\text{DetPts}_{\mathcal{I} \cup \mathcal{J}}(\mathcal{S}) = \text{DetPts}_{\mathcal{I}}(\mathcal{S}) \cap \text{DetPts}_{\mathcal{J}}(\mathcal{S})$ and using that in general $\text{Conv}(\mathbb{P} \cup \mathbb{Q}) \subseteq \text{Conv}(\mathbb{P}) \cap \text{Conv}(\mathbb{Q})$, we find that the partially deterministic polytopes form a hierarchy in the sense that

$$\mathbb{D}_{\mathcal{I} \cup \mathcal{J}}(\mathcal{S}) \subseteq \mathbb{D}_{\mathcal{I}}(\mathcal{S}) \cap \mathbb{D}_{\mathcal{J}}(\mathcal{S}). \quad (\text{A.80})$$

Projection and relation to local polytope

Expressing $\mathbb{D}_{\mathcal{I}}(\mathcal{S})$ in the product form (A.77) this way allows a few properties of the partially deterministic polytopes to be derived from properties of the behaviour product and projection operations derived earlier. For instance, it is sufficient to impose determinism on all but one of only (say) Alice's inputs in order to recover the local polytope. For an excluded input $A \in \mathcal{A}$,

$$\begin{aligned}\mathbb{D}_{\mathcal{A} \setminus A}(\mathcal{A} \cup \mathcal{B}) &= \mathbb{L}(\mathcal{A} \setminus A) *_c \mathbb{NS}(A \cup \mathcal{B}) \\ &= \mathbb{L}(\mathcal{A} \setminus A) *_c \mathbb{L}(A \cup \mathcal{B}) \\ &= \mathbb{L}(\mathcal{A} \cup \mathcal{B}),\end{aligned}\tag{A.81}$$

where we used that, with only one input A on Alice's side, the distinction between the no-signalling and local polytopes $\mathbb{NS}(A \cup \mathcal{B})$ and $\mathbb{L}(A \cup \mathcal{B})$ collapses. Projections of partially deterministic polytopes are also partially deterministic polytopes. For $\mathcal{I}, \mathcal{J} \subseteq \mathcal{S}$,

$$\begin{aligned}\Pi_{\mathcal{J}} \mathbb{D}_{\mathcal{I}}(\mathcal{S}) &= \Pi_{\mathcal{J}}(\mathbb{L}(\mathcal{I}) *_c \mathbb{NS}(\mathcal{S} \setminus \mathcal{I})) \\ &= (\Pi_{\mathcal{J} \cap \mathcal{I}} \mathbb{L}(\mathcal{I})) *_c (\Pi_{\mathcal{J} \cap (\mathcal{S} \setminus \mathcal{I})} \mathbb{NS}(\mathcal{S} \setminus \mathcal{I})) \\ &= \mathbb{L}(\mathcal{J} \cap \mathcal{I}) *_c \mathbb{NS}(\mathcal{J} \cap (\mathcal{S} \setminus \mathcal{I})) \\ &= \mathbb{D}_{\mathcal{J} \cap \mathcal{I}}((\mathcal{J} \cap \mathcal{I}) \cup [\mathcal{J} \cap (\mathcal{S} \setminus \mathcal{I})]) \\ &= \mathbb{D}_{\mathcal{J} \cap \mathcal{I}}(\mathcal{J} \cap \mathcal{S}).\end{aligned}\tag{A.82}$$

For a subset $\mathcal{A}' \subset \mathcal{A}$ of Alice's inputs, projecting with an additional input $A \in \mathcal{A} \setminus \mathcal{A}'$ on Alice's side always results in a local polytope:

$$\begin{aligned}\Pi_{A \cup \mathcal{A}' \cup \mathcal{B}} \mathbb{D}_{\mathcal{A}'}(\mathcal{S}) &= \mathbb{D}_{\mathcal{A}'}(A \cup \mathcal{A}' \cup \mathcal{B}) \\ &= \mathbb{L}(\mathcal{A}') *_c \mathbb{NS}(A \cup \mathcal{B}) \\ &= \mathbb{L}(A \cup \mathcal{A}' \cup \mathcal{B}).\end{aligned}\tag{A.83}$$

Relevance to device-independent randomness

The partially deterministic polytopes $\mathbb{D}_{A_0}$ and $\mathbb{D}_{A_0 B_0}$ are precisely the sets of behaviours for which no randomness, as measured by the respective guessing probabilities, can be certified device independently. Specifically,

$$\mathbb{G}_{A_0}(\mathcal{S})_1 = \mathbb{D}_{A_0}(\mathcal{S}),\tag{A.84}$$

$$\mathbb{G}_{A_0 B_0}(\mathcal{S})_1 = \mathbb{D}_{A_0 B_0}(\mathcal{S}).\tag{A.85}$$

The equivalence is easily seen simply by expressing that, for instance for $\mathbf{P} \in \mathbb{G}_{A_0}(\mathcal{S})_1$, $\mathbf{P}$ by definition admits a decomposition $\{p_e, \mathbf{P}^{(e)}\}$ such that

$$G_{A_0}(\mathbf{P}) = \sum_a p_a P_A^{(a)}(a | 0) = 1, \quad (\text{A.86})$$

which necessarily implies that $P_A^{(e)}(a | 0) = \delta_{ae}$. This implies that any $\mathbf{P} \in \mathbb{G}_{A_0}$ can be expressed as a convex sum of deterministic points $\mathbf{P}^{(e)} \in \text{DetPts}_{A_0}(\mathcal{S})$, i.e., that $\mathbf{P} \in \text{Conv DetPts}_{A_0}(\mathcal{S}) = \mathbb{D}_{A_0}(\mathcal{S})$. Conversely, if $\mathbf{D} \in \mathbb{D}_{A_0}(\mathcal{S})$ is a deterministic point on A_0 , then $1 \geq G_{A_0}(\mathbf{D}) \geq \max_a D_A(a | 0) = 1$. Since $\mathbb{G}_{A_0}(\mathcal{S})$ is convex, $\mathbf{P} \in \mathbb{D}_{A_0}(\mathcal{S}) \Rightarrow \mathbf{P} \in \mathbb{G}_{A_0}(\mathcal{S})$.

The equivalence $\mathbb{G}_{A_0 B_0}(\mathcal{S})_1 = \mathbb{D}_{A_0 B_0}(\mathcal{S})$ can be shown in an analogous manner.

A.3.2 Local projections

One fixed input

Earlier, for $\mathcal{A}' \subset \mathcal{A}$, we showed as (A.83) that projections of the type $\Pi_{A \cup \mathcal{A}' \cup \mathcal{B}} \mathbb{D}_{\mathcal{A}'}(\mathcal{S})$ for $A \in \mathcal{A} \setminus \mathcal{A}'$ are local polytopes. In the special case where only one input is fixed, for instance $\mathcal{A}' = \{A_0\}$, we will show that this fully characterises the partially deterministic polytope, i.e., we demonstrate the equivalence

$$\mathbf{P} \in \mathbb{D}_{A_0}(\mathcal{A} \cup \mathcal{B}) \quad \Leftrightarrow \quad \Pi_{A_0 \cup A_\xi \cup \mathcal{B}} \mathbf{P} \in \mathbb{L}(A_0 \cup A_\xi \cup \mathcal{B}), \quad \forall A_\xi \in \mathcal{A} \setminus A_0. \quad (\text{A.87})$$

To this end, suppose $\mathbf{P} \in \text{NS}(\mathcal{A} \cup \mathcal{B})$ is such that each projection $\Pi_{A_0 \cup A_\xi \cup \mathcal{B}} \mathbf{P}$ for $A_\xi \in \mathcal{A} \setminus A_0$ is local. Applying the expression (A.21) for a local behaviour for each input A_ξ , it follows that the elements of $\mathbf{P}$ admit a decomposition of the form

$$P(ab | xy) = \sum_{a_0, a_\xi} p^{(\xi)}(a_0, a_\xi) \delta_{a, a_x} P_B^{(\xi)}(b | y; a_0, a_\xi), \quad (\text{A.88})$$

with the index ξ indicating that the local model may depend on the input A_ξ . Setting $p^{(\xi)}(a_0, a_\xi) = p^{(\xi)}(a_0) p^{(\xi)}(a_\xi | a_0)$,

$$P(ab | xy) = \sum_{a_0} p^{(\xi)}(a_0) \sum_{a_\xi} p^{(\xi)}(a_\xi | a_0) \delta_{a, a_x} P_B^{(\xi)}(b | y; a_0, a_\xi). \quad (\text{A.89})$$

The marginal $P_A(a \mid 0)$ evaluates to $p^{(\xi)}(a_0)$. Since this should be independent of ξ , the index can be discarded. Evaluating (A.89) separately for $x = 0$ and $x = \xi$, we find

$$P(ab \mid 0y) = p(a) \sum_{a_\xi} p^{(\xi)}(a_\xi \mid a) P_B^{(\xi)}(b \mid y; a, a_\xi), \quad (\text{A.90})$$

$$P(ab \mid \xi y) = \sum_{a_0} p(a_0) p^{(\xi)}(a \mid a_0) P_B^{(\xi)}(b \mid y; a_0, a), \quad (\text{A.91})$$

with $p(a) = P_A(a \mid 0)$. Inserting the deterministic distributions $D_A^\lambda \in \text{Vert } \mathbb{L}(A_0)$ of elements $D_A^\lambda(a \mid 0) = \delta_{a,\lambda}$ into (A.90) and renaming some of the indices,

$$P(ab \mid 0y) = \sum_{\lambda} p(\lambda) D_A^\lambda(a \mid 0) \sum_{a'} p^{(\xi)}(a' \mid \lambda) P^{(\xi)}(b \mid y; \lambda, a'), \quad (\text{A.92})$$

$$P(ab \mid \xi y) = \sum_{\lambda} p(\lambda) p^{(\xi)}(a \mid \lambda) P^{(\xi)}(b \mid y; \lambda, a). \quad (\text{A.93})$$

We set $\mathbf{P}^\lambda \in \mathbb{NS}(\mathcal{A} \cup \mathcal{B} \setminus A_0)$ to be the behaviour of elements

$$P^\lambda(ab \mid \xi y) = p^{(\xi)}(a \mid \lambda) P_B^{(\xi)}(b \mid y; \lambda, a). \quad (\text{A.94})$$

That $\mathbf{P}^\lambda$ is a no-signalling distribution can be verified by checking the marginals. We find that $P_A^\lambda(a \mid \xi) = p^{(\xi)}(a \mid \lambda)$, which is independent of y . The marginal $P_B^\lambda(b \mid y) = \sum_a p^{(\xi)}(a \mid \lambda) P_B^{(\xi)}(b \mid y; \lambda, a)$ is likewise independent of ξ , as can be seen from its appearance in (A.90) which does not depend on ξ . The elements of $P(ab \mid xy)$ thus simplify to

$$P(ab \mid 0y) = \sum_{\lambda} p(\lambda) D_A^\lambda(a \mid 0) P_B^\lambda(b \mid y), \quad (\text{A.95})$$

$$P(ab \mid \xi y) = \sum_{\lambda} p(\lambda) P^\lambda(ab \mid \xi y). \quad (\text{A.96})$$

More compactly, we have shown that $\mathbf{P}$ can be expressed in the form

$$\mathbf{P} = \sum_{\lambda} p(\lambda) D_A^\lambda * \mathbf{P}^\lambda, \quad (\text{A.97})$$

with

$$\begin{aligned} D_A^\lambda * \mathbf{P}^\lambda &\in \text{Vert } \mathbb{L}(A_0) * \mathbb{NS}(\mathcal{A} \cup \mathcal{B} \setminus A_0) \\ &= \text{DetPts}_{A_0}(\mathcal{A} \cup \mathcal{B}), \end{aligned} \quad (\text{A.98})$$

confirming that $\mathbf{P} \in \mathbb{D}_{A_0}(\mathcal{A} \cup \mathcal{B})$.

Counterexample for two fixed inputs

The previous result does not generalise for projections $\Pi_{A \cup A' \cup B} \mathbb{D}_{A'}(\mathcal{A} \cup \mathcal{B})$ for which $A' \subset \mathcal{A}$ contains more than one of Alice's inputs. We demonstrate this by an explicit counterexample. The smallest candidate case, which proved to be sufficient, is the $\mathbb{D}_{22}(4322) = \mathbb{D}_{A_0 A_1}(4322)$ polytope in the scenario $\mathcal{S} = 4322 = \{A_0, A_1, A_2, A_3, B_0, B_1, B_2\}$ in which all the measurements are dichotomic, i.e., $A_x, B_y \simeq \{0, 1\}$. We demonstrate this by showing that $\mathbb{D}_{22}(4322)$ has facets that are not facets of the $\mathbb{L}(3322)$ local polytope, indicating that, in order to tell whether a behaviour $\mathbf{P}$ is in $\mathbb{D}_{22}(4322)$, it is not sufficient to check that the two projections $\Pi_{\mathcal{S} \setminus A_2} \mathbf{P}$ and $\Pi_{\mathcal{S} \setminus A_3} \mathbf{P}$ of $\mathbf{P}$ are in the corresponding local polytopes $\mathbb{L}(\mathcal{S} \setminus A_2)$ and $\mathbb{L}(\mathcal{S} \setminus A_3)$.

The facets of $\mathbb{L}(3322)$ and $\mathbb{L}(4322)$ are all known and are listed for convenience in appendix A.A. Deriving all the facets of $\mathbb{D}_{22}(4322)$ given its vertices $\text{Vert } \mathbb{L}(22) * \text{Vert } \mathbb{NS}(2322)$ proved to be infeasible. Instead, we exhibited the counterexample in the following way: we first used `polymake` [109] (which itself uses `cddlib` [110]) to generate all the facets of $\mathbb{L}(4322)$, which are listed in appendix A.A.2. We then generated all the vertices of $\mathbb{D}_{22}(4322)$, tested each vertex against each facet of $\mathbb{L}(4322)$, and removed those facets which were violated by a partially deterministic vertex. We finally grouped the remaining facets into equivalence classes, where we consider two facets to be equivalent if one can be obtained from the other by arbitrary permutations of outputs and arbitrary permutations within the sets $\{A_0, A_1\}$, $\{A_2, A_3\}$, and $\{B_0, B_1, B_2\}$ of inputs.

Representative facets from each remaining class, listed with multiplicity, are:

$$1536 \times \begin{array}{c|ccc} & 1 & 1 & 1 \\ \hline 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ \hline & 0 & 0 & 0 \end{array} \leq 5, \quad (\text{A.99})$$

$$768 \times \begin{array}{c|ccc} & 2 & 1 & 1 \\ \hline 0 & 0 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 \\ \hline & 1 & 1 & 0 \end{array} \leq 6, \quad (\text{A.100})$$

$$768 \times \begin{array}{c|ccc} & -1 & -1 & 1 \\ \hline 0 & 1 & -1 & 0 \\ 1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \leq 4, \quad (\text{A.101})$$

$$\begin{array}{c|cccc}
& & 1 & 1 & 0 \\
\hline
384 \times & 1 & -1 & -1 & 1 \\
& 1 & -1 & -1 & -1 \\
& 0 & 1 & -1 & 0 \\
& 0 & 0 & 0 & 0 \\
& & 0 & 0 & 0 \\
\hline
\end{array} \leq 4, \quad (\text{A.102})$$

$$\begin{array}{c|cccc}
& & 1 & 1 & 0 \\
\hline
96 \times & 0 & 1 & 1 & 0 \\
& 0 & 0 & 0 & 0 \\
& 0 & 1 & -1 & 0 \\
& 0 & 0 & 0 & 0 \\
& & 0 & 0 & 0 \\
\hline
\end{array} \leq 2, \quad (\text{A.103})$$

$$\begin{array}{c|cccc}
& & 1 & 1 & 0 \\
\hline
24 \times & 0 & 1 & -1 & 0 \\
& 0 & 0 & 0 & 0 \\
& 0 & 0 & 0 & 0 \\
& & 1 & 0 & 0 \\
\hline
\end{array} \leq 2, \quad (\text{A.104})$$

$$\begin{array}{c|cccc}
& & 1 & 0 & 0 \\
\hline
24 \times & 0 & 0 & 0 & 0 \\
& 0 & 0 & 0 & 0 \\
& 1 & -1 & 0 & 0 \\
& 0 & 0 & 0 & 0 \\
& & 1 & 0 & 0 \\
\hline
\end{array} \leq 1, \quad (\text{A.105})$$

$$\begin{array}{c|cccc}
& & 1 & 0 & 0 \\
\hline
24 \times & 1 & -1 & 0 & 0 \\
& 0 & 0 & 0 & 0 \\
& 0 & 0 & 0 & 0 \\
& 0 & 0 & 0 & 0 \\
& & 1 & 0 & 0 \\
\hline
\end{array} \leq 1. \quad (\text{A.106})$$

In particular, the facet classes (A.99) and (A.100) involve all four of Alice's inputs and are not $\mathbb{L}(3322)$ facets.

A.3.3 The $\mathbb{D}_{1122}(3322)$ polytope

Earlier we showed that, for input sets $\mathcal{I}, \mathcal{J} \in \mathcal{S}$, the partially deterministic polytopes form a hierarchy in that $\mathbb{D}_{\mathcal{I} \cup \mathcal{J}}(\mathcal{S}) \subseteq \mathbb{D}_{\mathcal{I}}(\mathcal{S}) \cap \mathbb{D}_{\mathcal{J}}(\mathcal{S})$. Generally, the inclusion is strict, i.e., generally, $\mathbb{D}_{\mathcal{I} \cup \mathcal{J}}(\mathcal{S}) \neq \mathbb{D}_{\mathcal{I}}(\mathcal{S}) \cap \mathbb{D}_{\mathcal{J}}(\mathcal{S})$. We have already given an example in the previous subsection, in that we found that $\mathbb{D}_{A_0 A_1}(4322)$ possesses facets which are not facets of either $\mathbb{D}_{A_0}(4322)$ or $\mathbb{D}_{A_1}(4322)$ (the latter possess only CHSH-type facets). Here, we show that $\mathbb{D}_{A_0 B_0}(3322) \neq \mathbb{D}_{A_0}(3322) \cap \mathbb{D}_{B_0}(3322)$. This is done in two ways: we first used `polymake` to determine the vertices of $\mathbb{D}_{A_0}(3322) \cap \mathbb{D}_{B_0}(3322)$. The first vertex $\mathbf{P}_*$ found this way – a behaviour for which $G_{A_0}(\mathbf{P}) = 1$ and $G_{B_0}(\mathbf{P}) = 1$ but $G_{A_0 B_0}(\mathbf{P}) \neq 1$ – was not a vertex of $\mathbb{D}_{A_0 B_0}(3322)$. We also give a complete characterisation of $\mathbb{D}_{1122}(3322)$ in terms of its facets.

Vertex of $\mathbb{D}_{A_0} \cap \mathbb{D}_{B_0}$

We determined the vertices of $\mathbb{D}_{A_0} \cap \mathbb{D}_{B_0}$ by taking the union of the facets of $\mathbb{D}_{A_0}$ and $\mathbb{D}_{B_0}$ and solving for the vertices. The first vertex found was the behaviour

$$\mathbf{P}_* = \left[\begin{array}{cc|cc|cc} 1/3 & 1/3 & 1/3 & 1/3 & 0 & 2/3 \\ 1/3 & 0 & 0 & 1/3 & 1/3 & 0 \\ \hline 1/3 & 0 & 0 & 1/3 & 0 & 1/3 \\ 1/3 & 1/3 & 1/3 & 1/3 & 1/3 & 1/3 \\ \hline 0 & 1/3 & 0 & 1/3 & 0 & 1/3 \\ 2/3 & 0 & 1/3 & 1/3 & 1/3 & 1/3 \end{array} \right]. \quad (\text{A.107})$$

It is easy to verify that this point is in (for instance) $\mathbb{D}_{A_0}$ analytically. Specifically, it can be uniquely decomposed as

$$\frac{2}{3} \left[\begin{array}{cc|cc|cc} 1/2 & 1/2 & 1/2 & 1/2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 1/2 & 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 & 0 & 0 & 1/2 \\ \hline 0 & 1/2 & 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 & 1/2 & 0 \end{array} \right] + \frac{1}{3} \left[\begin{array}{cc|cc|cc} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \end{array} \right]. \quad (\text{A.108})$$

Since both terms are deterministic on the input A_0 , $G_{A_0}(\mathbf{P}_*) = 1$. $\mathbf{P}_*$ is symmetric under exchange of the two parties and admits a similar decomposition for which the input B_0 becomes deterministic, i.e., we also have $G_{B_0}(\mathbf{P}_*) = 1$. In either case, $\mathbf{P}_*$ cannot be further decomposed (the first term above contains a PR box which fixes the rest of the table), which implies that $G_{A_0 B_0}(\mathbf{P}_*) \neq 1$.

Facets of $\mathbb{D}_{1122}(3322)$

We used `polymake` to compute the facets of $\mathbb{D}_{1122}(3322)$ given its vertices $\text{Vert } \mathbb{L}(1122) * \text{Vert } \mathbb{N}\mathbb{S}(2222)$. Grouped into equivalence classes, the full list of facets is

$$256 \times \left[\begin{array}{c|ccc} & 1 & 1 & 0 \\ 1 & -1 & -2 & 0 \\ 1 & -2 & 2 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right] \leq 6, \quad (\text{A.109})$$

$$128 \times \left[\begin{array}{c|ccc} & 1 & 0 & 0 \\ 1 & -1 & 2 & 0 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & -1 \end{array} \right] \leq 5, \quad (\text{A.110})$$

$$128 \times \begin{array}{c|ccc} & 1 & 1 & 1 \\ \hline 1 & -1 & -1 & -1 \\ 0 & 2 & -1 & -1 \\ 0 & 0 & 1 & -1 \end{array} \leq 5, \quad (\text{A.111})$$

$$256 \times \begin{array}{c|ccc} & 1 & 1 & 0 \\ \hline 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & 0 \end{array} \leq 4, \quad (\text{A.112})$$

$$64 \times \begin{array}{c|ccc} & 0 & 1 & 1 \\ \hline 0 & 0 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & -1 \end{array} \leq 4, \quad (\text{A.113})$$

$$32 \times \begin{array}{c|ccc} & 0 & 0 & 0 \\ \hline 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \leq 2, \quad (\text{A.114})$$

$$16 \times \begin{array}{c|ccc} & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \leq 2, \quad (\text{A.115})$$

$$16 \times \begin{array}{c|ccc} & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \leq 2, \quad (\text{A.116})$$

$$4 \times \begin{array}{c|ccc} & 1 & 0 & 0 \\ \hline 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \leq 1, \quad (\text{A.117})$$

$$8 \times \begin{array}{c|ccc} & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \leq 1, \quad (\text{A.118})$$

$$8 \times \begin{array}{c|ccc} & 0 & 1 & 0 \\ \hline 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \leq 1, \quad (\text{A.119})$$

$$16 \times \begin{array}{c|ccc} & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \leq 1. \quad (\text{A.120})$$

In total there are 932 facets, of which a total of 512 facets in 3 equivalence classes [equations (A.109), (A.110), and (A.111)] are not local facets.

A.A Relevant known local facets

A.A.1 Facets of the 3322 local polytope

The 3322 local polytope was first fully characterised in terms of its facets by Froissart [111]. It has a total of 684 facets. Up to permutations of inputs and outputs these form three equivalence classes (the Froissart/ $\mathbf{I}_{3322}$, CHSH, and positivity facets). In the expectation value projection these are:

$$576 \times \begin{array}{c|ccc} & 1 & 1 & 0 \\ \hline 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & -1 \\ 0 & 1 & -1 & 0 \end{array} \leq 4, \quad (\text{A.121})$$

$$72 \times \begin{array}{c|ccc} & 0 & 0 & 0 \\ \hline 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \leq 2, \quad (\text{A.122})$$

$$36 \times \begin{array}{c|ccc} & 1 & 0 & 0 \\ \hline 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \leq 1. \quad (\text{A.123})$$

A.A.2 Facets of the 4322 local polytope

Collins and Gisin gave the full list of facets for the 4322 local polytope in [106]. In total it has 12480 facets in 6 distinct equivalence classes. In the expectation value projection, these are:

$$2304 \times \begin{array}{c|ccc} & 2 & 0 & 0 \\ \hline 1 & -1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & -1 & -1 \end{array} \leq 6, \quad (\text{A.124})$$

$$\begin{array}{c}
4608 \times \\
\begin{array}{c|ccc}
& 0 & 0 & 0 \\
\hline
1 & 1 & 1 & 1 \\
1 & -1 & -1 & -1 \\
0 & 2 & -1 & -1 \\
0 & 0 & 1 & -1 \\
\hline
& 1 & 1 & 1
\end{array}
\end{array} \leq 6, \quad (\text{A.125})$$

$$\begin{array}{c}
3072 \times \\
\begin{array}{c|ccc}
& -1 & -1 & -1 \\
\hline
1 & -1 & -1 & -1 \\
0 & 1 & -1 & 0 \\
0 & 1 & 0 & -1 \\
0 & 0 & 1 & -1 \\
\hline
& 1 & 1 & 0
\end{array}
\end{array} \leq 5, \quad (\text{A.126})$$

$$\begin{array}{c}
2304 \times \\
\begin{array}{c|ccc}
& -1 & -1 & 1 \\
\hline
1 & -1 & -1 & 1 \\
1 & -1 & -1 & -1 \\
0 & 1 & -1 & 0 \\
0 & 0 & 0 & 0 \\
\hline
& 0 & 0 & 0
\end{array}
\end{array} \leq 4, \quad (\text{A.127})$$

$$\begin{array}{c}
144 \times \\
\begin{array}{c|ccc}
& 1 & 1 & 0 \\
\hline
0 & 1 & 1 & 0 \\
0 & 1 & -1 & 0 \\
0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 \\
\hline
& 1 & 0 & 0
\end{array}
\end{array} \leq 2, \quad (\text{A.128})$$

$$\begin{array}{c}
48 \times \\
\begin{array}{c|ccc}
& -1 & 0 & 0 \\
\hline
1 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0
\end{array}
\end{array} \leq 1. \quad (\text{A.129})$$

A.B Example **polymake** sessions

A collection of Perl subroutines that were run from **polymake** to generate the vertices and facets in this section are listed here, along with a few example sessions illustrating their use. Between them, the subroutines in question can:

- generate vertices and facets for the local and no-signalling polytopes in $m_A m_B d_A d_B$ Bell scenarios,
- combine vertices (e.g. to compute the vertices of a partially deterministic polytope),
- group both vertices and facets in $m_A m_B 22$ scenarios into equivalence classes, defined by lists of allowed permutations of Alice's and Bob's

inputs (e.g., $\{\{A_0\}, \{A_1, A_2\}\}$ and $\{\{B_0\}, \{B_1, B_2\}\}$).

- print and generate L^AT_EX output for vertices and (for dichotomic outputs) facets.

The code in question is not intended to be particularly elegant or efficient. A worthwhile longer-term approach may be to incorporate some of the needed functionality described here into the `faacets` project [112, 113]. The remainder of this section gives a few example `polymake` sessions illustrating the use of the subroutines.

The CHSH and positivity facets of $\mathbb{L}(2222)$ can be obtained, sorted into equivalence classes, and printed out with the following sequence of commands:

```
polytope > $vL22 = vert_L_CG(2, 2, 2, 2) * prob_CGtoX(2, 2);
polytope > $L22 = new Polytope(VERTICES=>$vL22);
polytope > $fL22 = $L22->FACETS;
polytope > $orbits = vec_classes_X($fL22, [[0, 1]], [[0, 1]]);
polytope > $reps = fac_reps_X($orbits, 2, 2);
polytope > print_ineqs_X($reps, 2, 2);

<A_0> + <B_0> - <A_0 B_0> <= 1

<A_0 B_0> + <A_0 B_1> + <A_1 B_0> - <A_1 B_1> <= 2
```

The following set of commands obtains and sorts the vertices of $\mathbb{NS}(2222)$ from its facets and prints out one of the PR boxes:

```
polytope > $fNS22 = fac_NS_CG(2, 2, 2, 2) * transpose(prob_XtoCG(2, 2));
polytope > $NS22 = new Polytope(FACETS=>$fNS22);
polytope > $vNS22 = $NS22->VERTICES;
polytope > $orbits = vec_classes_X($vNS22, [[0, 1]], [[0, 1]]);
polytope > $reps = prob_reps_X($orbits, 2, 2);
polytope > $P = $reps * prob_XtoP(2, 2);
polytope > print $P;
1 1 0 1 0 0 0 0 0 1 0 1 0 0 0 0 0
1 1/2 0 1/2 0 0 1/2 0 1/2 1/2 0 0 1/2 0 1/2 1/2 0
polytope > table_prob_P($P->row(1), 2, 2, 2, 2, "txt");
+-----+-----+
| 1/2   0 | 1/2   0 |
| 0    1/2 | 0    1/2 |
+-----+-----+
| 1/2   0 | 0    1/2 |
| 0    1/2 | 1/2   0 |
+-----+-----+
```

The following example finds the facets of $\mathbb{L}(3322)$, composes the list of facets of the intersection of $\mathbb{D}_{A_0}(3322)$ and $\mathbb{D}_{B_0}(3322)$, and displays one of the 1248 vertices:

```

polytope > $vL33 = vert_L_CG(3, 3, 2, 2) * prob_CGtoX(3, 3);
polytope > $fL33 = new Polytope(VERTICES=>$vL33)->FACETS;
polytope > $orbits = vec_classes_X($fL33, [[0], [1, 2]], [[0], [1, 2]]);
polytope > print fac_reps_X($orbits, 3, 3);
...
polytope > $f_ict = $orbits->[4] / $orbits->[5] / $orbits->[6] / $orbits
->[8] / $orbits->[9] / $orbits->[10] / $orbits->[11];
polytope > $v_ict = new Polytope(FACETS=>$f_ict)->VERTICES;
polytope > print $v_ict->rows;
1248
polytope > $orbits = vec_classes_X($v_ict, [[0], [1, 2]], [[0], [1, 2]]);
polytope > $reps = prob_reps_X($orbits, 3, 3);
polytope > table_prob_P($reps->[0] * prob_XtoP(3, 3), 3, 3, 2, 2, "txt");
+-----+-----+-----+
| 1/3  1/3 | 2/3   0 | 1/3  1/3 |
| 1/3   0 | 0    1/3 | 1/3   0 |
+-----+-----+-----+
| 2/3   0 | 1/3  1/3 | 1/3  1/3 |
| 0    1/3 | 1/3   0 | 1/3   0 |
+-----+-----+-----+
| 1/3  1/3 | 1/3  1/3 | 1/3  1/3 |
| 1/3   0 | 1/3   0 | 1/3   0 |
+-----+-----+-----+

```

Finally, the following sequence sorts and prints L^AT_EX source for the 932 facets of the $D_{1122}(3322)$ polytope:

```

polytope > $vL11 = vert_L_CG(1, 1, 2, 2) * prob_CGtoX(1, 1);
polytope > $fNS22 = fac_NS_CG(2, 2, 2, 2) * transpose(prob_XtoCG(2, 2));
polytope > $vNS22 = new Polytope(FACETS=>$fNS22)->VERTICES;
polytope > $vD33 = combine_all($vL11, $vNS22, 1, 1, 2, 2, 2, 2);
polytope > $fD33 = new Polytope(VERTICES=>$vD33)->FACETS;
polytope > print $fD33->rows;
932
polytope > $orbits = vec_classes_X($fD33, [[0], [1, 2]], [[0], [1, 2]]);
polytope > $reps = fac_reps_X($orbits, 3, 3);
polytope > table_ineqs_X($reps, 3, 3, "tex");
\begin{array}{c|ccc}
& \backslash\mathrm{phneg}\,1 & \backslash\mathrm{phneg}\,1 & \backslash\mathrm{phneg}\,0 & \backslash\backslash\,\mathrm{hline} \\
\backslash\mathrm{phneg}\,1 & \& -\,1 & \& -\,2 & \backslash\mathrm{phneg}\,0 & \backslash\backslash \\
\backslash\mathrm{phneg}\,1 & \& -\,2 & \backslash\mathrm{phneg}\,2 & \backslash\mathrm{phneg}\,1 & \backslash\backslash \\
\backslash\mathrm{phneg}\,0 & \backslash\mathrm{phneg}\,0 & \backslash\mathrm{phneg}\,1 & \& -\,1 & \backslash\backslash \\
\end{array} \backslash\mathrm{leq}\,6
...

```

(in the L^AT_EX source for this document, $\backslash\mathrm{phneg}$ is a macro defined by $\backslash\mathrm{newcommand}\{\backslash\mathrm{phneg}\}\{\backslash\mathrm{phantom}\{-\}\}$.)

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